GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VIRGINIA 24153

INSTRUCTIONS GEK-40624A
MK II MS5002/3002
TURBINE CONTROL PANEL

GENERAL@ ELECTRIC

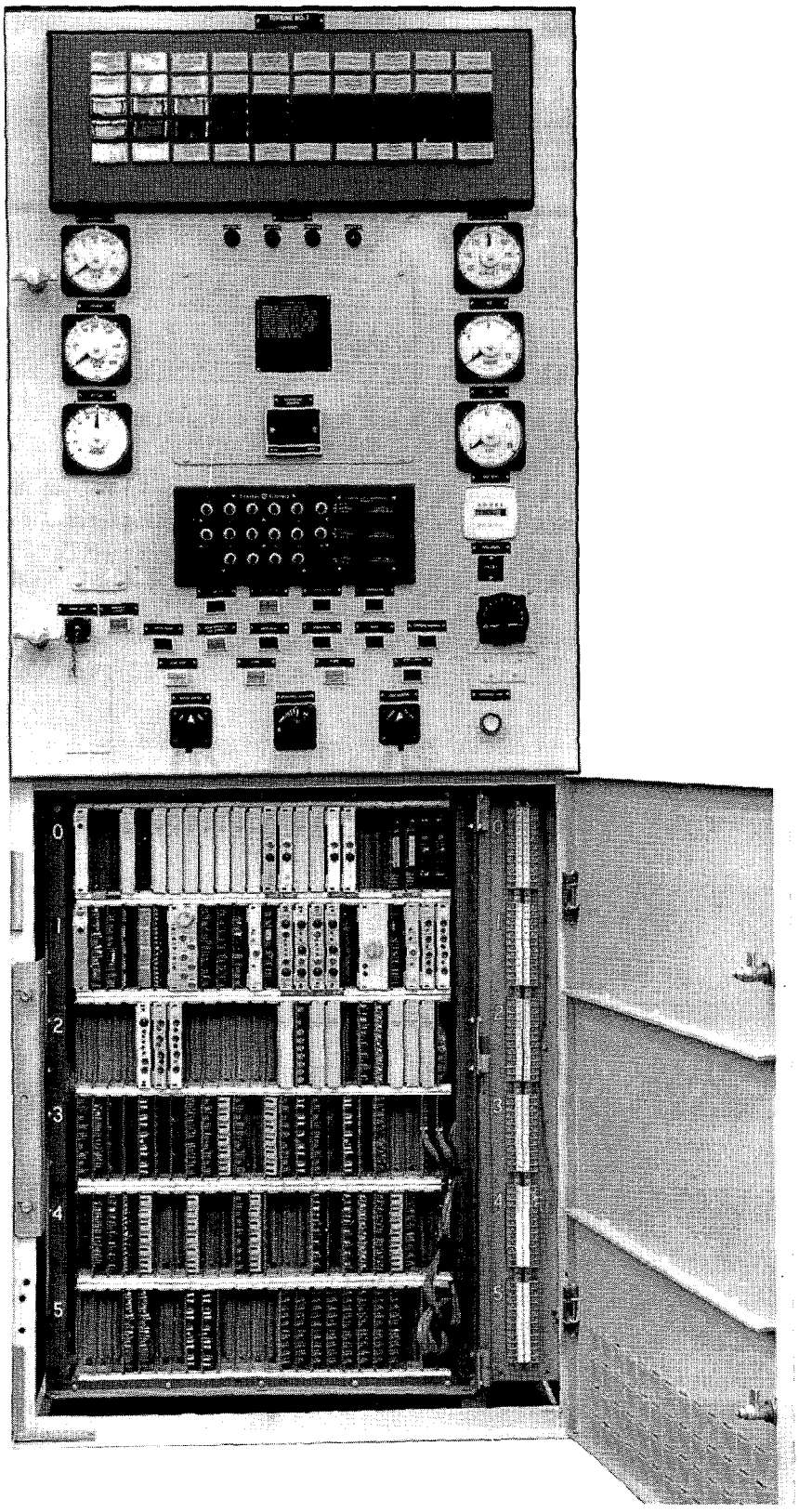


Fig. 1. Turbine control panel (MK II MS 5002/3002)

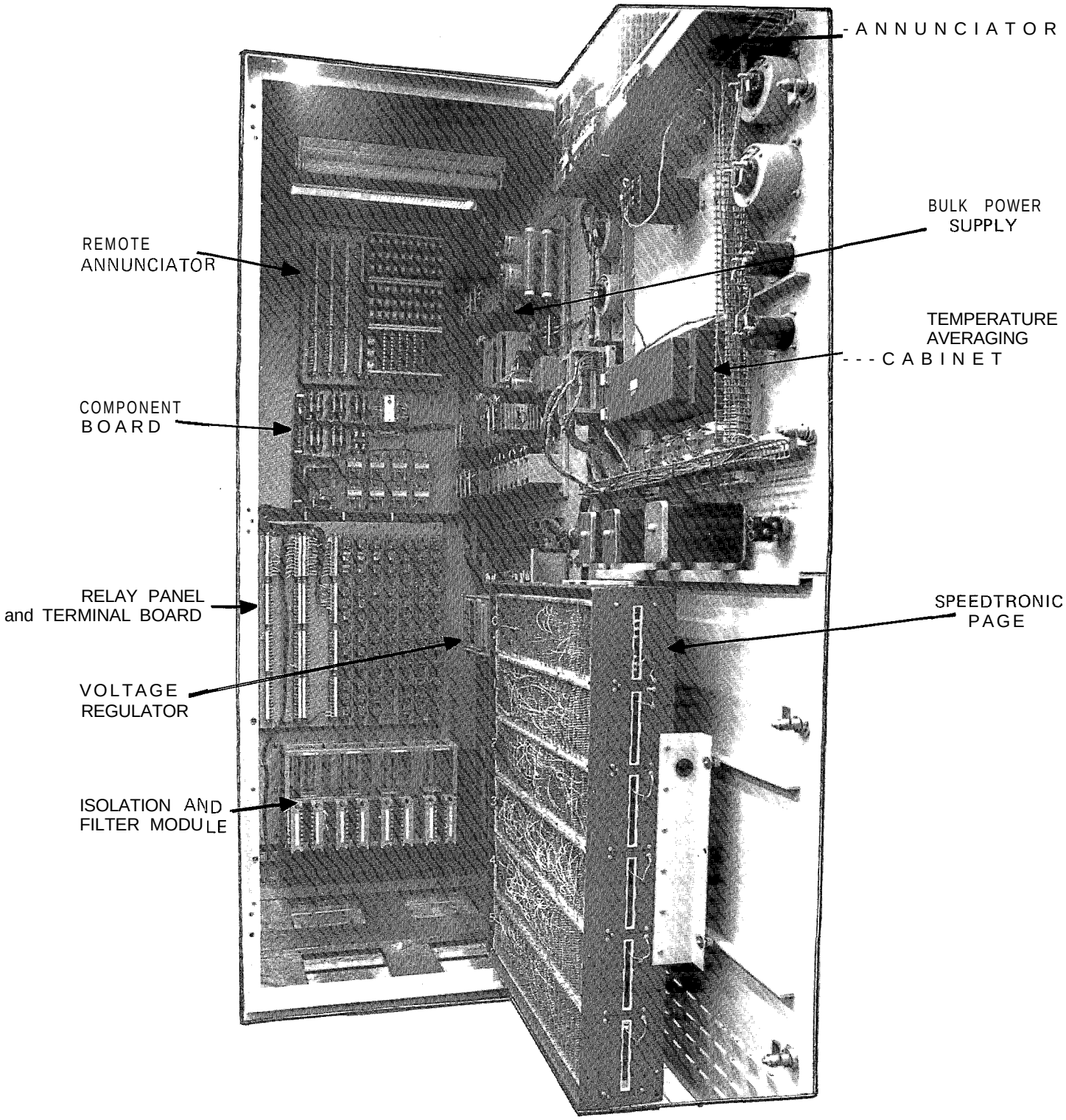


Fig. 3. Panel interior

CARD PIN NOMENCLATURE

(The following list will be of assistance in "reading" the card symbols. The abbreviations used are intended as cryptic notations and are not necessarily rigorous.)

- AL - Alarm
- ALM - Alarm
- ALR - Acceleration Limit Release
- AB - Ambient Bias
- ACC - Acceleration
- ANN - Annunciator
- ACC - Acceleration Control
- , BOV - Blow Off Valve
- C - Counter or Carry
- CALIB - Calibrate
- CL - Clear
- COMP - Compare
- DA - Differential Alarm
- DN - Down
- DO - Drop Out
- EXP - Exponential
- ESTOP - Emergency Stop
- FB - Feedback
- FFN - Feedback, Fuel Flow
- FG - Fully on Gas
- FL - Fully on Liquid
- FLT - Fault
- FS - Fast Start
- FS5 - Frame Size 5
- G -Gas
- GND - Ground
- HP - High Pressure Turbine
- HR - High Range
- INC - Increase
- INH - Inhibit
- INN - Input, Negative
- INP - Input, Positive
- IS0 - Isochronous
- LIQ - Liquid
- LP - Low Pressure Turbine
- LSB - Least Significant Bit
- LVDT - Linear Variable Displacement Transformer
- M - Meter
- MIN - Minimum
- MPU - Magnetic Pick-up
- MSB - Most Significant Bit
- MT - Minimum Test
- N - Speed
- NC - Speed Control/Nozzle Control
- NDO - Normal Drop Out
- NHP - Speed, High Pressure Turbine
- NLP - Speed, Low Pressure Turbine
- NOZ - Nozzle
- o c - Inverse Counter Input
- OFBF - Inverse Feedback Fault
- OFLT - Inverse Fault
- OR-EXO - "OR" expander
- OS - Overspeed
- OTA, B - Overtemperature A, B
- OUTA, B -Alarm Output A, B
- P - Purge
- PCD - Pressure, Compressor discharge
- PCDN - Pressure, Compressor discharge normalized
- PK - Peak
- PKOP - Peak Operation
- PR/A - Pulse Rate to Analog
- PR - Peak Reserve
- PRST - Preset
- PROT - Peak Reserve Over-temperature
- PU - Pick Up
- PUL - Pulse
- RA, B - Range A, B, etc.
- RD - Relay Driver
- REF - Reference
- RM - Run Mode
- RST - Reset
- SB - Shutdown Bias
- SC - Start-up Control
- SCA, B - Servo, Current Loop A, B, etc.
- SER - Servo
- SFT - System fault trip
- SIG - Signal
- SJ - Summing Junction
- SM - Start-up Mode
- SRT - Serto-Trip
- SP - Set Point
- SRV - Speed Ratio Valve
- ST - Steer
- TC - Thermocouple
- TC ADJ - Time Constant Adjust
- TP - Tie Point
- TP - Test Point
- TX - Average Exhaust Temperature
- UFD - Under Frequency Drop Out
- u s - Underspeed
- VCE - Variable Control Voltage
- VXDR - Voltage Transducer

SECTION A
INTRODUCTION

The MK II turbine control panel along with sensors and output servo valves controls the start-up, protection, and operation of the General Electric heavy-duty gas turbine. The primary functions provided by the SPEEDTRONIC* MK II Control are shown in Fig. 4.

Control System

The fuel control system consists of three major loops. These are startup, LP speed, and temperature. These loops are illustrated in Figure 5. These loops are tied together into a minimum value gate such that the loop calling for the least amount of fuel will be in command. In addition to the fuel control loops, a nozzle control loop is provided which positions the angle of the nozzle to control HP speed. The control system is designed to provide maximum running reliability. Redundant sensors are frequently used to supply feedback signals. Reliability is further enhanced by "redundancy by association." For example: If the system is operating on speed control, the temperature control will back it up.

Sequential

Sequential circuitry is provided to monitor the turbine, starting means, and auxiliaries to determine if all systems are normal. The return signal from these devices serve as permissives to operate the turbine. Abnormal conditions are annunciated via local and remote (optional) annunciators. The sequential circuitry also generates logic signals to facilitate starting and stopping the turbine *in* a prescribed manner. These logic signals include speed signals, digital setpoint control, starting means control, timers, and others.

Protection

Protection circuitry is provided to trip the turbine when critical parameters are exceeded or critical equipment fails. Alarm annunciation is provided to indicate when critical levels are being approached and to indicate abnormal equipment operation.

Power Supply

The control panel contains its own highly reliable solid-state power supply. The power supply consists of a d-c to d-c converter and final regulators for P28, P12, P5.3, and N12 volts. The converter will operate satisfactorily with input voltage of 95 to 140 volts d-c. A source selector which will permit operating from a 95- to 140-volt d-c battery, commercial power, or inductor alternator is optional.

Publications and Repair Notes

The intent of this instruction book is for use in conjunction with the (1) Turbine Control Specification, (2) Control Panel Elementary Diagram, (3) Schematic Piping Diagram, (4) Operating Instructions, and (5) Turbine Control Elementary Reading Aid. This instruction book is typical and does not purport to cover all details or variations that may arise on a particular application.

When troubleshooting the Turbine Control Panel, the test points provided on the calibrator (if available) and card fronts should be used whenever possible to avoid damage to the cards that often results when connecting test probes to the pins on back of the SPEEDTRONIC nage. If it is absolutely necessary to troubleshoot on the back of the page, insulated test probes should be used. Recommended procedures concerning removing or replacing cards with power applied is given in Technical Bulletin 060 TB l-01-001.

Failed cards and modules should be returned to the General Electric Company for repairs. No attempt should be made to repair the card in the field. The procedure for returning cards and modules can be obtained from your nearest General Electric representative.

*Trademark of General Electric Company.

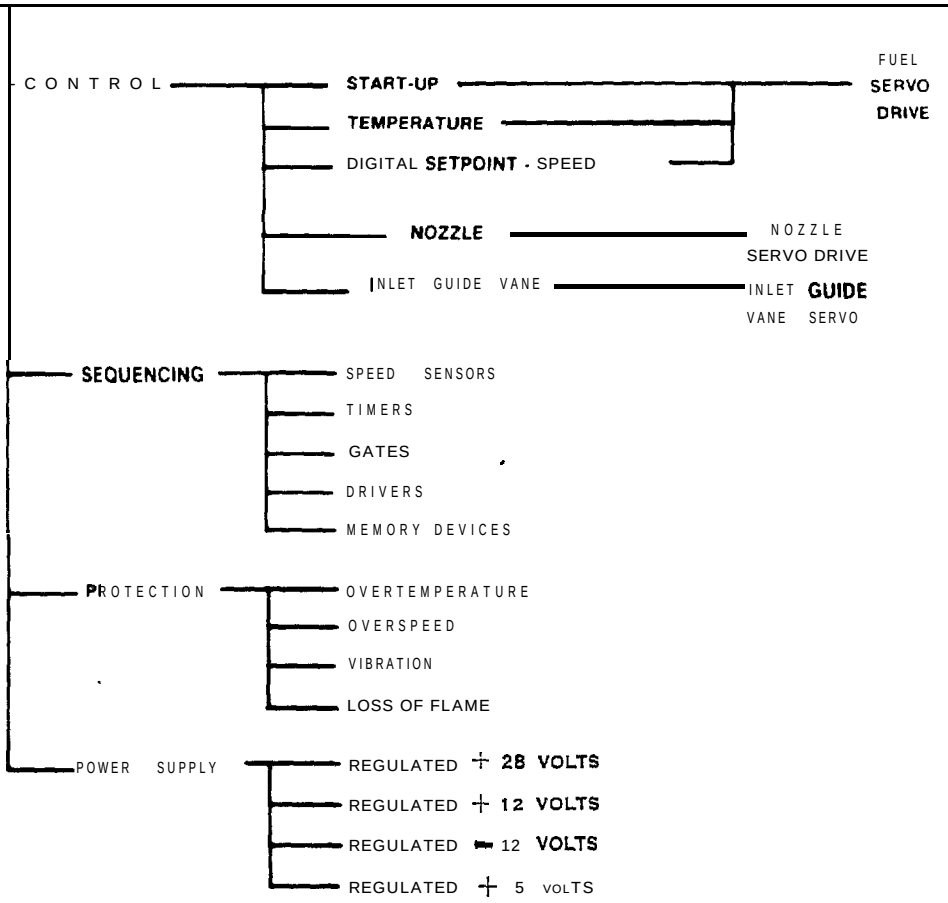


Fig. 4. SPEEDTRONIC MK II*

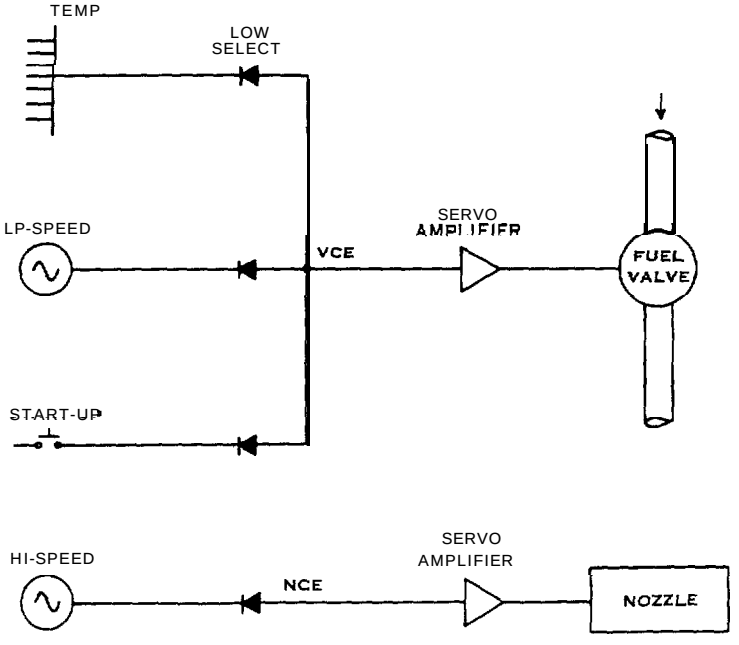


Fig. 5. Fuel control system

*Trademark of General Electric Company.

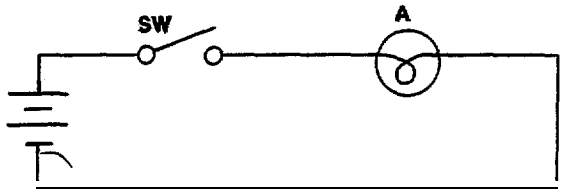
SECTION B
SEQUENTIAL CONTROL

B-I LOGIC INTERPRETATION

A brief discussion of logic signals is presented to aid in understanding the sequential circuitry. A comparison is made between relay logic and solid-state logic to help those familiar with relay logic, but not familiar with solid-state logic, to read solid-state logic.

A logic signal has only two possible states. These states have various names in different literature. In this instruction book the two states will be defined as "one" ("1") and "zero" ("0"). A "1" will be used to indicate that some event has occurred or is occurring. A "0" will indicate that some event has not occurred or is not occurring. In the MK II system a "1" state is usually represented by +5 volts and a "0" state is usually represented by zero volts.

If the state of some event is designated by using the variable A, then when A equals "1" the event is occurring or has occurred and when A equals "0" the event is not occurring or has not occurred. For example, in the circuit below, the event is defined as the illumination of the light, and the variable A is used to represent the state of the event. When the switch is closed, A is equal *to* "1' (lamp illuminated) and when the switch is open, A is equal to "0" (lamp is out).



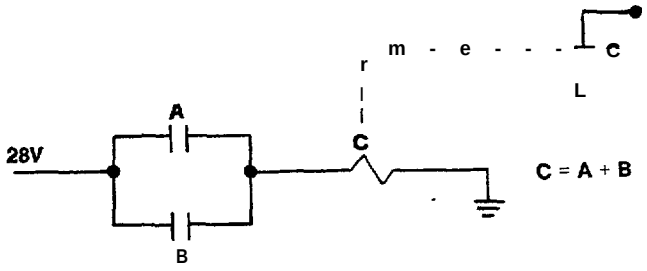
The mathematics of logic signals is referred to as Boolean Algebra. A very brief discussion of Boolean Algebra is given subsequently. For each Boolean operation, both relay logic and Directo-Matic* II logic to perform the operation is given.

There are three basic operations associated with Boolean Algebra. These are:

(1) LOGIC ADDITION **("OR" FUNCTION):** Logic addition is designated by the symbol "+". The addition operation is defined as:

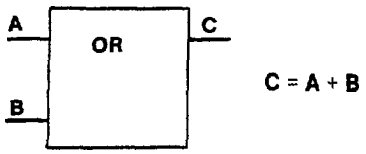
$$1 + 1 = 1$$
$$1 + 0 = 1$$
$$0 + 1 = 1$$
$$0 + 0 = 0$$

(a) Relay logic for addition (OR).



Relay C will be picked up if either contact A or contact B is closed.

(b) Directo-Matic II solid-state logic for addition (OR).

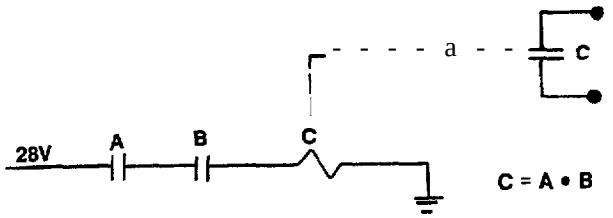


Output C will be "1" (+5 volts) if either input A or B is a "1" (+5 volts).

(2) LOGIC MULTIPLICATION ("AND" FUNCTION): Logic multiplication is designated by the symbol "•" The multiplication operation is defined as:

$$1 \cdot 1 = 1$$
$$1 \cdot 0 = 0$$
$$0 \cdot 1 = 0$$
$$0 \cdot 0 = 0$$

(a) Relay implementation.

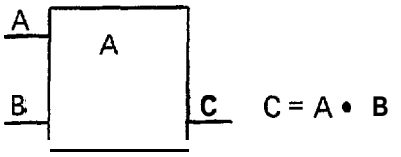


If A and B are made, relay C will be picked up.

NOTE: The symbol "•" is often omitted, i.e. A•B = AB.

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(b) Directo-Matic* II logic implementation.

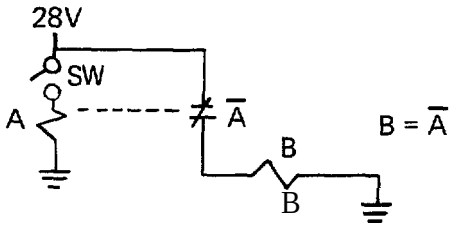


If A = "1" and B = "1" then C will be "1." For any other combination of A and B, C is "0."

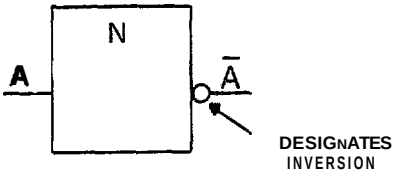
(3) LOGIC NEGATION ("NOT" FUNCTION): Logic negation (also called inverse or complement) is designated by placing a bar above the variable. (Example: If A is the variable, then $\bar{A}$ is the inverse of A.) The inverse of A is defined as follows:

When A = "1", $\bar{A}$ = "0"
When A = "0", $\bar{A}$ = "1"

(a) Relay implementation.



(b) Directo-Matic II implementation.



Logic operations can be combined to give logic equations such as:

$$A = (C \bullet B) + D$$

NOTE: ***The priority of the operations are •, +, NOT. Therefore, if no parentheses are included, all the "•" operations are completed first; then the "+" and the negation is last. Example: A = C + B•D is understood to be A = C + (B•D).***

Logic equations containing terms such as A = B•D can be simplified using the following technique.

To simplify an expression such as $\overline{B \bullet D}$, substitute "+" for "•" and complement each individual variable. For example, the expression $\overline{B \bullet D}$ simplifies to $\bar{B} + \bar{D}$. Likewise $\overline{B+D}$ simplifies to B • D. This procedure can be extended to more complex expressions. The following example illustrates the technique used.

Example: $E = \overline{\overline{A \bullet B} \bullet (C+D)}$

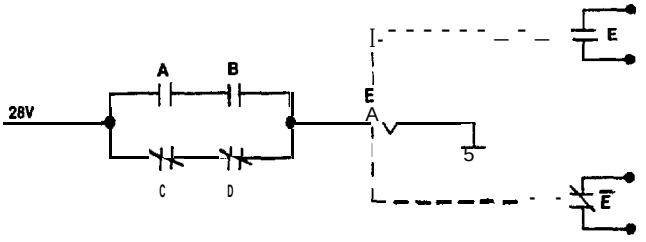
(1) $E = \overline{\overline{A \bullet B}} + \overline{(C+D)}$

(2) $E = A \bullet B + \overline{(C+D)}$

(3) $E = A \bullet B + (\bar{C} \bullet \bar{D})$

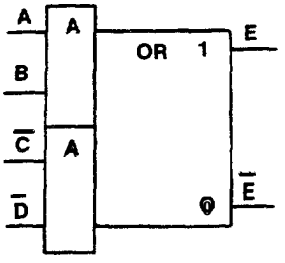
The implementation of logic equations using both relay and Directo-Matic II logic is illustrated below.

(a) Relay:



$E = (A \bullet B) + (\bar{C} \bullet \bar{D})$

(b) Directo-Matic II

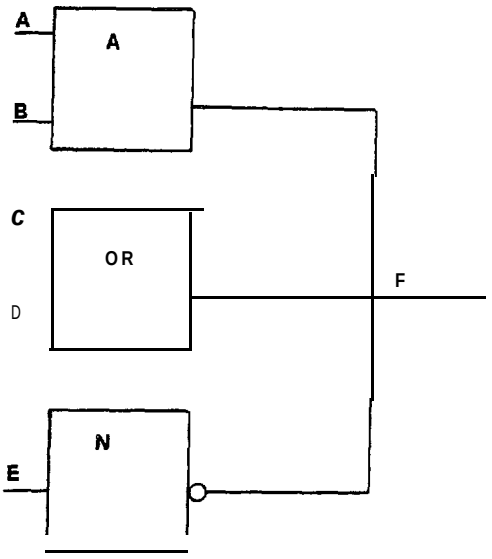


$E = A \bullet B + \bar{C} \bullet \bar{D}$

The 1 and 0 located inside the symbol adjacent to outputs E and $\bar{E}$ respectively are often used to designate the normal and inverse outputs. The circuit is said to be in the true state when the output adjacent to 1 is a logic one. It is said to be in the false state when the output adjacent to 0 is a logic one. It should be noted here that in the case where

*Trademark of General Electric Company,

the outputs of more than one logic element are tied together, a "0" will dominate over-a "1." The following circuit illustrates this.



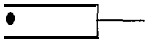
F is equal to "0" if any of the following conditions are true.

(1) A + B= 0
(2) C . D = 0
(3) E = 1

PRECAUTION: ***A "I " can be forced to a "0" by jumpering the signal to common; however, trying to force a "0" to*** a ***"1" by jumpering the signal to +5.3 volts could result in damage to the circuit.***

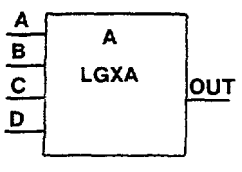
B-2 DIRECTO-MATIC* II LOGIC CARDS AND SYMBOLS†

†This section is merely to familiarize the reader with Directo-Matic II logic cards. For specific information consult the Directo-Matic II Card Application Data book.

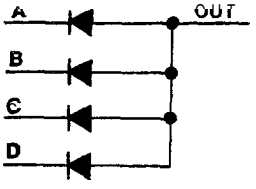


The presence of the "•" at the input terminal of an element denotes the ability to accept either normal logic outputs or the output of a LGXA.

IC3600LGXA - "AND" Gate Expander



9 CIRCUITS PER CARD

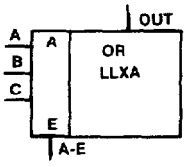


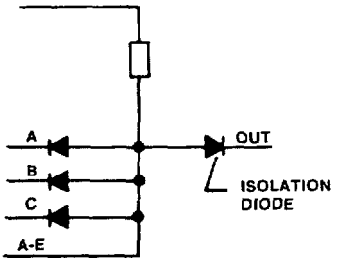
CIRCUIT

*Trademark of General Electric Company

The LGXA provides a cluster of diodes for expanding the "AND" inputs of many Directo-Matic II logic cards. An illustration of how the LGXA is used is given in the LCDA discussion. When the OUT terminal of the LGXA is connected to the A-E input of another logic element, a "0" on any input will pull OUT to "0." A "1" on all inputs will allow OUT to go to a "1."

IC3600LLXA - "OR" Logic Expander



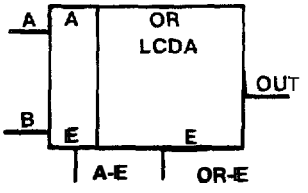


9 CIRCUITS PER CARD

The LLXA provides circuitry for input expansion of logic elements having an OR-E input. If all inputs are "1," the isolation diode will be forward biased and pull OR-E input of connected logic element to a "1" which makes the normal output of that element a "1." If any input is a "0," the isolation diode will be reversed biased and allow the output of the connected element to be controlled by its normal inputs. An A-E input is provided for expanding the "AND" inputs using a LGXA. The use of the LLXA will be illustrated in the LCDA discussion.

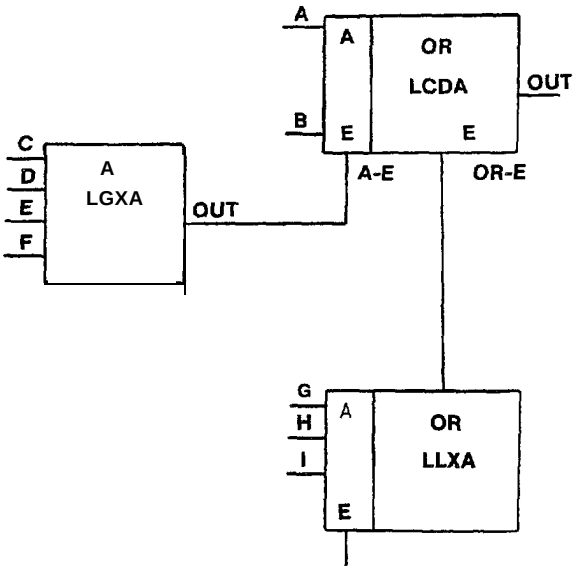
NOTE: For most Directo-Matic II cards, an unused input ***left open*** *is recognized* as a *"1."*

IC3600LCDA - Clock Driver



9 CIRCUITS PER CARD

The LCDA clock driver contains a power amplifier which will drive 32 "L" logic loads. In addition to the two normal "AND" inputs, A-E and OR-E inputs are provided. If no connection is made to A-E or OR-E, the circuit performs as a normal two input "AND" gate with power amplifier. An illustration of how the A-E and OR-E inputs could be used is illustrated and followed by its truth table.



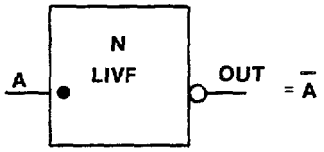
OUT = **(A • B • C • D • E • F) + (G • H • I)**

TRUTH TABLE*

A	B	C	D	E	F	G	H	I	OUT
0	0	0	1	1	1	0	1	1	0
ϕ	ϕ	ϕ	1	1	1	1	1	1	1
0	0	1	1	1	1	0	1	1	0
0	1	1	1	1	1	0	1	1	0
1	0	0	1	1	1	0	1	1	0
1	0	1	1	1	1	0	1	1	0
1	1	0	1	1	1	0	1	1	0
1	1	1	1	1	1	0	1	1	1
1	1	1	1	1	1	1	1	1	1

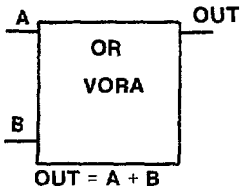
*In order to simplify table, assume inputs D, E, F, H and I are always a "1." For example, only inputs A, B, C and G can change states.
9 equals "don't care" if a "1" or "0."

IC3600LIVF - Inverter

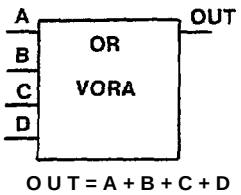


The LIVF provides the function of logic inversion. A "1" in will give a "0" out and a "0" in will give a "1" out.

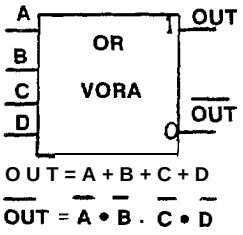
IC3600VORA Logic OR



8 CIRCUITS PER CARD



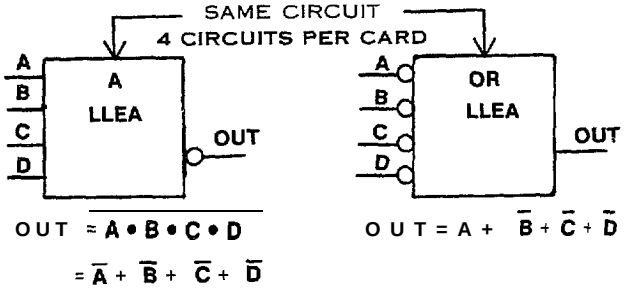
2 CIRCUITS PER CARD



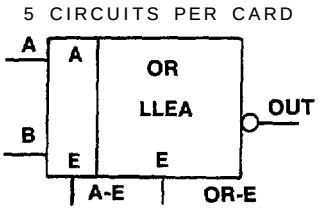
2 CIRCUITS PER CARD

This card provides a true OR function. A "1" on any input will give a "1" on OUT. Two circuits provide an inverse output OUT. For this circuit a "1" on any input will give a "1" on OUT and a "0" on OUT. If all inputs are "0," OUT is "0" and OUT is "1."

IC3600LLEA • Logic Element



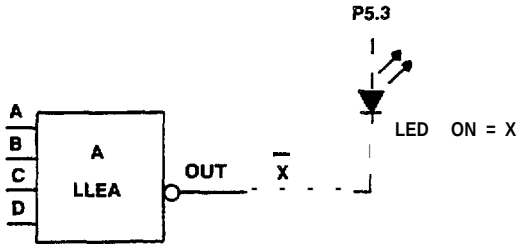
5 CIRCUITS PER CARD



OUT= A . B . (A-E) + (OR-E)
OUT = [Ā + B̄ + (A-E)] • (OR-E)

The LLEA performs the NAND (NOT AND) function. Five circuits per card have A-E and OR-E expander inputs for connection to LGXA and LLXA respectively. The remaining 4 circuits per card are typical NAND gates. Either of the first two LLEA symbols shown on page 13 may be used for these circuits. The symbol used is determined by the logic level desired at OUT to represent the active condition of the inputs. The desired logic level is defined by the output signal nomenclature. The following examples will illustrate this.

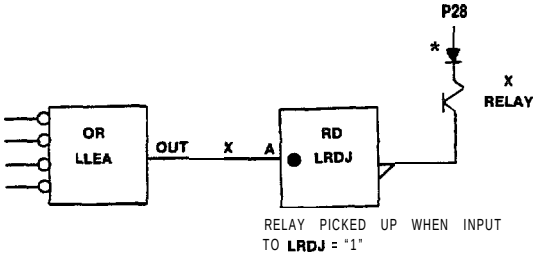
Example 1:



*Used for illustration only; not recommended method.

The active condition is when the LED is on. If a "0" is desired to represent the active condition, the above symbol is used and the output is labeled $\overline{X}$. The desired logic level is dictated by circuit that is being driven by the LLEA.

Example 2:



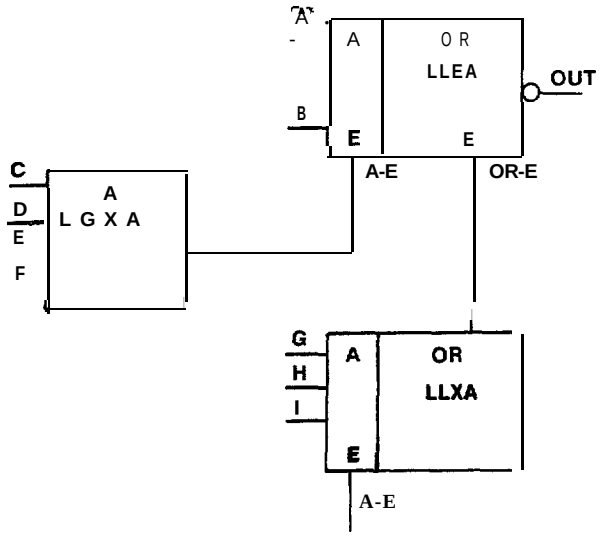
*Used for illustration only; not recommended method.

The active condition is relay X energized. If a "1" is desired at OUT to represent active condition and pick up relay, this symbol is used and output of LLEA labeled X.

LLEATRUTHTABLE
FOR CIRCUITS WITHOUT EXPANDER INPUTS

A	B	C	D	OUT
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

The following circuit illustrates the use of the expander inputs. The use of both A-E and OR-E are illustrated; however, either may be used separately.

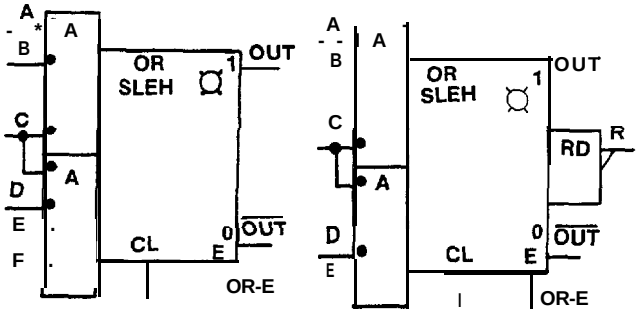


TRUTH TABLE FOR EXPANDER CIRCUIT
(Assume D, E, F, H, and I are "1" at all times)

A	B	C	G	OUT
∅	∅	3	0 1 0	1
0	0	1	0	1
0	∅	0	0	1
0				
∅	‖	0 1	0 0 1	1
∅	0	10	0	1
1	1	0	∅	1
1	1	1		0

∅ equals "Don't care" if "1" or "0."

IC3600SLEH - Logic Element



2 CIRCUITS PER CARD

$OUT = A \cdot B \cdot C + D \cdot E \cdot F \cdot C$

$\overline{OUT} = (\bar{A}+\bar{B}+\bar{C}) \cdot (D+E+F+C)$

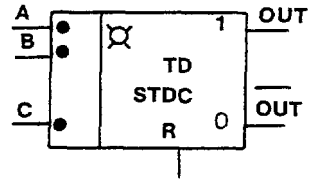
3 CIRCUITS PER CARD

$OUT = A \cdot B \cdot C+D-E*C$

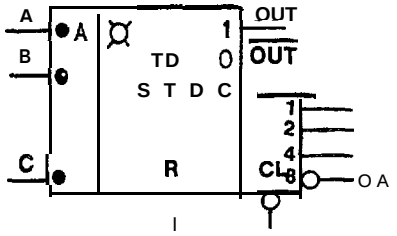
$\overline{OUT} = (\bar{A}+\bar{B}+\bar{C}) \cdot (D+E+C)$

The basic SLEH card consists of an "OR" logic element gated by two "AND" gates. If either "AND" gate has all "1" inputs, OUT will be a "1." An OR-E expander input is provided for use with the LLXA. Three circuits per card provide a relay driver output. The relay driver provides a closure to common when OUT is a "1."

IC3600STDC - Time Delay



3 CIRCUITS PER CARD

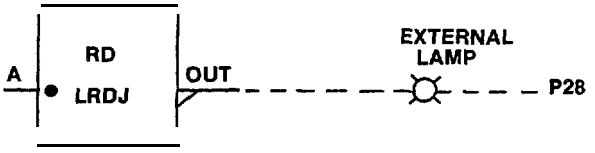


1 CIRCUIT PER CARD

This card provides a time-delay function for relatively long periods of one second to five minutes. The timing is initiated by all inputs being a "1." If all inputs remain a "1," OUT will become a "1" and $\overline{OUT}$ a "0" after the time delay set by adjustment potentiometer. OUT will go from a "1" to a "0" immediately, if any input goes to a "0." The LED will be on from the instant timing is initiated until OUT becomes a "1."

The time delay provided is divided into two ranges *to* give *more* adjustment resolution. The short range (pin R open) is adjustable from 1 to 25 seconds. The long range (pin R to N12) is adjustable from approximately 20 seconds to 5 minutes. There are four circuits per card. One circuit provides an eight-count counter for additional time delay. By connecting OUT of this circuit to one of the inputs, and OA to another, the timer will cycle eight times before stopping with OA equal "0." This will provide a delay up to 40 minutes.

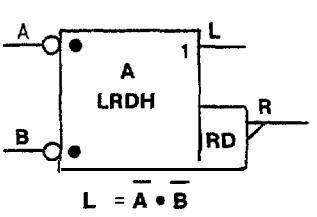
IC3600LRDJ - Relay Driver



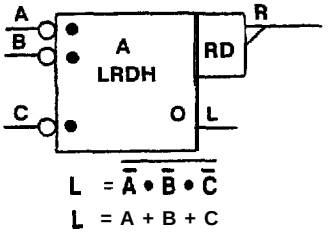
16 CIRCUITS PER CARD

This circuit provides a closure to common in response to a "1" input, an open circuit in response to a "0" input. The circuit is intended to drive a load such as a relay connected between OUT and positive supply voltage not greater than P28.

IC3600LRDH - Relay Driver



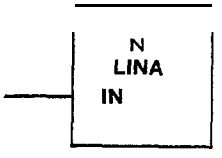
5 CIRCUITS PER CARD



2 CIRCUITS PER CARD

These circuits provide a closure to common for a load between P28 and the "R" output in response to proper input logic. In addition, they also provide a logic output signal at "L" in response to the input logic. Circuits 1 thru 5 require both inputs to be a logic "0" to provide closure to common and to generate a logic "1" at the output. Circuits 6 and 7 are similar to 1 thru 5 except that a 3-input "AND" gate is provided instead of a 2-input gate and the logic output is a "0" when closure to common is provided.

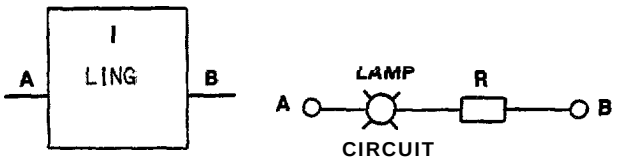
IC3600LINA - Light Indicator



16 CIRCUITS PER CARD

The LINA includes a light amplifier and indicating light. The light will be illuminated when the input is a "1." Inputs for unused circuits must be tied to common to prevent light staying on.

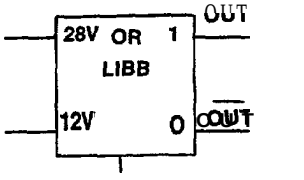
IC3600LING - Indicator



16 CIRCUITS PER CARD

The LING provides indicating lamps to be driven by display drivers, relay drivers, or contact closing. Each circuit is completely independent. This card cannot be operated by a normal logic signal.

IC3600LIBB -&put Buffer

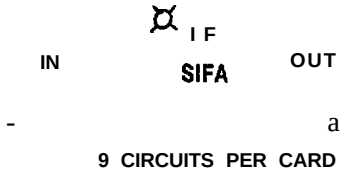


9 CIRCUITS PER CARD

The LIBB converts relay contact inputs to logic levels. Inputs are provided for contact closure to either 28 volts or 12 volts. When contact to supply is closed, OUT is a "1" and $\overline{OUT}$ is a "0." The 28-volt input should not be used off page, When 28 volts is connected to an off-page contact, the signal must be conditioned by an input filter circuit.

NOTE: LIBA card works the same as LZBB but has different rating *and* timing. *DO NOT interchange cards.*

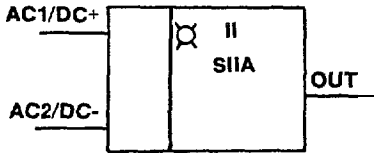
IC3606SIFA - Non-isolated Input Filter



9 CIRCUITS PER CARD

The SIFA is designed to receive a 28-volt signal, filter the signal, and provide drive to the 12-volt input of the LIBB card. An LED is on when input is at 28 volts.

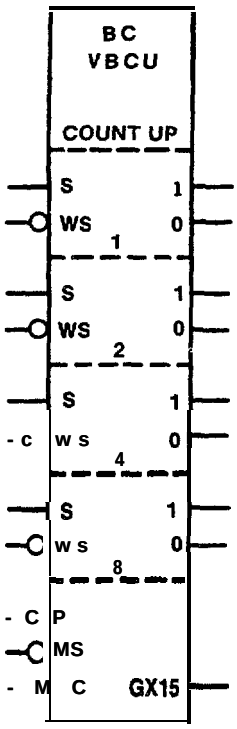
IC3606SIIA - Isolated Input Filter



9 CIRCUITS PER CARD

The SIIA provides isolation and filtering of high level (125 volts d-c or 115 volts a-c) signals. The output is designed to drive the 12-volt input of the LIBB card.

IC3600VBCU -Binary Counter



2 CIRCUITS PER CARD

The VBCU card contains two 4bit counters. The counter will count up one count each time the signal on input P changes from a "0" to a "1." Each bit has a "normal" and "inverse" output. A logic "1" on MC will clear all bits. A logic "0" on MS will set all bits whose S input is a "1." Input WS is provided for wired OR expansion of set operation. S must be a "0," or MS must be a "1" for this function. Output GX15 will be a "1," if all four bits are set. The two 4-bit counters may be operated as an 8-bit counter by connecting the "0" output of the lower order Bit 8 count to the P input of the second counter.

B-3 ANNUNCIATOR

The annunciator on the Directo-Matic* II MS-5002/3002 turbine control panel provides the circuitry to annunciate up to 60 alarm conditions. The alarms are latching and will remain, even if the alarm condition is momentary, until cleared by a push button on the panel. An alarm may be annunciated by the sounding of a bell and the flashing of the alarm's associated Iights.

The alarm indicators are referred to as drops. Special features which will be discussed subsequently are provided to handle multiple alarms. A block diagram is given in Fig. 7, page 19. It consists of one annunciator master card (VANB) and up to 15 annunciator cards (VANA). Each VANA card contains four circuits; therefore, up to 60 drops are available. The standard system normally consists of 40 drops.

There are three operator push buttons associated with the annunciator. (1) The "lamp test" push button provides a means to check for burned-out bulbs and faulty lamp drivers. (2) The "alarm reset" push button will silence the alarm bell and cause the flashing light to become steady on. It also enables an additional alarm to come on flashing. (3) The "lamp reset" push button will clear the light if the alarm condition has been removed. An automatic lamp reset is available to automatically reset the lamp when the alarm condition is removed.

The flow chart (See Fig. 6, page 18) is included to illustrate some typical sequences that may occur in the case of multiple alarm conditions. Alarms must occur at least 36 milliseconds apart for the annunciator to determine which occurred first. If there is less than 36 milliseconds between alarms, both will come on flashing.

The VANB card provides the control signals to the VANA card. (See Fig, 7). It also contains the driver for relay 30X whose contacts are used to drive the alarm bell. VANB output FL provides the flashing signal to all VANA cards. It is present at all times.

The signal flow for a typical alarm condition will be given to aid in explaining the annunciator.

Assuming that no alarm condition is present, output AHD of VANB will be a "1." This is a flashing permissive to all VANA cards. Now assume that a fault occurs (a "1" on L). On the VANA card, the flashing signal appears on LT and the corresponding indicator light flashes, and output AH becomes a "0." This "0" goes to VANB input AH and performs two functions. It places a "0" on HRD and picks up relay 30X which causes the bell to sound, and output AHD becomes a "0" which removes the flashing permissive signal.

The first fault remains flashing but any more faults occurring will come on steady. Suppose that the "alarm reset" push button is now depressed, the VANB output ACK will become a "1" which also performs two functions in the VANA card. It removes the flashing signal from LT and replaces it with a steady "0" which makes the light remain steady on. It also causes output AH to become a "1." This "1" on AH goes to VANB and silences the bell. It also allows VANB output AHD to become a "1." This "1" on $\overline{\text{AHD}}$ is a permissive to all other VANA cards to flash if an additional alarm occurs.

Now, assume that no other alarm condition occurs, and the present fault condition is cleared. Depressing push button "lamp reset" places a "0" on the $\overline{\text{CL}}$ bus and the LT signal on VANA becomes a "1" which turns the lamp off.

*Trademark of General Electric Company.

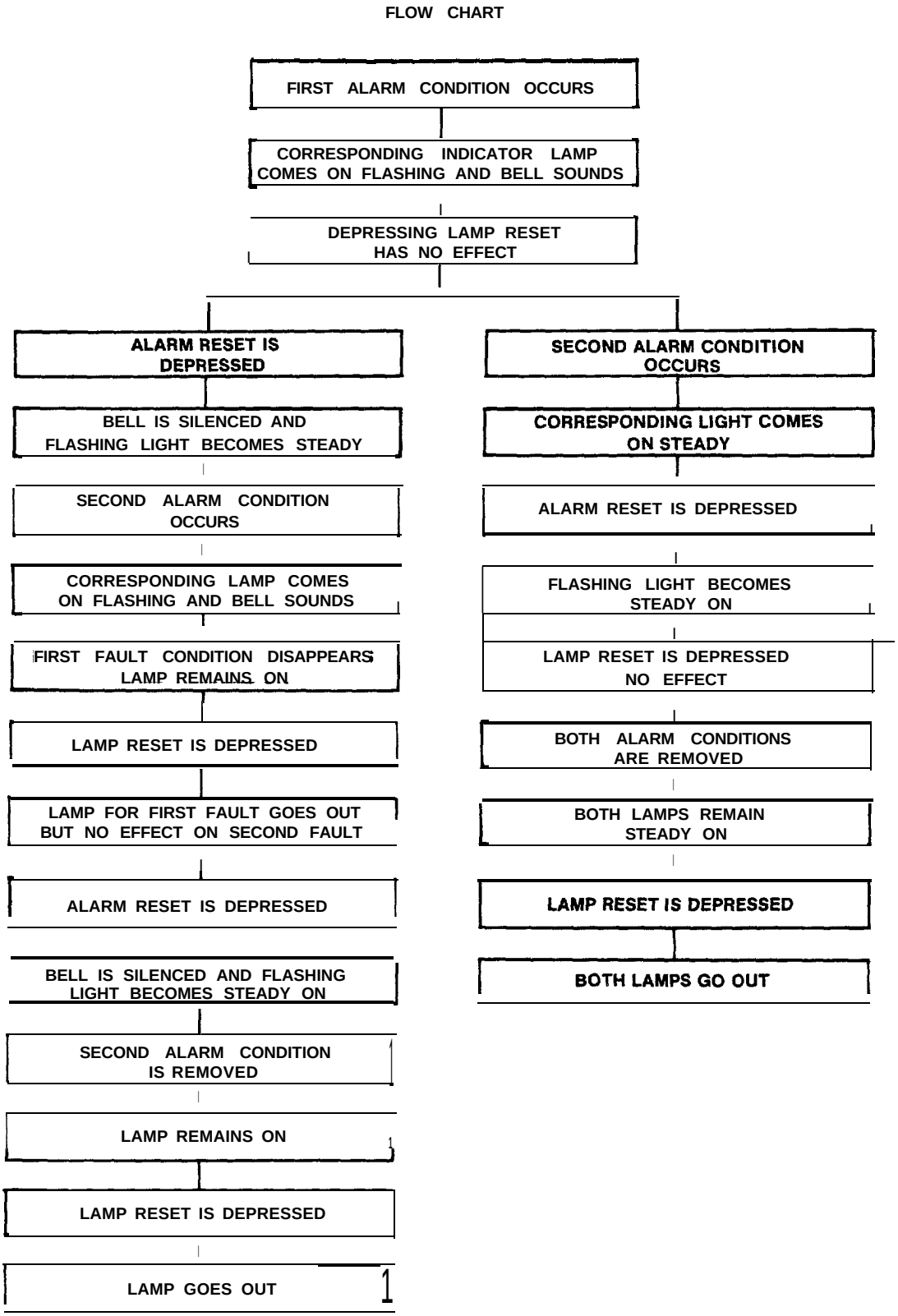


Fig. 6. Flow chart

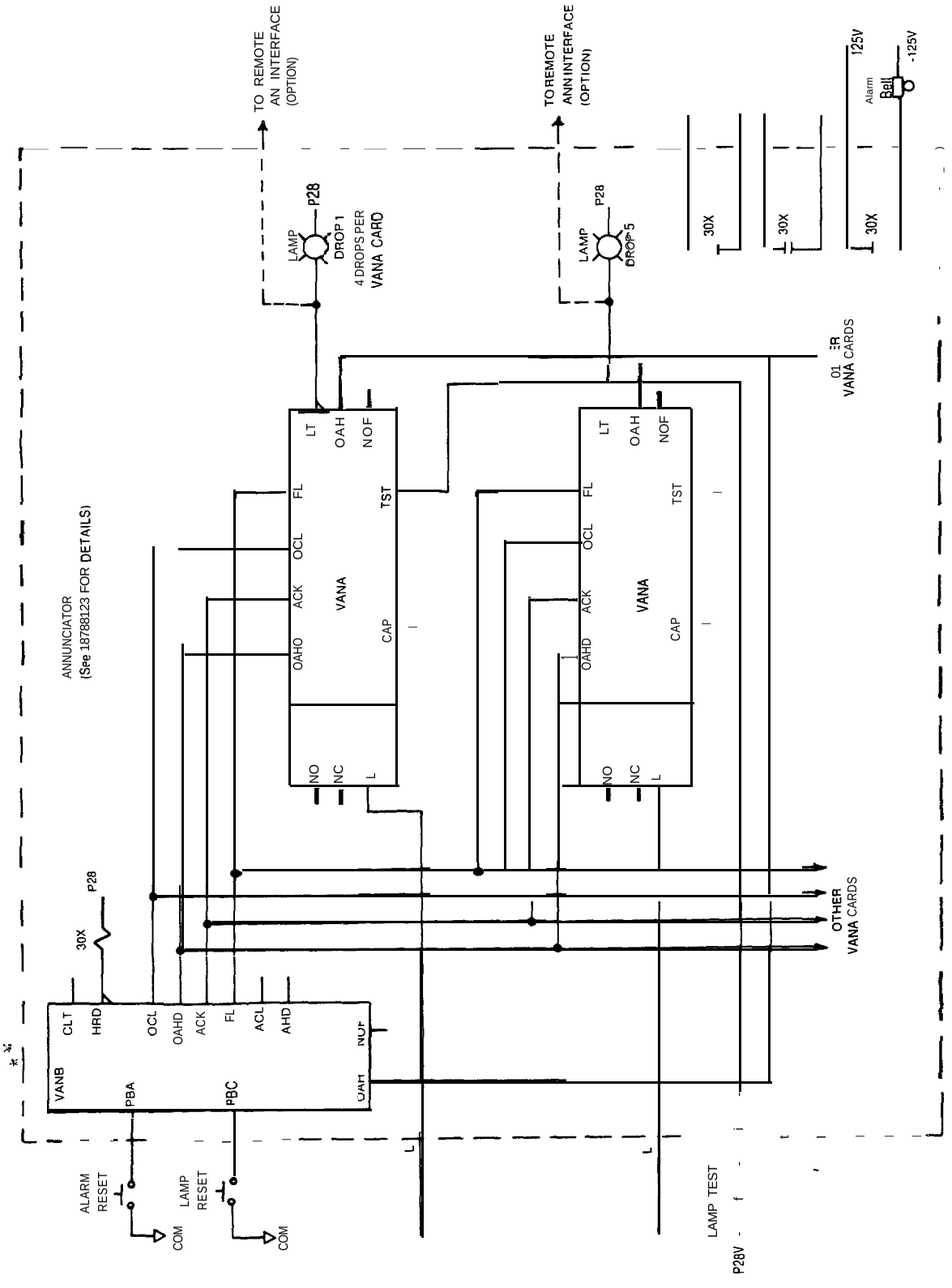


Fig. 7. Typical annunciator

B-4 REMOTE ANNUNCIATOR INTERFACE MODULE (OPTION)

A remote annunciator interface module is available to provide alarm indications at a remote location. The remote terminal is completely isolated from the turbine control panel. The remote terminal provides its own power-supply voltage and alarm indication is via relay contact closure in the remote annunciator interface module. Refer to turbine control panel elementary sheets 18A and 18B for connectron between remote interface module and local annunciator. Provision is made for three operator push buttons to be located at the remote terminal to perform the same function as the three local push buttons. The remote push buttons pick up relays in the interface module. The contacts of these relays are either paralleled or connected in series (whichever is appropriate) with the local push buttons.

When an alarm occurs, the local annunciator operates as discussed previously. The output of the VANA card that drives the alarm lamp is also routed to a corresponding relay in the interface module. Therefore, the relay in the remote annunciator interface module is picked up when the lamp in the local annunciator is on. The relay is normally connected such that it is latching and does not follow the flashing light. However, a flashing remote alarm is available. In the case where the relay is latching, when the "lamp reset" push button is depressed the latching path is removed and the relay will drop out if the alarm condition has been removed.

B-5 SPEED SENSORS

The speed sensor cards receive analog NHP and NLP signals from the PR/A converter on the SSZB cards, and frequency inputs from magnetic pickups. These sensor cards provide logic signals to indicate that the turbine shaft, or other rotating device has reached a certain speed.

SVSE Card (See Fig. 8, 9)

This card contains five speed sensors with given (fixed or adjustable) pickup and dropout characteristics. These are:

14HM - minimum speed
14HA - accelerate speed
14HS - operating speed
14HR - zero speed
NMR - speed monitor

The first three sensors operate from a common input signal NHP. The zero-speed sensor operates directly from an electromagnetic pickup. The speed monitor receives a 0- to 10-volt signal and gives a logic output at MR and MR for a preset input level.

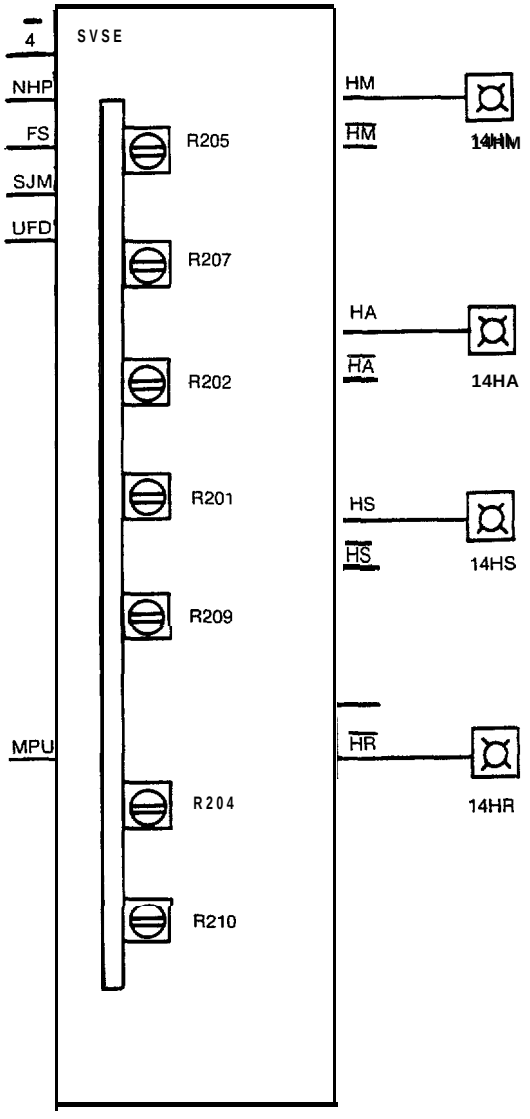


Fig. 8. HP Speed sensors

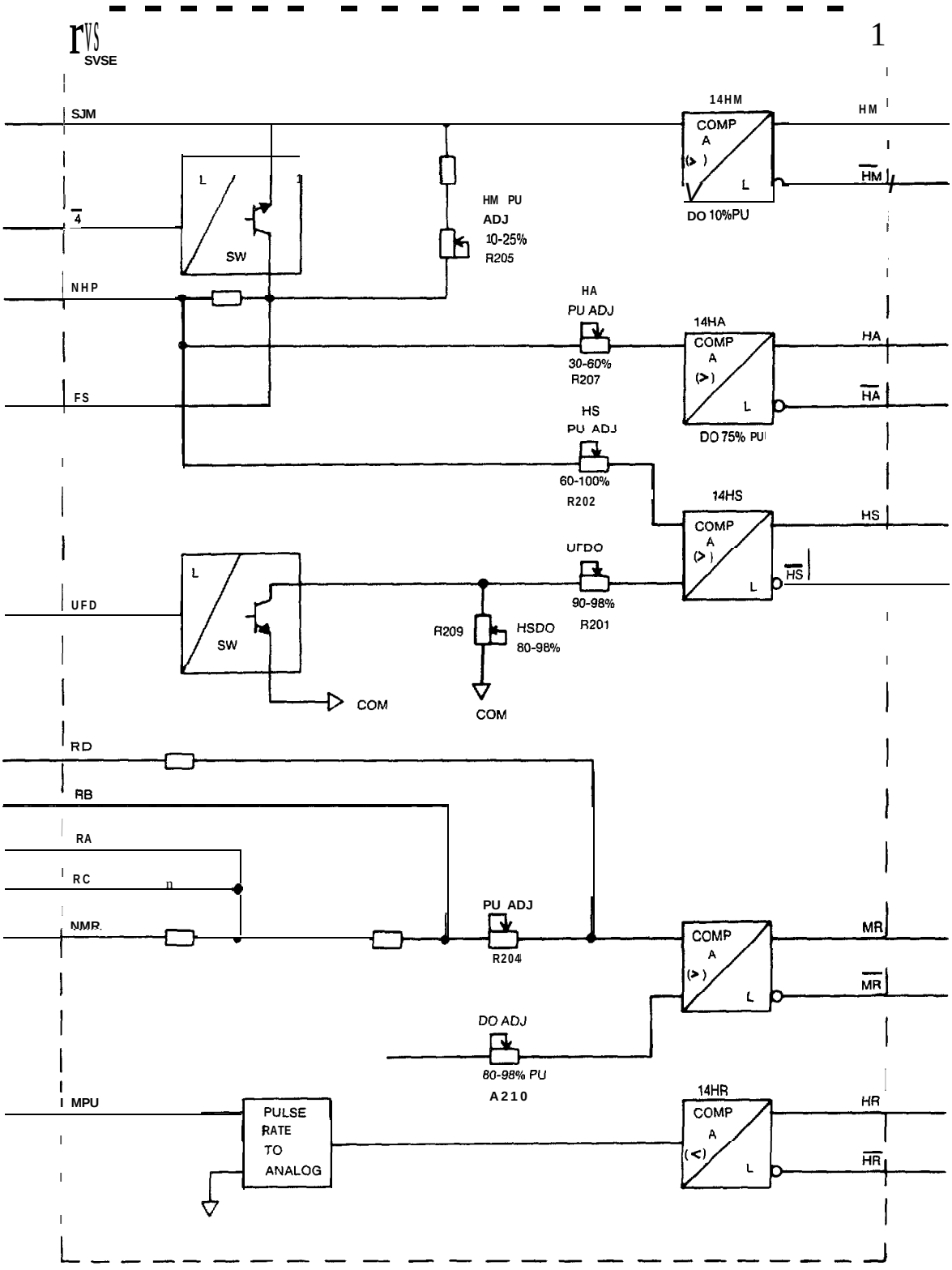


Fig. 9. SVSE card

The 14HM pickup and dropout points are programmable by logic inputs on $\overline{4}$, FS and SJM. A "1" on 4 reprograms the pickup point to 2% and dropout to 1.8%. Resistors can be switched between FS and SJM to reprogram the pickup and dropout. The dropout of 14HM is fixed at 90% of pickup. No separate adjustment is provided. The 14HM sensor outputs are on pins HM and $\overline{HM}$.

The 14HA sensor looks at NHP and provides logic outputs on HA and $\overline{HA}$ for a preset level of NHP. The dropout point is dependent on the pickup setting and does not have a separate adjustment. The dropout is 75% of pickup value.

The 14HS sensor looks at NHP and provides logic signals at HS and $\overline{HS}$. The pickup and dropout level is adjustable. The dropout is also programmable with a logic signal on input UFD. For UFD equal "0," dropout is 80 to 98% of pickup. For UFD equal "1," dropout is 90 to 98% of pickup. Potentiometer R201 (UFDO) is used to set dropout level when UFD is a "1." After R201 is set, then R209 (NDO) can be set for normal dropout. (See control specification for adjustments)

The speed monitor sensor has adjustable pickup ranges that are jumper selectable. The pickup ranges and jumpers are:

Range (% Speed)	Jumper
20 - 60	RA to RB
50 - 90	RC to RB
80 - 110	RD to RB

The 14HR sensor receives a frequency on MPU proportional to RPM and generates an analog signal. This analog signal is compared to a fixed reference and gives a "1" on HR if analog signal is less than reference. Unlike the other sensors, 14 HR is picked up at zero speed and drops out as speed is increased.

ADJUSTMENT	POTENTIOMETER	Range (% Speed)
14HM PU	R205	10 - 25
14HA PU	R207	30 - 60
14HS PU	R202	60 - 100
14HS DO	R209	48 - 9%
14HS UFDO	R201	54 - 98

SLPA Card (See Fig. **10, 11).**

This card contains three speed sensors with given (fixed or adjustable) pickup and dropout characteristics. These are:

14LS - Operating Speed
14SR - Zero Speed
14ST - Speed Monitor

The 14LS sensor operates from the input signal, NLP. The 14LS pickup is adjustable by jumper selected ranges. These ranges are:

Range (% Speed)	Jumper
20 - 60	RA to RB
50 - 90	RC to RB
80 - 110	RD to RB

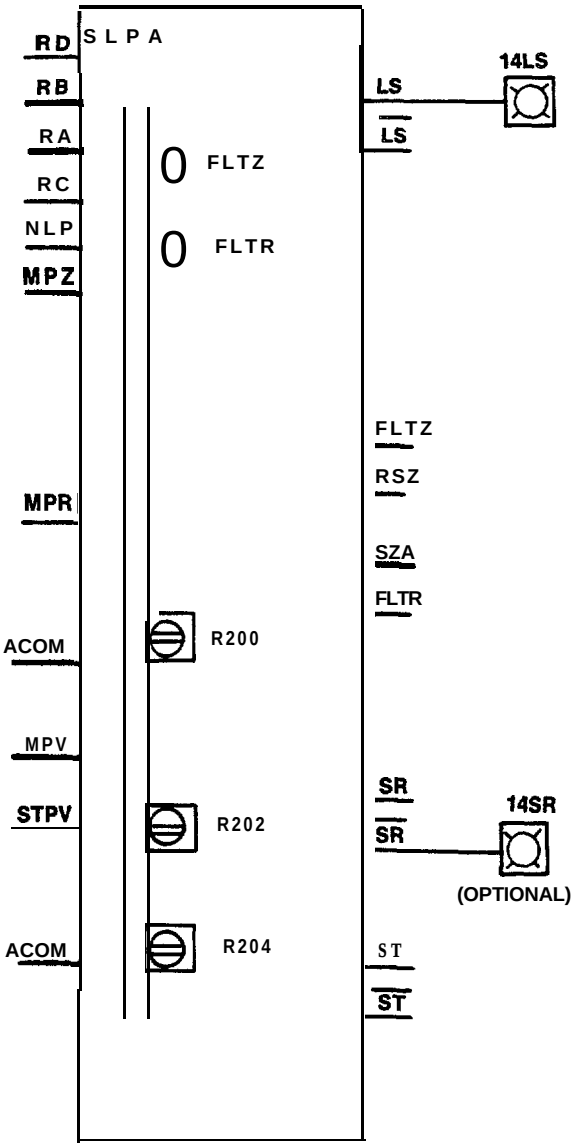


Fig. IO. LP Speed sensors

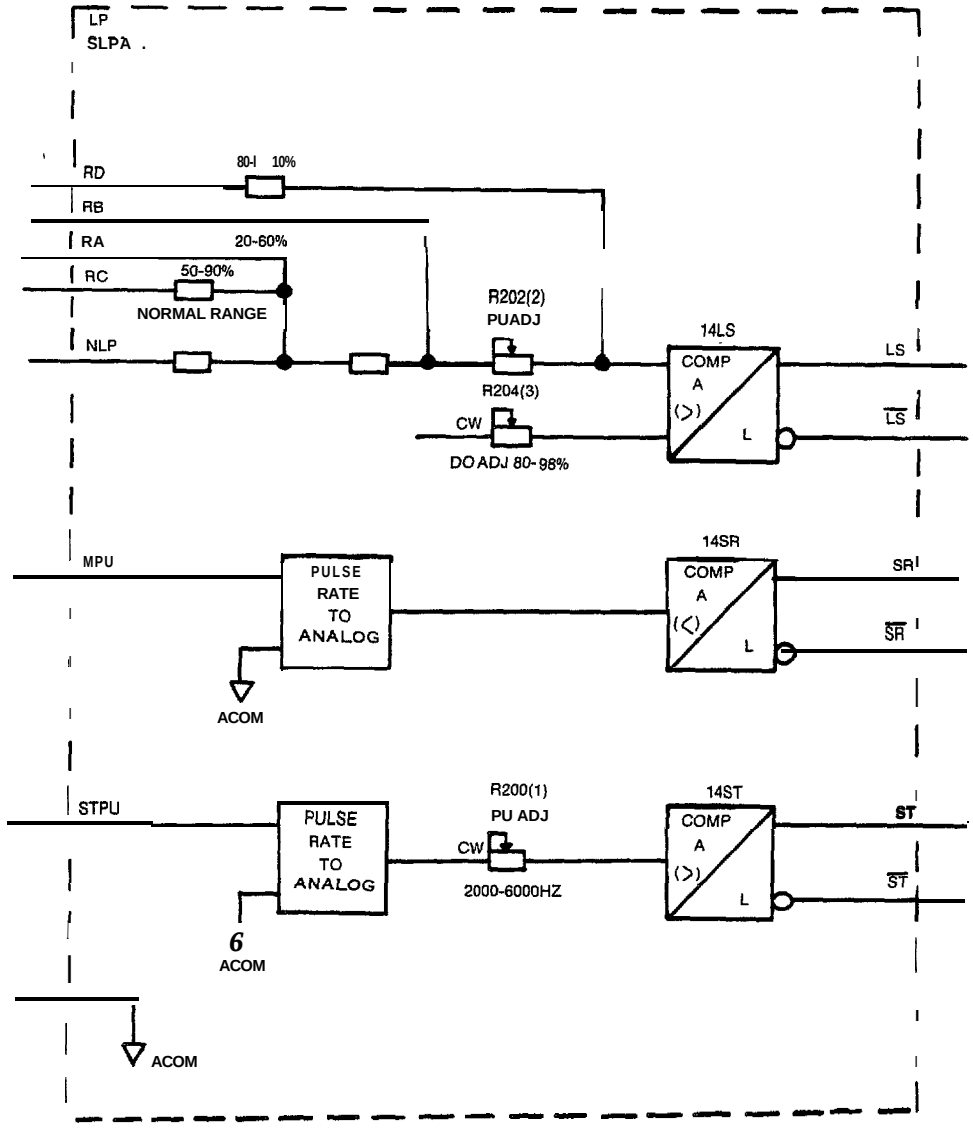


Fig. 11. SLPA card

Potentiometer R202 is used to set the exact pickup value within the selected range. The dropout value is adjustable from 80 to 98% of the pickup value, using potentiometer R204. This sensor provides logic signals at points LS and $\overline{\text{LS}}$ for use in sequencing turbine operation.

The 14SR sensor inputs a frequency proportional to RPM, on terminal MPU, and generates a corresponding analog signal. This signal is then compared to a fixed reference. Terminal SP goes to a "1" if the signal is less than the reference. This sensor differs from the others in that it is picked up at zero speed, and drops out as speed increases.

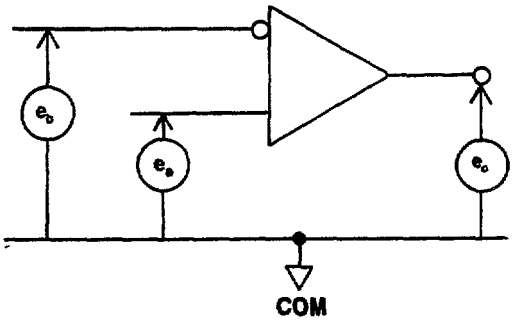
The 14ST sensor also inputs a frequency proportional to RPM, on terminal STPU, and generates an analog signal. This signal is then compared to an adjustable reference. Terminal ST goes to a "1" if the signal level exceeds the reference. The reference is adjustable from 2000 to 6000 Hz., using potentiometer R200. The dropout value is fixed ***at*** less than 90% of the pickup value.

SECTION C
CONTROL

C-I OPERATIONAL AMPLIFIERS

The Directo-Matic* II SPEEDTRONIC* control uses the operational amplifier (normally referred to as op amp) extensively; therefore, to understand the control panel, one must have a basic understanding of the op amp. A brief discussion of the op amp enhanced with several examples is given subsequently. The intent of this text is not to present a rigorous analytical description of the op amp, but an introduction to the op amp in simplified terms. It is hoped that from this groundwork the user can follow the turbine control elementary and gain an understanding of the SPEEDTRONIC control.

The operational amplifier is generally considered a high gain, high input impedance, low output impedance amplifier capable of amplifying d-c as well as a-c signals. The standard symbol for the op amp is shown below.



The input with the circle is the inverting input and the other input is non-inverting. The output is

$$e_o = K(e_a - e_b)$$

where K is the gain of the amplifier. As mentioned previously, the op amp has a very high gain (typically 10^5 in Directo-Matic II); therefore, a very small difference between e_a and e_b will give a large e_o. However e_o is limited by the supply voltages of the op amp. When e_o reaches approximately the supply voltage, it is said to be in saturation. Typical supply voltages in Directo-Matic II are +12 volts.

Using a +12-volt supply and a gain of 10^5, the value of $e_a - e_b$ for saturation is

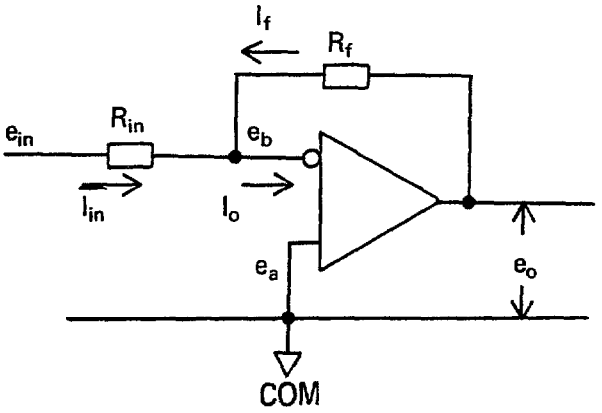
$$12 = 10^5(e_a - e_b)$$

$$(e_a - e_b) = \frac{12}{10^5} = 12 \times 10^5$$

$$(e_a - e_b) = 120 \text{ MICROVOLTS}$$

Because of this high gain the op amp is practically useless when used alone; however, with negative feedback its uses are limited only by the designer'simagination.

The most common circuit of the op amp is shown below:



In this case the non-inverting input e_a is tied to common. The useful operating *region* of the op amp is when it is not in saturation; therefore, e_b is approximately equal to e_a. In this case e_a is tied to common; therefore, e_b is approximately zero volts. Operating in this mode e_b is often referred to as virtual common. To determine e_o as a function of e_{in}, the current at e_b are summed together. This junction is often referred to as the summing junction.

$$I_{in} + I_f - I_f = 0$$

*Trademark of General Electric Company.

The op amp has a very high input impedance and I, can be considered zero. Then

$$I_{in} + I_f = 0$$

The op amp has a very low output impedance and will supply the necessary current to make the above equation true.

I_{in} and I_f can be found as follows:

$$I_{in} = \frac{e_{in}}{R_{in}} \qquad I_f = \frac{e_o}{R_f}$$

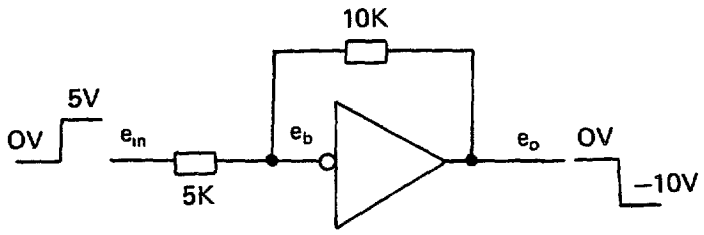
Substitutinggives:

$$\frac{e_{in}}{R_{in}} + \frac{e_o}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\frac{e_{in}}{R_{in}}$$

$$e_o = -e_{in}\frac{R_f}{R_{in}}$$

This equation applies as long as the op amp is not in saturation. If it goes into saturation, e_b will no longer be at virtual common and the equation loses validity. To further illustrate the inverter, the circuit is repeated below with values assigned. The connection of e_a to common on the symbol is often omitted with the understanding that it exists.

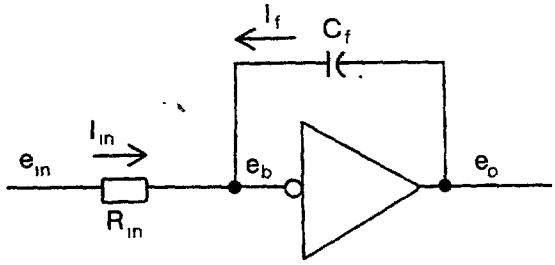


$$e_o = -e_{in}\frac{10K}{5K} = -2e_{in}$$

If e_{in} is a +5V signal, then

$$e_o = -2(5)V = \text{-10 VOLTS}$$

The second op amp circuit discussed is the integrator. The following is an integrator circuit:



Again, summing the currents at e_b gives:

$I_{in} + I_f = 0$ WHERE

$I_{in} = \frac{e_{in}}{R_{in}}$ AND

$$I_f = C_f \frac{d(e_o)}{dt}$$

$$\frac{e_{in}}{R_{in}} + C_f\frac{d(e_o)}{dt} = 0$$

$$C_f \frac{d(e_o)}{dt} = -\frac{e_{in}}{R_{in}}$$

$$d(e_o) = -\frac{e_{in}}{R_{in}C_f}\,dt$$

$$\int d(e_o) = -\int \frac{e_{in}}{R_{in}C_f}\,dt$$

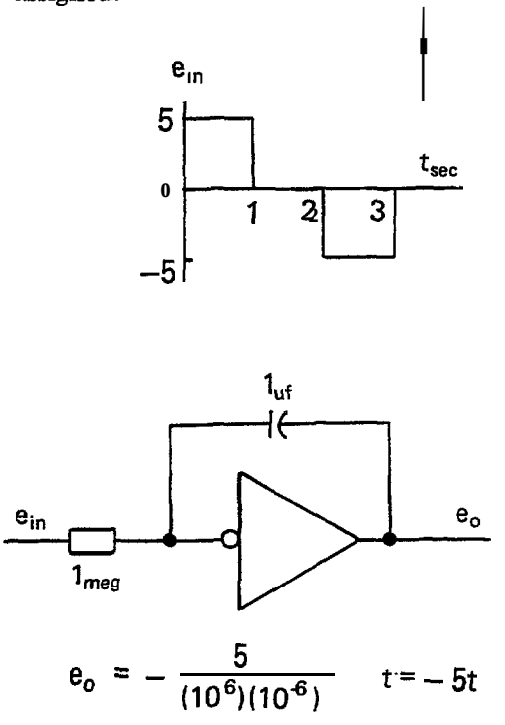
$$e_o = -\frac{1}{R_{in}C_f}\int e_{in}\,dt$$

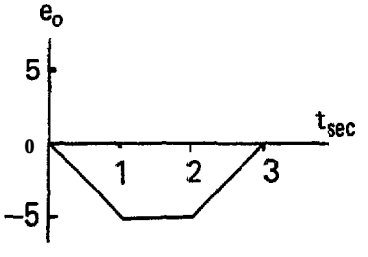
If e_{in} is a constant or step input, then

$e_o = -\frac{e_{in}}{R_{in}C_f}t$ OR

$e_o = -\frac{I_{in}}{C_f}t$ t = SECONDS

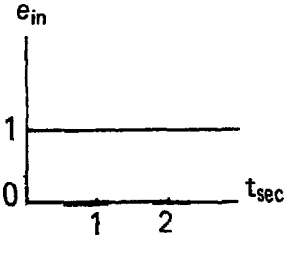
The integrator circuit is redrawn below with values assigned.

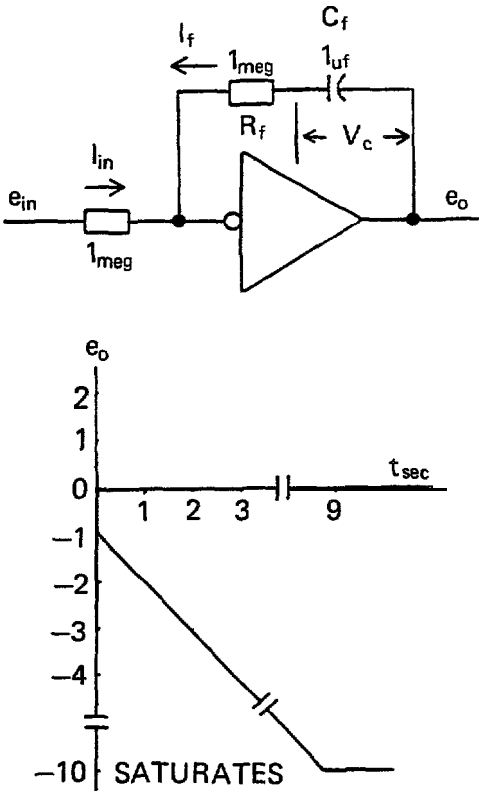




As can be seen from the curves adjacent to the circuit, in response to a step input the output increases on a ramp. If the input returns to zero, the capacitor remains charged and e_c remains at the level prior to input going to zero. If a step input of opposite polarity is applied, the capacitor will begin discharging.

For the next circuit, the input-output relationship will not be developed mathematically; instead an intuitive approach will be presented. The circuit is often referred to as a proportional plus integral.





To begin, assume that e_{in} and e_o are zero (Vc=0). Now assume a step input of 1 volt is applied at e_{in}.

$$I_{in} = \frac{e_{in}}{R_{in}} = \frac{1V}{1M} = 1_{ua}$$

$$I_f = -I_{in} = -1_{ua}$$

Now e_o is equal to the voltage across R_f and C_f. The voltage across R_f is

$$V_t = I_f R_f = -I_{in}R_f$$

$$V_t = -1 \text{ VOLT}$$

The voltage across a capacitor subjected to a constant current is

$$V_c = \frac{I}{C} t$$

Therefore, if e_{in} is a step input

$$e_o = -I_{in}R_f - \frac{I_{in}}{C} t$$

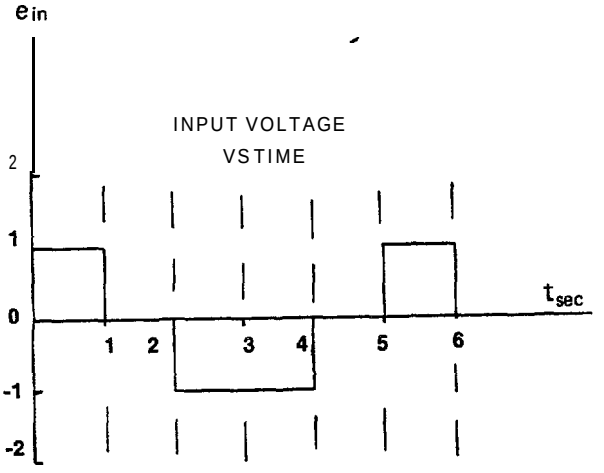
at the instant that the step is applied (t = 0)

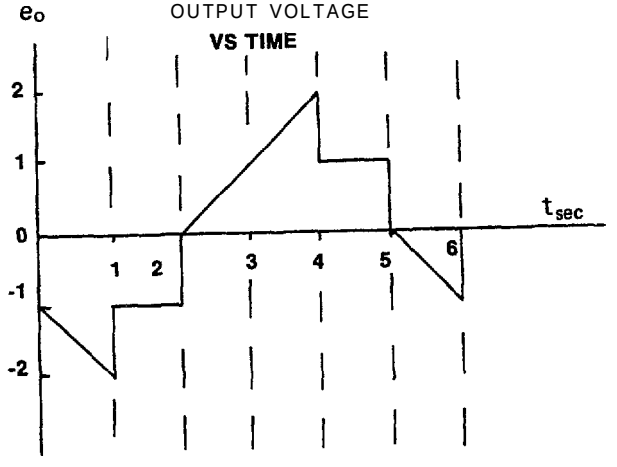
$$e_o = -I_{in}R_f$$

$$e_o = -(1_{ua})(1_{meg})$$

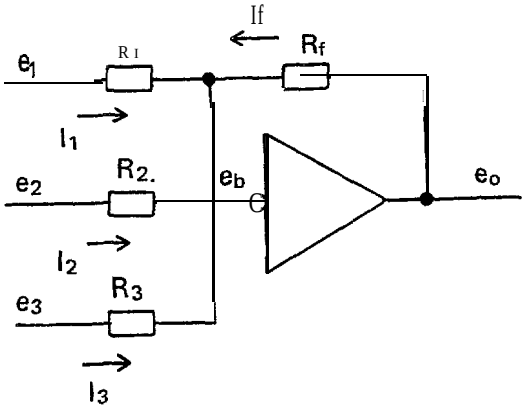
$$e_o = -1 \text{ VOLT}$$

As time increases, the capacitor charges up as shown in the graph adjacent to circuit. Following are curves showing output response to a varying input voltage.





The next op amp circuit discussed is the summer. The circuit is shown below:



Summing the currents at e_b yields:

$$I_1 + I_2 + I_3 + If = 0$$

$$\frac{e_1}{R_1} + \frac{e_2}{R2} + \frac{e_3}{R3} + \frac{e_0}{R_f} = 0$$

$$\frac{e_o}{R_f} = -\left(\frac{e_1}{R_1} + \frac{e_2}{R_2} + \frac{e_3}{R_3}\right)$$

$$e_o = -\left(\frac{R_f\, e_1}{R_1} + \frac{R_f\, e_2}{R_2} + \frac{R_f\, e_3}{R_3}\right)$$

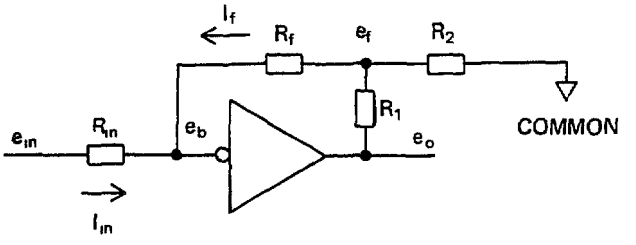
As implied by its name, the circuit sums the inputs with each input weighted by the ratio of its input resistor to the feedback resistor, R_f.

The multiple input configuration can also be applied to the integrator. Referring back to the integrator equation of a step input

$$e_o = -\frac{I_{in}}{C_f}t$$

For multiple inputs, the algebraic sum of the input currents is substituted for I_{in}.

Occasionally the feedback voltage is not derived directly from e_o. Following is an example of such a circuit:



Again, summing the currents at e_b gives

$$I_{in}'' + I_f = 0 \quad \text{(SUMMING JUNCTION EQUATION) WHERE}$$

$$I_{in} = \frac{e_{in}}{R_{in}}$$

$$I_f = \frac{e_f}{R_{in}} \quad \text{(If EQUATION) BUT}$$

$$e_f = e_o \left(\frac{\frac{R_f R_2}{R_f + R_2}}{R_1 + \frac{R_f R_2}{R_f + R_2}} \right) \quad \text{LET}$$

$$R_{eq} = \frac{R_f R_2}{R_f + R_2} \quad \text{THEN}$$

$$e_f = e_o \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

Substituting this in I_f equation gives

$$I_f = \frac{e_o}{R_f} \left(\frac{R_{eq}}{R_1 + R_{eq}} \right)$$

The summing junction equation now becomes

$$\frac{e_o}{R_f} \left(\frac{R_{eq}}{R_1 + R_{eq}} \right) = \frac{e_{in}}{R_{in}}$$

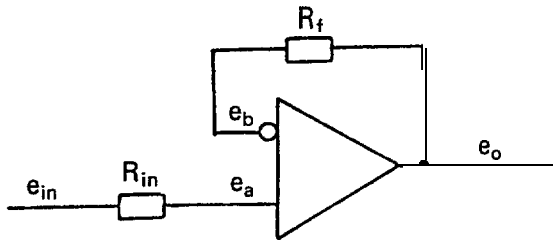
$$e_o = -e_{in} \frac{R_f}{R_{in}} \left(\frac{R_1 + R_{eq}}{R_{eq}} \right)$$

Up to this point, we have only discussed the inverting mode of the op amp. Before going into the non-inverting mode, two important features of the op amp discussed previously will be repeated.

1. The op amp has very high input impedance; therefore, it is safe to assume that it draws zero input current.

2. A very small difference between e_a and e_b causes the op amp to go into saturation; therefore, in the active region, it is assumed that e_a equals e_b .

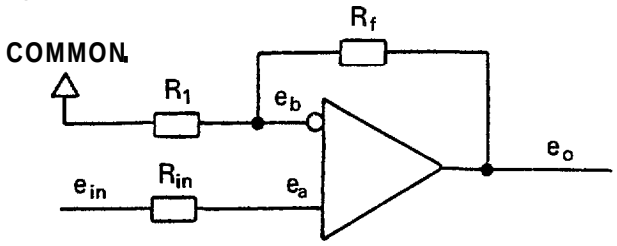
The first non-inverting op amp circuit to be discussed is the voltage follower. The circuit is shown below:



Since the op amp input draws no current, e_a is equal to e_{in}. Per feature number 2, e_b equals e_a and no current is drawn by the op amp; therefore, e_o equals e_b which in turn is equal to e_{in}. The input-output equation is

$$e_o = e_{in}$$

To obtain a gain greater than one, the following circuit is used:



As in previous circuit e_b equals e_a which equals e_{in} because no current flows into the op amp. e_b is also equal to

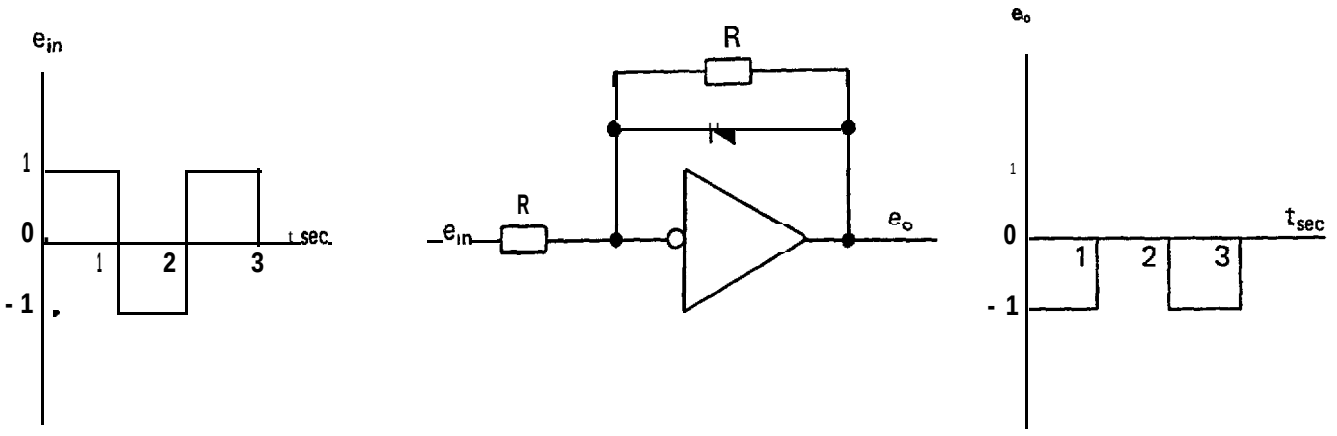
$$e_b = e_o \frac{R_1}{R_1 + R_f} \quad \text{OR}$$

$$e_o = \left(\frac{R_1 + R_f}{R_1} \right) e_b$$

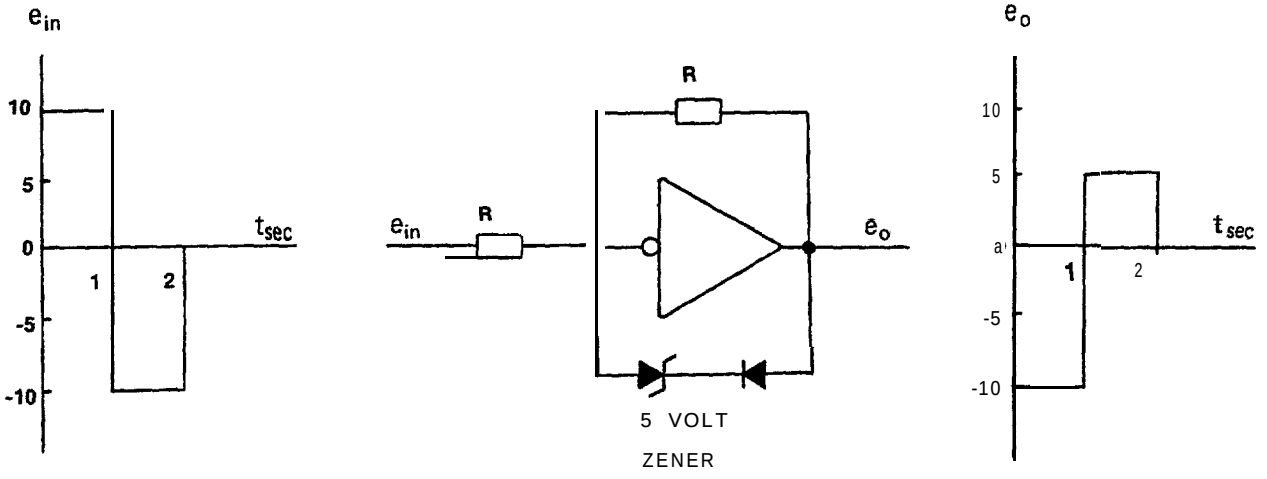
$$e_o = \left(\frac{R_1 + R_f}{R_1} \right) e_{in} \qquad e_b = e_{in}$$

$$e_o = \left(1 + \frac{R_f}{R_1} \right) e_{in}$$

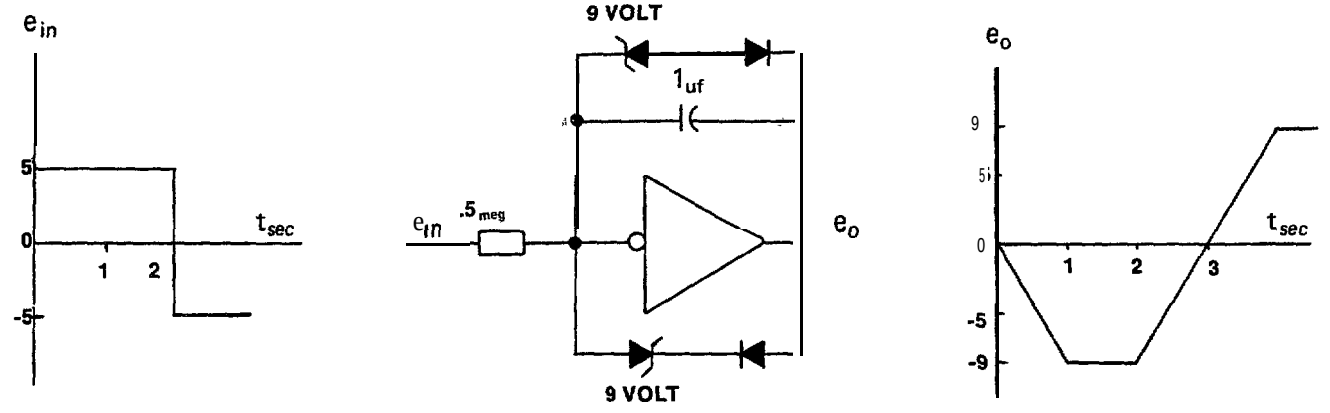
As seen from this equation the gain of this circuit is always greater than one. It is often desired to clamp the op amp output at some value. The following circuits illustrate the use of diodes and zener diodes to clamp the output.



Above circuit limits positive swing to approximately zero.



Above circuit limits positive swing to approximately 5 volts.



Above circuit limits positive and negative swings to 9 volts. This circuit is often used to prevent op amp from saturating and summing junctions from rising above common.

C-2 START-UP

The turbine control panel contains start-up, L.P. speed, and temperature control loops which modulate VCE. VCE serves as a reference to the fuel control loops which regulate fuel flow to the turbine. A nozzle control loop modulates NCE to control H.P. speed. A typical control system is shown in Fig. 12, page 31.

The start-up control is an open loop system that operates in conjunction with the sequential-control circuitry to provide preset fuel levels to the turbine during startup. In addition to the above, it also sets the maximum and minimum limits on VCE after the start-up period is over.

A typical VCE start-up curve is shown in Fig. 13. A typical control system is shown in Fig. 14, page 32.

SSKC Card (See Fig. 15, page 33)

A "1" on input OFF or input FIRE, will clamp output VCE to zero volts. If OFF is a "0" then a "0" occurring on $\overline{\text{FIRE}}$ will remove the clamp on VCE. To facilitate describing the function of the card, assume that OFF is a "0," ALR is a "0," and $\overline{\text{FIRE}}$, $\overline{\text{WU}}$ and ACC are "1." Under these conditions, VCE is clamped to zero volts. Now assume that a "0" occurs on input FIRE. VCE will jump to FIRE level as set by R102. A "0" occurring now on WU will cause VCE to drop from FIRE to WARM-UP level which is set by R101. VCE will remain at this level until ACC becomes "0" or VCE is pulled down by some other control loop. Upon receiving a "0" on $\overline{\text{ACC}}$, VCE will be released to increase on an exponential curve until a limit set by R100 is reached. This limit on VCE will remain until input ALR becomes a "1." A "1" on ALR will remove the limit and allow VCE to increase to a maximum limit set by R106.

A minimum VCE circuit monitors VCE and provides a negative bias at output MIN when VCE decreases below a MIN LEVEL set by R104.

Typically, the bias out of MIN is connected to the speed and acceleration controls and prevents them from driving VCE below the MIN level. A test circuit is provided that supplies a positive bias out of MT when MIN push button is depressed. This can be used as an input to the speed card to simulate speed feedback which tells the speed card to decrease VCE. The MIN circuit should then prevent VCE from going below the preset level.

Push buttons (MAX, MIN, FIRE, WU, ACC) are provided on the card for testing and adjusting their corresponding preset levels. Depressing either the MIN, FIRE, or WU push buttons will set VCE to its corresponding voltage level (See ranges below). Depressing the MAX push button will set VCE at its maximum value, only if the "43" switch is in the off or crank position (See range below). Depressing the ACC push button will allow VCE to rise exponentially to its accelerate limit. Depressing the TCC and ACC push buttons simultaneously, will reduce the exponential time constant by about a factor of 10. With any of the previous ranges selected, the actual value of VCE can be adjusted by the potentiometers listed below.

Potentiometer RH1 is provided for manual control of VCE. VCE can be adjusted from MAX VCE to zero using RH1 if some other control loop is not pulling VCE down. A limit switch on the potentiometer ties pins SW1 and SW2 together when RH1 is not fully counter-clockwise.

ADJUSTMENT	POTENTIOMETER	RANGE (volts)
MAX	R106	8.00 to 10.50
MIN	R104	2.75 to 4.00
FIRE	R102	3.25 to 5.00
WU	R101	2.50 to 4.00
ACC	R100	3.50 to 6.50

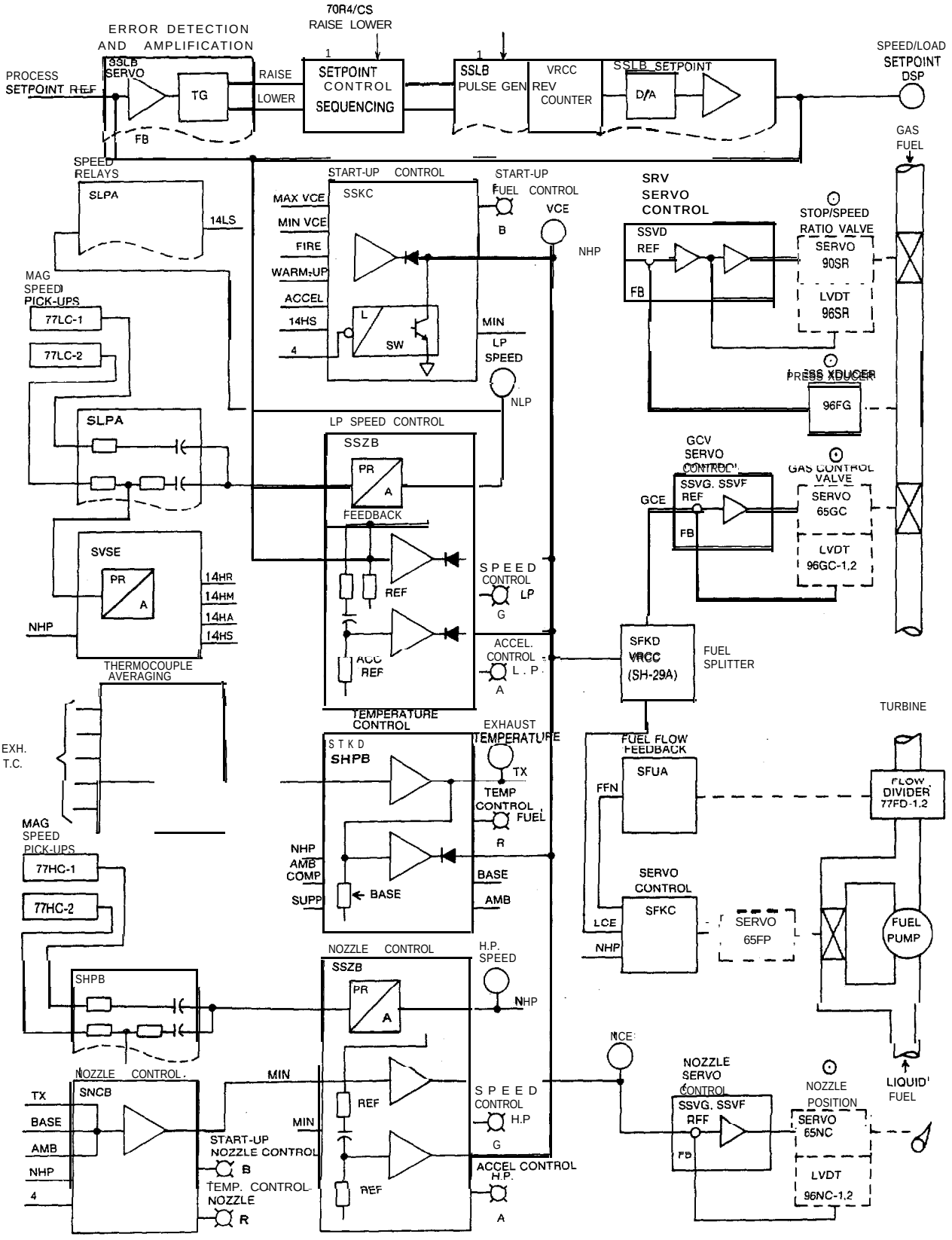
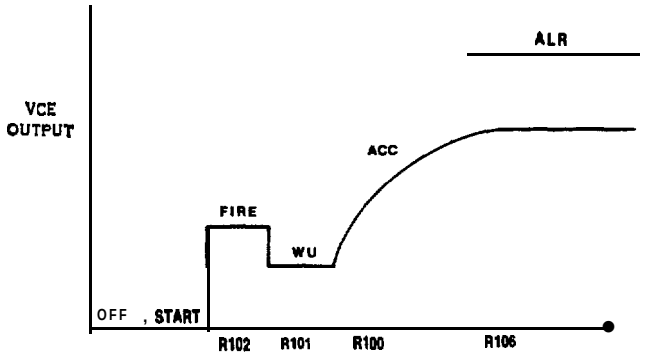


Fig. 12. Typical control system



OFF	1	0	0	0	0	0
$\overline{FIRE}$	Ø	1	0	0	0	0
$\overline{WU}$	Ø	1	1	0	0	0
$\overline{ACC}$	Ø	1	1	1	0	0
ALR	Ø	0	0	0	0	1

Ø= DON'T CARE "0" OR "1"

Fig. 13. Open loop start-up profile

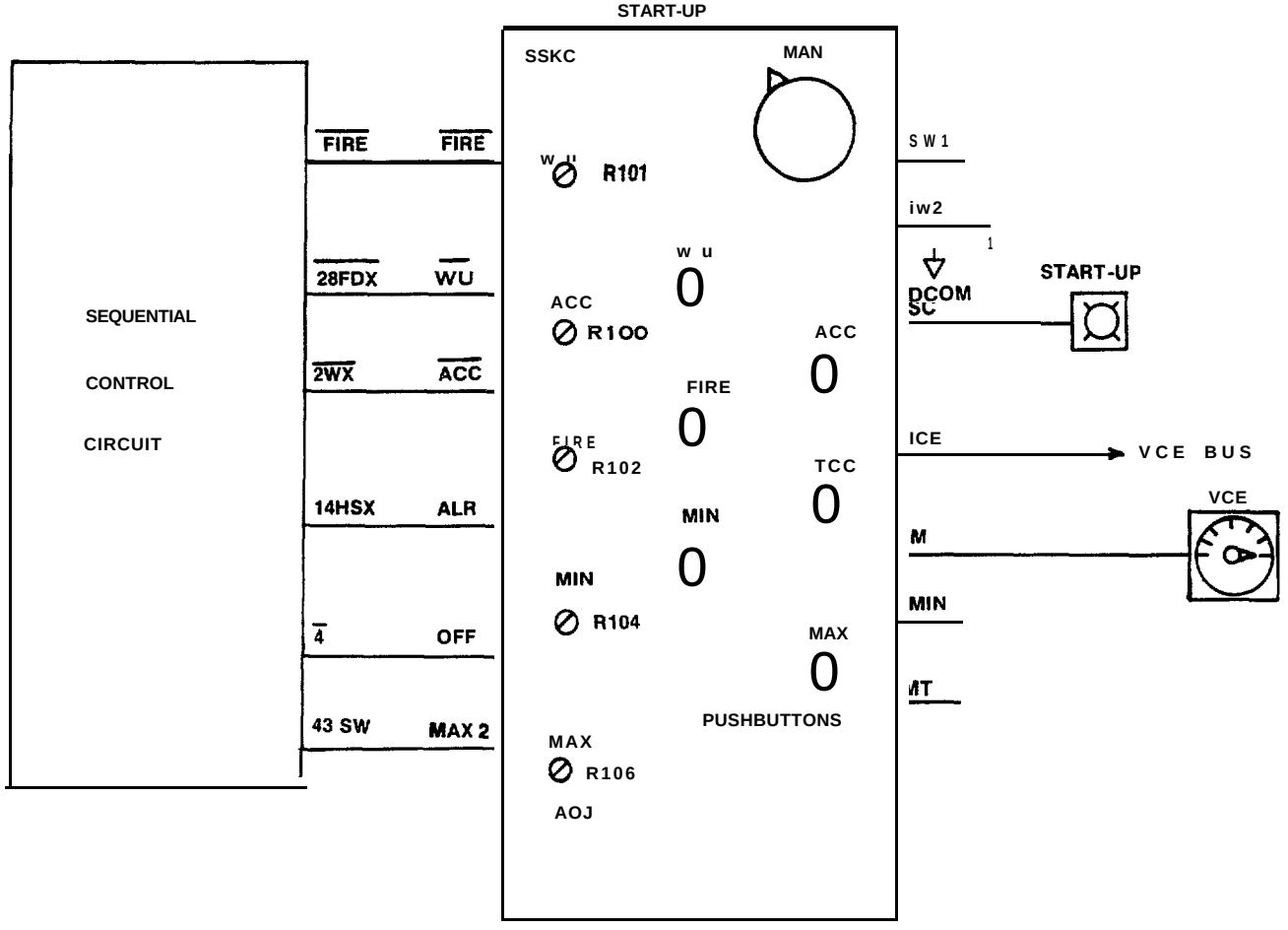


Fig. 14. Typical start-up control system

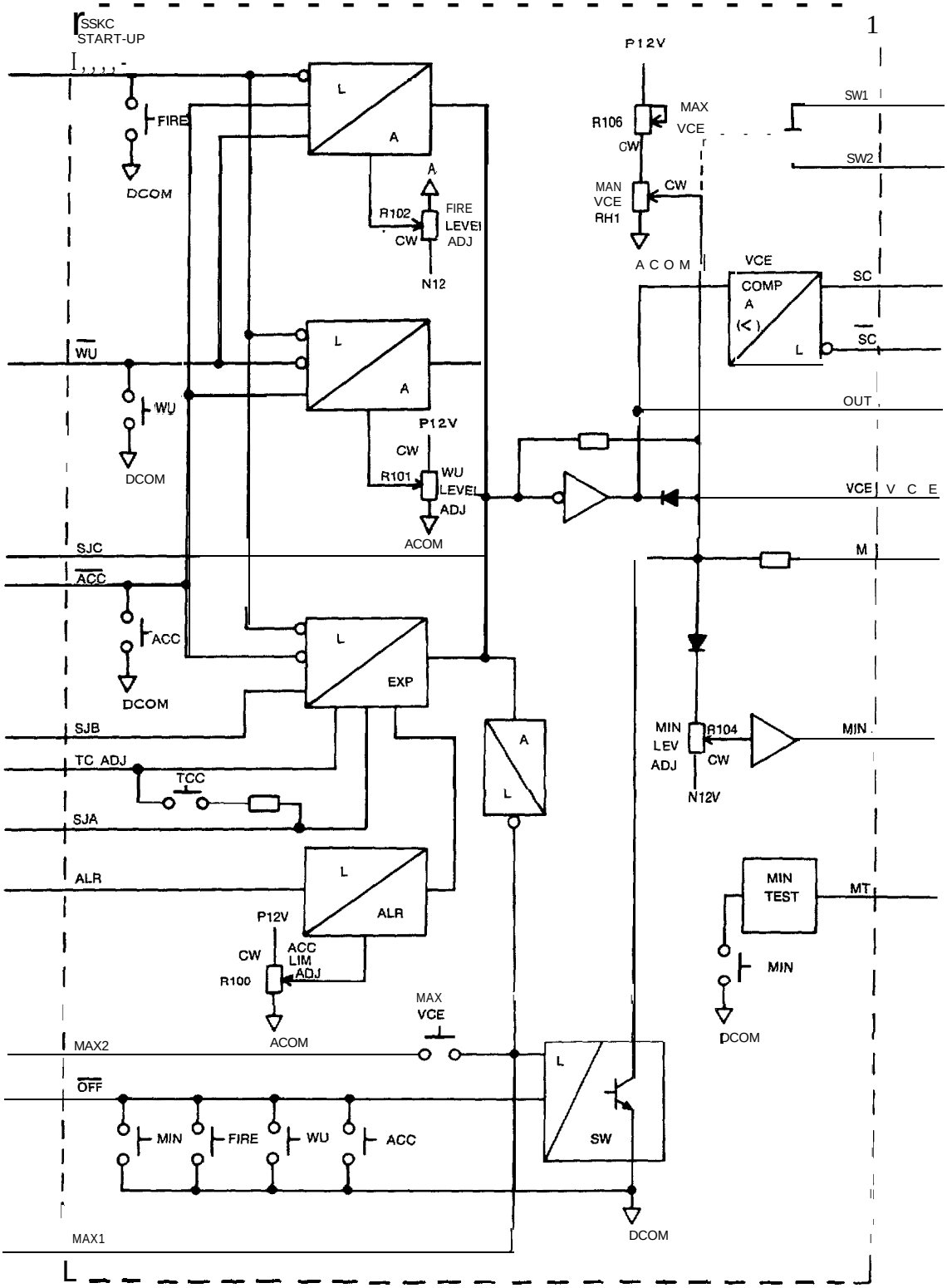


Fig. 15. SSKC Card

C-3 DIGITAL SETPOINT

The digital setpoint provides a programmable analog signal used as a called for speed or load signal to the speed control card. The system consists of cards SSLB, VRCC, AIAD and associated control signals.

VRCC-Card (See Fig. 16, page 36)

The VRCC card contains a 10-bit reversible counter. The rate of the counter is determined by the frequency of the pulses coming in on PUL. The direction of the count is controlled by inputs on ST1 and STO. A "1" on ST1 and a "0" on STO will steer the counter up. A "0" on ST1 and a "1" on STO will steer the counter down. If both ST1 and STO are "0" the counter will not count. If both ST1 and STO are "1," all bits will flip on the next count pulse. A "0" on $\overline{CL}$ will clear all bits. The states of the output bits are typically displayed on LINA cards to aid in identifying the counter contents.

SSLB-Card (See Fig. 16)

The SSLB card provides three functions:

(1) steering logic and count pulses for VRCC;
(2) a digital to analog converter
(3) a comparator circuit

Output ST1 is a "1" if input UP is a "1" and DN is "0." Output STO is a "1" if DN is "1" and UP is "0." Count pulses are present on output SFT if either UP or DN is a "0" and counter is not full or empty. If the counter is full, input MSB and C will be a "1" and inhibit the pulses and output UST will be a "0." If the counter is empty, input MSB and C will be a "1" and also inhibit the count pulse generator and output DST will be a "0." The frequency of the generator may be decreased by connecting a capacitor between input CAP and common.

The D/A converter receives its input from the reversible counter on pins CO-C5 and C6-C9. The D/A converter output pin SP voltage range provides the digital setpoint signal to the speed control card. Output SPM provides a 1-mA signal for DSP equal to 10 volts which can be used to drive a setpoint indicator meter. Zero- and span-adjustment potentiometers are provided to enable adjustment of SP over various ranges corresponding to counter-empty and counter-full conditions.

The comparator receives one of its inputs from the outputs of the D/A converter. This input is compared with signals on inputs SJ1, REF, and PRST. If SP is greater than the signal or signals being compared to it, output $\overline{DN}$ will be a "0." If SP is less than the signal or signals, output UP will be a "1." Servo zero and servo span potentiometers are provided to enable matching the SP voltage range to the range of the voltage that is being compared to it. The output SP can be made to servo to a reference on inputs SJ1 or REF by connecting the $\overline{DN}$ output through an inverter to input DN and connecting output UP to input UP.

Adjustments

POTENTIOMETER	ADJUSTMENT
R124	Setpoint span
R125	Setpoint zero
R126	Servo zero
R127	Servo span

AIAD-Card (See Fig. 16, page 36)

The AIAD card provides isolation between the process speed setpoint circuit and the SSLB card. The card will accept the following signals at the designated inputs by making the specified jumpers:

SIGNAL	INPUT CONNECTION	JUMPERS
0 to 5 volt	+DtoO	E to F
0 to 10 volt	+DtoO	E to L
0 to 25 volt	+M to 0	E to F, D to N
1 to 5 mA	+AtoO	EtoF,DtoA
4 to 20 mA	+B to0	E to F, D to B
10 to 50 mA	+c to 0	E to F, D to C

The following outputs are possible:

OUTPUT	CONNECTION	JUMPERS
0 to 10 volt	+GtoP	G to H, Q to I
1 to 5mA	+GtoQ	Q to I, S to T
4 to 20 mA	+GtoQ	Q to J, R to T
10 to 50 mA	+GtoQ	Q to K, R to T

Adjustments

Input Bias - R70—0 to 30% on any scale

Output Bias - R74—0 to 2.5 mA (S connected to T)
2.5 to 15 mA (R connected to T)

Gain - R71—0.5 to 2 when used as current-to-current converter
+15% MIN on any voltage range.

C-4 LPSPEED

The LP speed control loop modulates VCE to control the speed and acceleration of the low-pressure shaft. The reference for the speed regulator is the sum of a 100% speed reference and a digital setpoint signal. A typical speed loop consists of cards SLPA and SSZB (See Fig. 18).

SLPA Card (See Fig. 17 and 18, pages 37, 38)

The SLPA provides a cross-coupling network so that the sensor input signals on MPZ and MPR can be combined by connecting outputs RSZ and SZA together. Each speed sensor is also checked for an open or short. A short or open on sensor connected to MPZ will generate a "1" on FLTZ and illuminate lamp FLTZ. A short or open on the sensor connected to MPR generates a "1" on FLTR and illuminates lamp FLTR.

SSZB Card (See Fig. 18 and 19, page 38)

The SSZB card contains a pulse rate to analog (PR/A) converter which receives an alternating signal at input IN and provides a proportional signal at output N. This speed feedback signal is summed with the reference signal and other input signals to modulate NVCE to obtain the desired speed and/or load. Jumpers are made on FA, FB, FC and FD to program the PR/A converter to give the appropriate voltage for various machine speeds.

A "0" on SB puts a positive bias into the SJ and forces NVCE to MIN level if pin SDN is connected to SJ. When used for HP set fuel limit and SDN is not connected to SJ, a "0" on SB reprograms the HP fuel limit to 80 percent.

An overspeed test push button which will feed a negative bias into SJ and increase NVCE is provided for overspeed testing.

Jumpers are made between either pin CA, CB, CC and either pin CG, CF, CE or CD to select transfer function desired. Potentiometer R114 provides a gain adjustment. Inputs SPA and SPB provide inputs for the digital setpoint signal. SPA is used when 0.2 volts DSP is equal to 1% change in speed, and SPB is used when 1 volt DSP is equal to 1% change in speed. A differentiator circuit looks at the speed feedback signal N and modulates VCE to limit the accelerations of the turbine. Output AC will be a "1" when VCE is under acceleration control. Connecting HR to ACOM will halve the acceleration control rate.

Adjustments :

ADJUSTMENT	POTENTIOMETER	RANGE (% Rated Speed)
GAIN(PR/A)	R112	NA
GAIN (NVCE)	R114	NA
100%N	R111	96 to 104

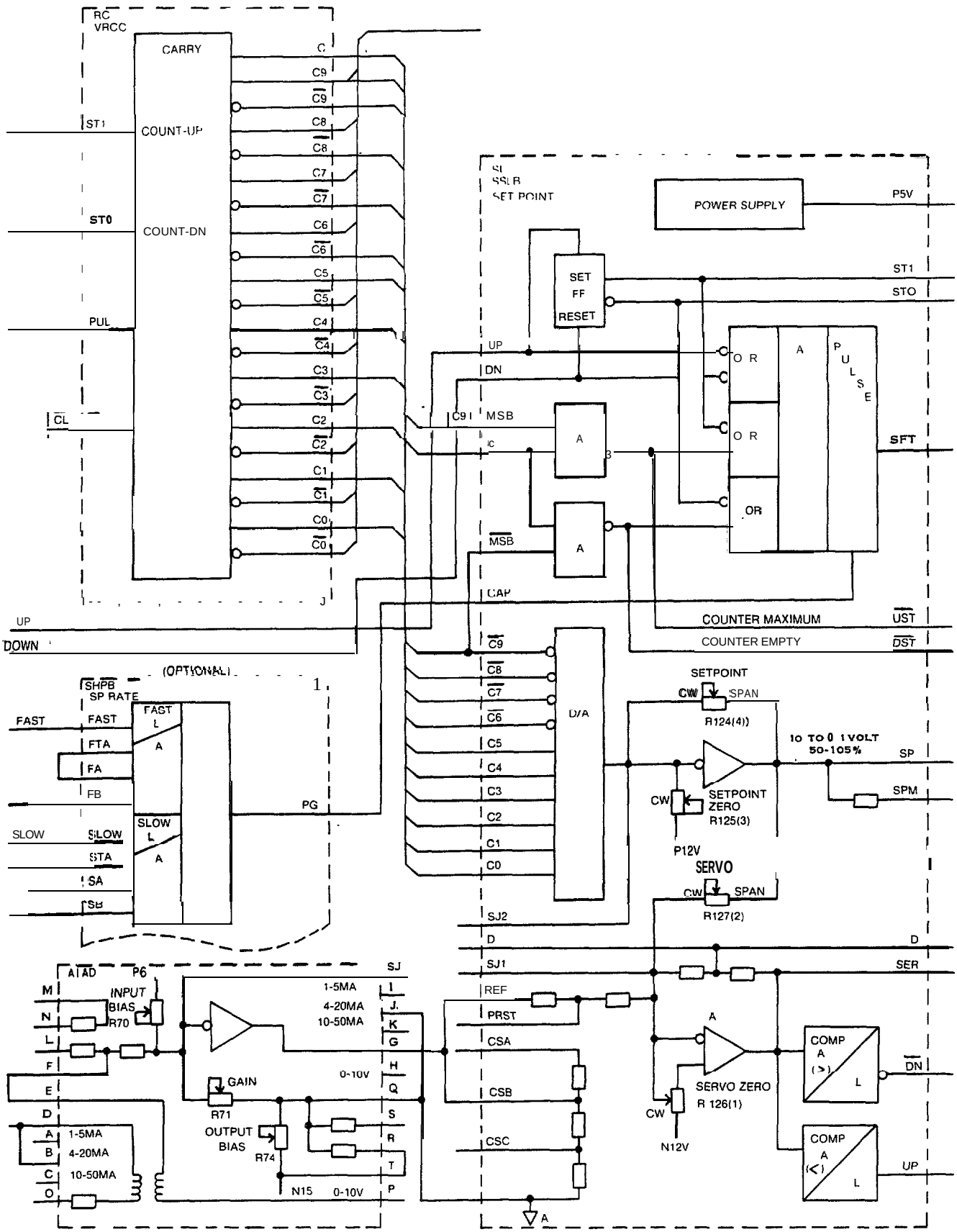


Fig. 16. Digital setpoint control

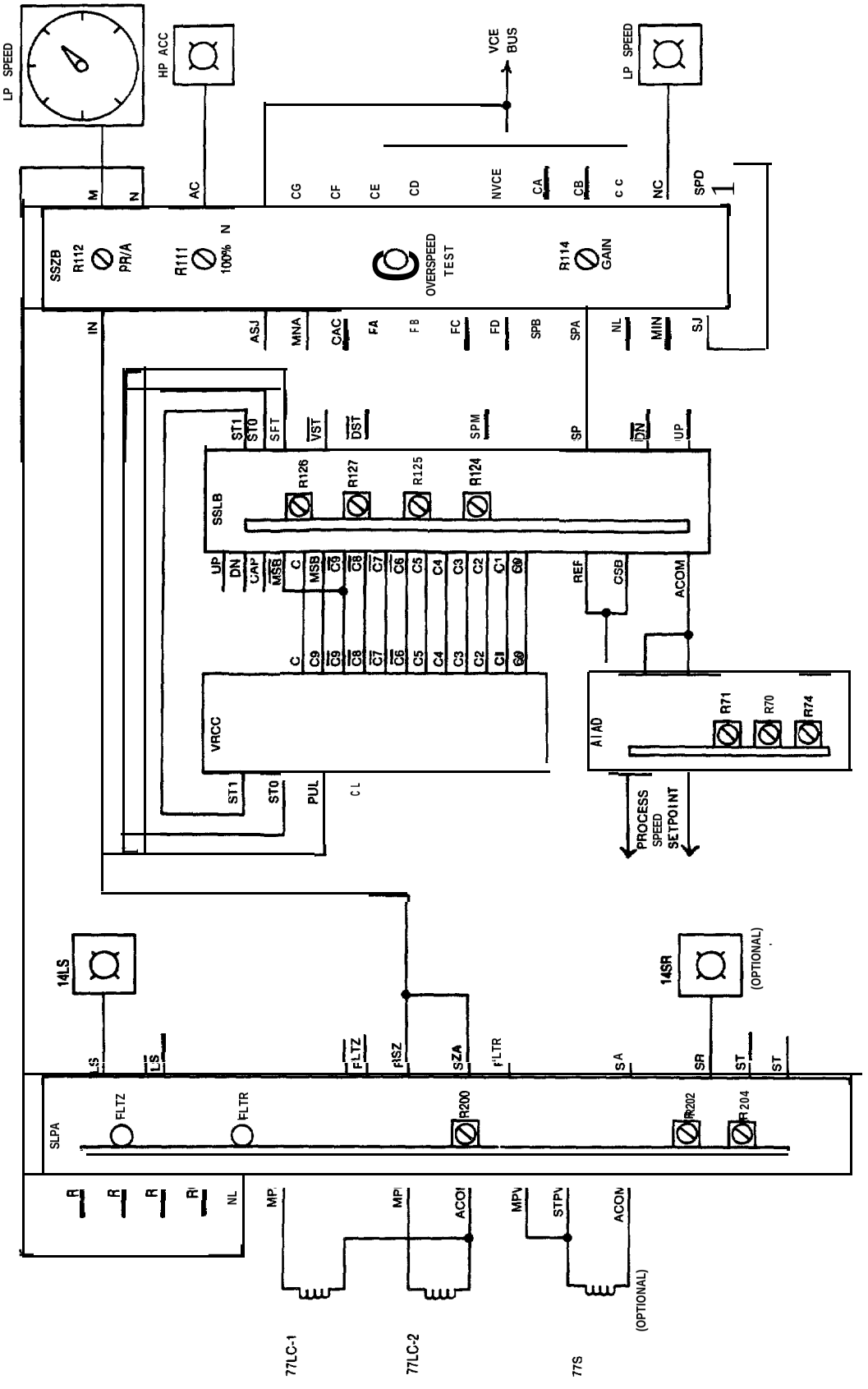


Fig. 17. LP speed control

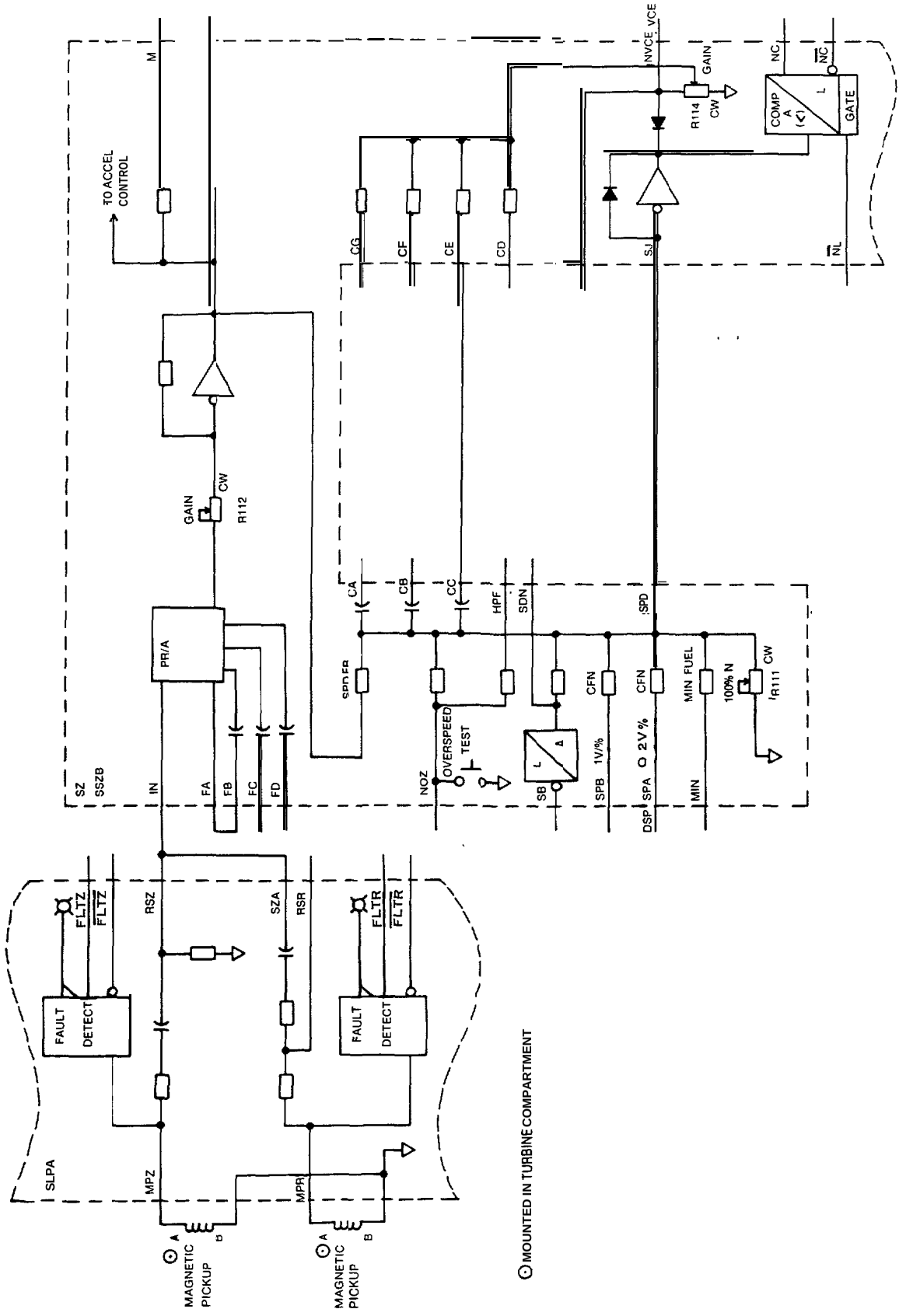


Fig. 18. LP speed control

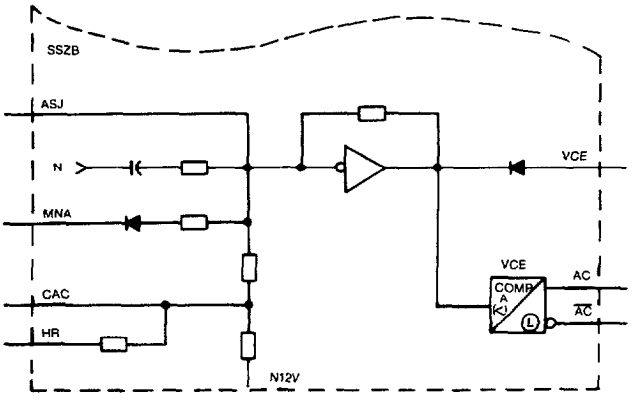


Fig. 19. Acceleration portion of SSZB.

C-5 NOZZLE CONTROL

The nozzle control is a closed-loop system which regulates the turbine's exhaust temperature by modulating NCE. NCE modulates the nozzle angle which regulates the exhaust temperature by varying the HP-shaft speed. The nozzle regulator consists of cards SHPB, SSZB, and SNCB. (See Fig. 21). The nozzle servo-valve control is similar to the gas control servo-valve control and will not be covered separately in this publication.

SHPB Card (See Fig. 20, 22 and 23, pages 40 and 43)

The SHPB card contains several circuits used in the nozzle control loop plus circuitry used elsewhere. Only the circuitry associated with the nozzle control is covered at this time.

The SHPB card provides a cross-coupling network so that the speed feedback signal from two magnetic pick-up heads can be combined such that a failure on either of the sensors will not result in a loss of the feedback signal. Outputs RSZ and SZA are tied together for this function. Pulses on input MPR also result in pulses out of RSR which is used for zero speed indication.

Short and open detectors are provided for each sensor. A short or open between MPZ and ACOM results in *a* "1" on FLTZ and illuminates light FLTZ. A short or open between MPR and ACOM results in a "1" on FLTR and illuminates light FLTR.

SSZB Card (See Fig. 19, 20, 21, pages 39, 40, 41)

The SSZB card contains a pulse rate to analog (PR/A) converter which receives an alternating signal at input IN and provides a proportional signal at output N. Jumpers are made on FA, FB, FC and FD to program the PR/A converter to give the appropriate voltage out for various speed machines.

Input SJ receives the sum of NHP and temperature reference signal from the SNCB card and modulates output NVCE. When SSZB is controlling NCE, output NC will be a "1" provided that permissive TC is a "1."

A differentiator circuit looks at the speed feedback signal N and modulates VCE to limit the accelerations of the turbine. Output AC will be a "1" when VCE is under acceleration control. Connecting HR *to* ACOM will halve the acceleration control rate.

Adjustments:

ADJUSTMENT	POTENTIOMETER
GAIN (PR/A)	R112
GAIN (NVCE)	R114

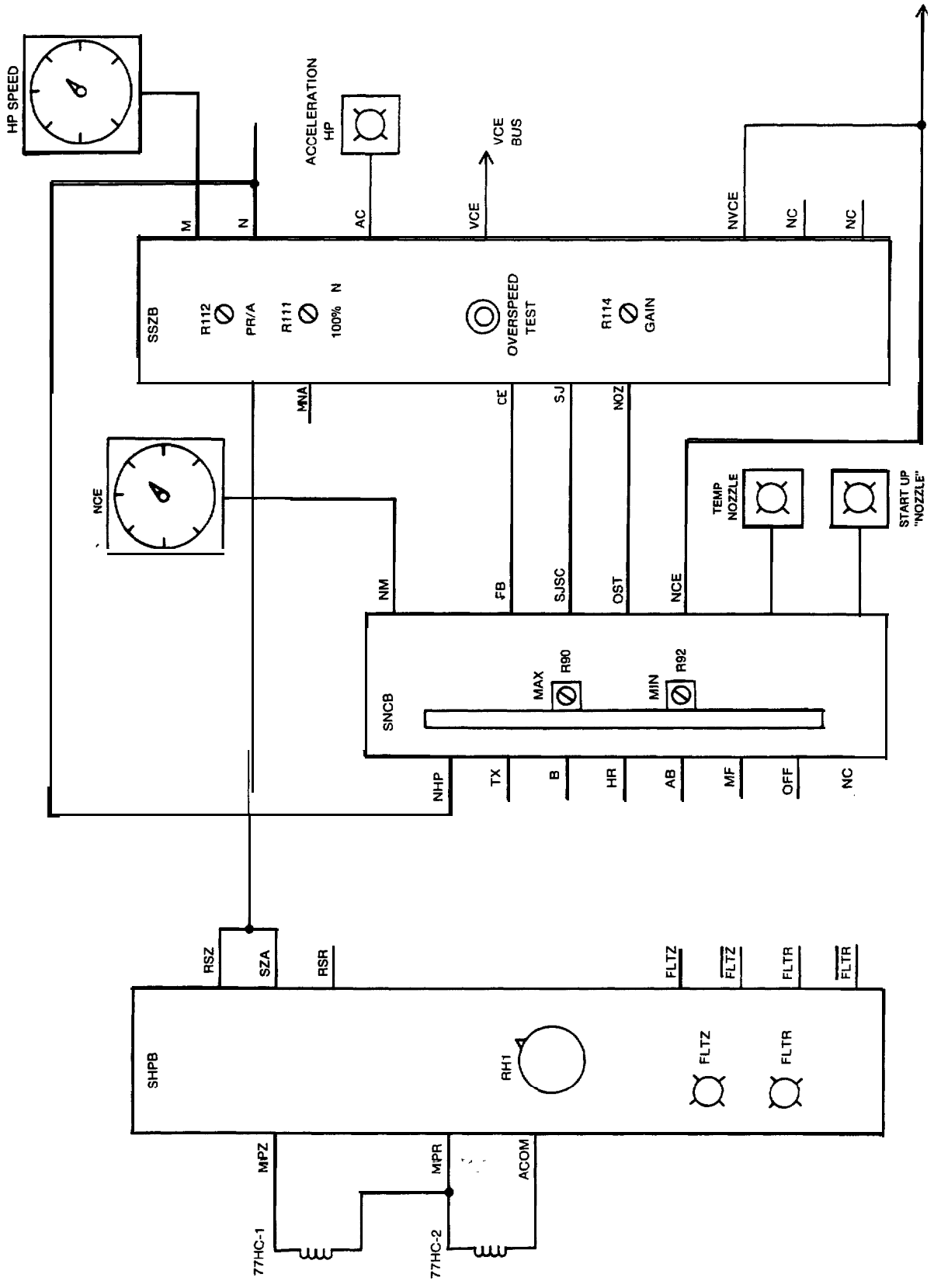
SNCB Card (See Fig. 20, 21)

The SNCB provides the drive for the NCE bus. The drive is developed similar to VCE on the SSKC card. However, the maximum limit is fixed and not adjustable as VCE is.

The SNCB card also receives the HP speed signal on input NHP and sums it with a speed reference and outputs a signal on SJSC to the regulator operational amplifier on the SSZB card to modulate NCE. The speed reference (temperature error) signal is the sum of the inputs on TX, B and AB. If TX plus AB is less than B, the reference changes in the positive direction to close the nozzle and decrease speed. If TX plus AB is greater than B, the reference changes in the negative direction to open the nozzle and increase HP speed. The temperature operating range can be altered via HR. The speed reference signal is limited to a minimum and maximum called for value. The nozzle may be forced into the open position by inserting a negative bias at MF. Typically MF is connected to MIN on SSKC card to open the nozzle at minimum VCE. A negative bias on $\overline{ST}$ will also force the nozzle open. Typically, $\overline{ST}$ is tied to the overspeed test push button on the SSZB card to facilitate overspeed testing of the turbine. Output TC will be a "1" when nozzle is on temperature control.

Adjustments:

ADJUSTMENT	POTENTIOMETER	RANGE (% Speed)
MAX SPEED	R90	80 to 105
MIN SPEED	R92	70 to 100



Fig, 20. Nozzle control

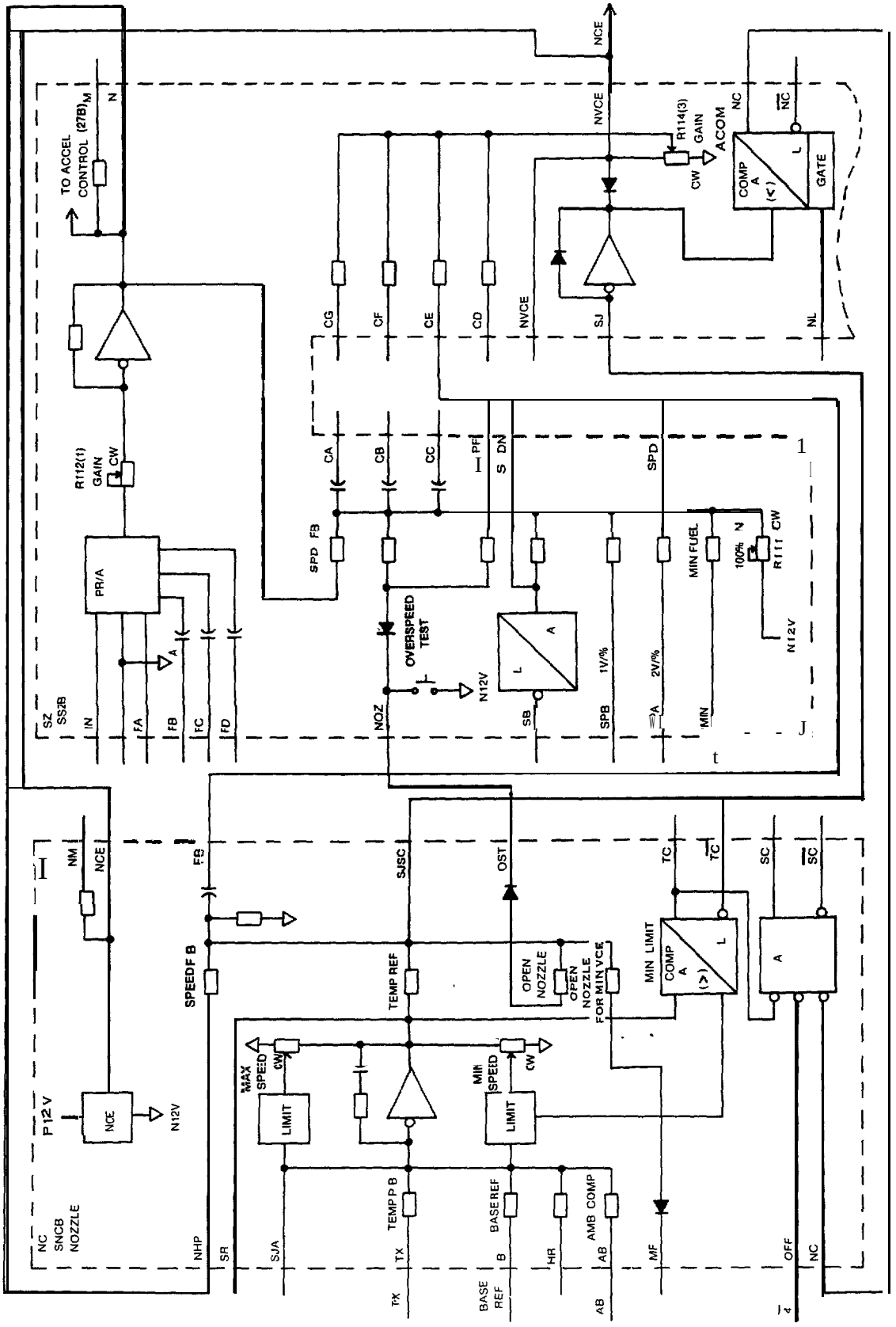


Fig. 21. Nozzle control loop

C-6 TEMPERATURE

The temperature feedback comes from up to 12 chromel-alumel thermocouples placed at selected points of the turbine exhaust duct. The thermocouple signals are averaged and cold junction compensated and then summed with reference signals to modulate VCE. The reference may consist of several signals. Included in these signals are base reference, start-up sunpression; ambient bias (AB) and high-pressure shaft speed bias. Compensated and uncompensated signals are available for temperature indication on various meters. A typical temperature control loop consists of cards STKD, SHPB and a temperature averaging module. (See Fig. 22, 24, page 43, 44).

Temperature Averaging Module (See Fig. 22, 23)

The input to the averaging cabinet is the twelve chromel-alumel thermocouples mounted in the turbine exhaust duct. The twelve thermocouples are connected in parallel to provide an average signal. A 50-ohm resistor is placed in each lead of the thermocouples to swamp out circulating currents due to differences in resistance of the thermocouple wires. Switches are provided so that each thermocouple can be individually checked. The toggle switch corresponding to the thermocouple to be checked must be placed in the CHECK position for this function. The check position of each thermocouple switch is tied to a common bus; therefore, if more than one switch is in the check position their average will be on the check bus. The thermocouple selector switch must be in position 11 to connect the digital indicator to the check bus. An individual check signal for connection to a recorder is available at +CR and -CR. A thermocouple can be removed from the average by placing its corresponding switch in the REJECT position.

The averaging cabinet provides a signal for the temperature regulator card which is the average of the twelve inputs and cold junction compensated using an RTD. A potentiometer is provided *to* adjust the voltage drop across the RTD; however, this is set up in the factory and should require no field adjustment. *Two* uncompensated **averages are** provided. The uncompensated average out of +UA and -UA goes to the thermocouple switch position 12. Uncompensated average out of +UAR and -UAR is available for connecting to a recorder. Calibration and testing jacks are accessible to the front of the module by removing the cover plate.

STKD **Card (See Fig. 23, 24,** pages **43, 44)**

The STKD card receives the millivolt signal from the temperature averaging cabinet between pins INP and INN. This signal is offset and amplified to give an output on TX. Both offset and gain adjustments are provided. Output TM provides an output for the TX meter indicator. TX is summed with the reference signals and modulates output VCE to control fuel flow. When TX is in control of the VCE bus, output TC is a **"1"** and $\overline{TC}$ is **"0."**

The base reference for the control is generated on the card and is adjustable with R86. The base reference is also available at output B. The ambient bias reference is also developed on the card. The ambient bias circuit consists of an operational amplifier with an offset and gain adjustment. The input to the ambient bias circuit can be a 0 to 10V signal on BVCE, a 0 to 5V signal on PCD5, or a 0 to 2.8V signal on PCD. A 10-50 mA input signal may be used by connecting to PCD and jumpering PCD to PTA. The output of the ambient bias amplifier is available at output pin AB. Additional reference signals may be connected to input SJ. Typically the start-up suppression bias is connected to SJ. Input HR is available for changing the range of the base adjustment.

Adjustments:

ADJUSTMENT	POTENTIOMETER	DESCRIPTION
OFFSET	R87	MV/V amplifier offset adjustment
GAIN	R83	MV/V amplifier gain adjustment
BASE	R86	Base reference adjustment
AB OFFSET	R84	AB offset adjustment
AB GAIN	R89	AB gain adjustment

SHPB Card (See Fig. **23, 24)**

The SHPB card contains the start up suppression bias circuitry. If input $\overline{TR}$ is a "1," output TXB tracks input TXO. When $\overline{TR}$ goes to "0," the voltage on TXB at the instant $\overline{TR}$ changes to **"0"** and is removed at a fixed **rate. The** rate can be altered with a **"0"** on input $\overline{FST}$. The amount it is altered is determined by a jumper between AF and AFT, AT or ASF.

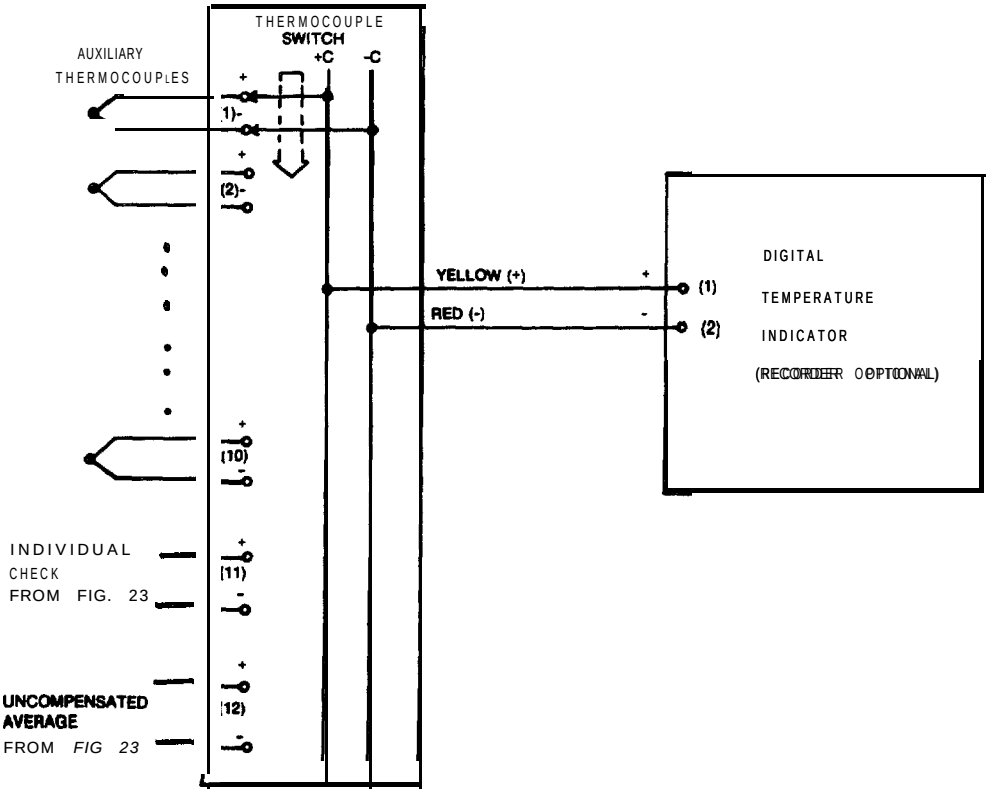


Fig. 22. Typical thermocouple switch

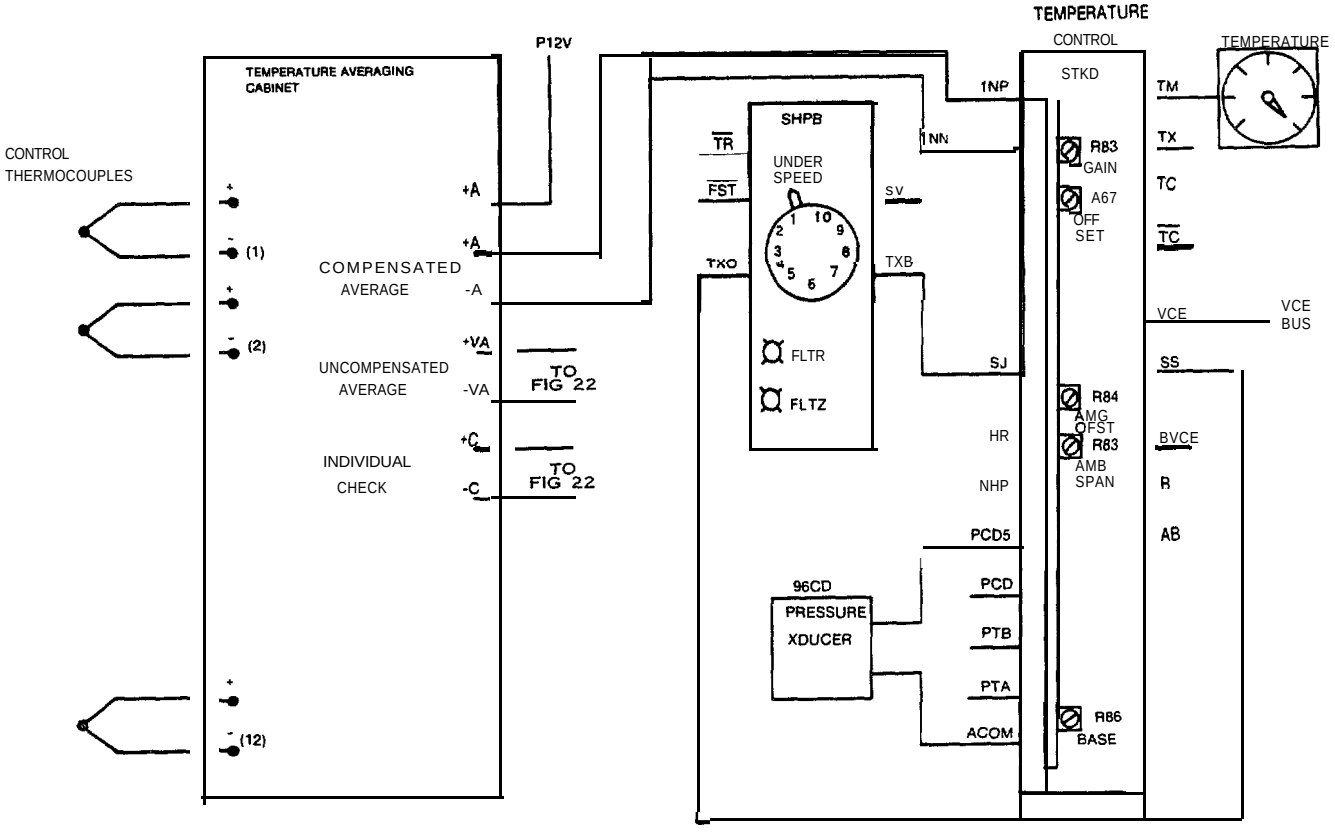


Fig. 23. Temperature control

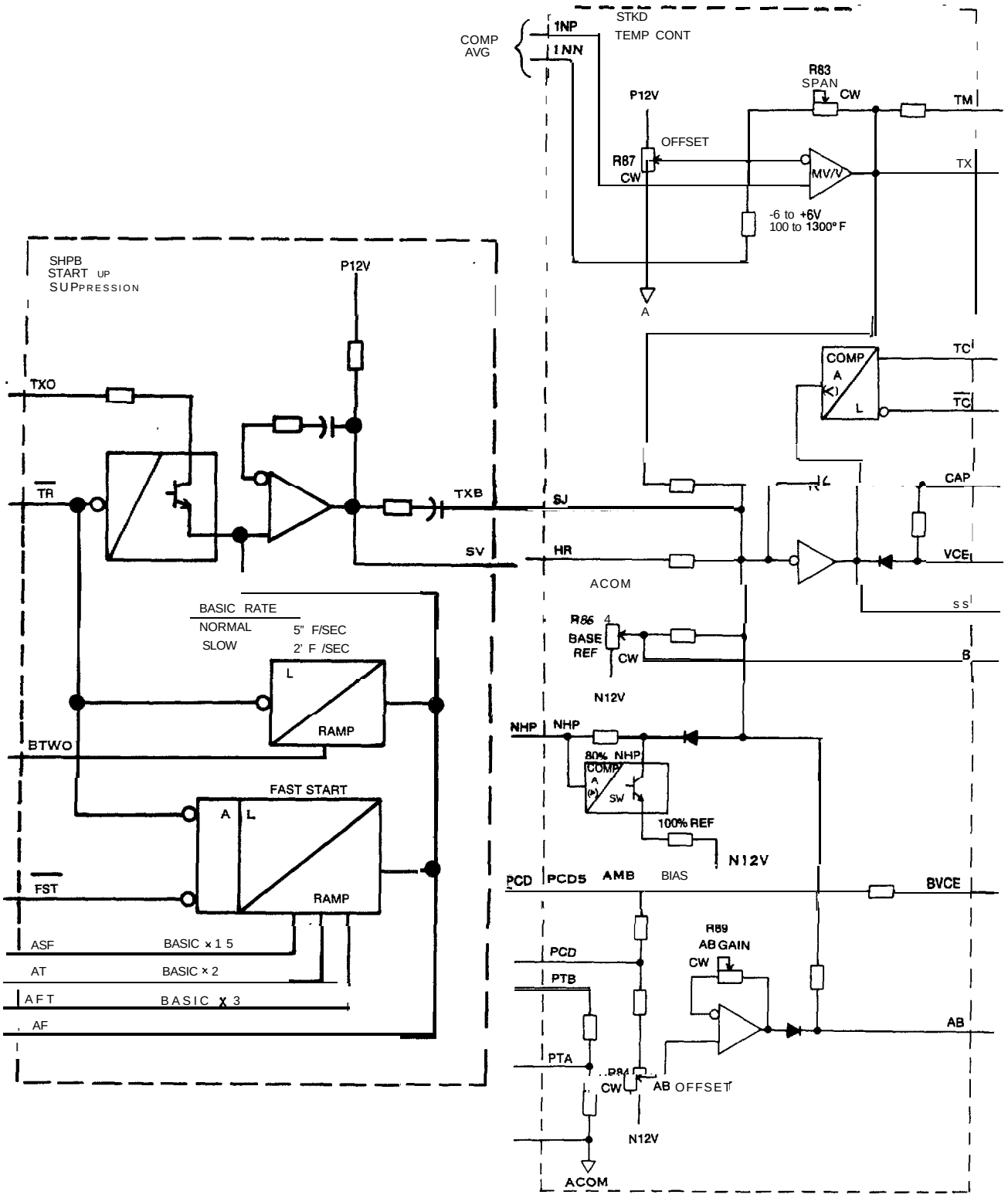


Fig. 24 Temperature control loop

C-7 FUEL SPLITTER (Optional)

A fuel splitter is provided for systems that have both gas and liquid fuel systems. The turbine control specification and control elementary diagram will define system that is used on a particular system. The fuel splitter is furnished to permit transfer from one type of fuel to another without interrupt of turbine operation.

The fuel splitter receives its command from either the local selector switch or a remote located switch. A purge period is available when going from full gas to liquid or from full liquid to gas. The fuel mixture can be stopped at any proportion, while in local control. The fuel splitter consists of cards SFKD and VRCC. The VRCC card was discussed in the digital setpoint section. The SFKD will be covered *now.*

SFKD Card (See Fig. 25, 26, page 46, 47)

The SFKD card generates the steer signals for the VRCC reversible counter on command of the fuel selector switch. A "1" on GAS will steer the counter **up (output** ST1 = "1") and a "1" on LIQ will steer the counter down (output STO = "1"). Output PUL provides count pulses to VRCC. Pulses are inhibited when any of the following are true.

1. LIQ and GAS are both "0" or "1."
2. Input CARRY is a "1."
3. DGAS is "0" and GAS is "1."
4. DLIQ is "1," NPG is "0" and LIQ is "1."

Inputs DLIQ and DGAS provide means of allowing a purge time after signals LIQ or GAS are received. NPG provides a way to bypass the purge when transferring from gas to liquid. Input CAP may be used to alter the frequency of the count pulses on PUL. Output $\overline{FG}$ is a "0" when fully on gas and $\overline{FL}$ is a "0" when fully on liquid.

The output of the VRCC counter is the input to the D/A converter on the SFKD card. The signal on input VCE is divided into two signals with their values depending on the output of the D/A converter. A simplified diagram follows. This diagram is used only as an example and does not include all aspects of the actual circuit.

For example, the signals VCE, GCE and LCE are each broken into active and inactive signals with the inactive signal being that portion required to crack the valve. All above this is considered to be the active part. The actual circuit is designed such that the active portion of LCE and GCE are always equal to the active portion of VCE.

Looking at the diagram and considering the active portion only, it is seen that

$$LCE = -(y)\,(VCE)$$

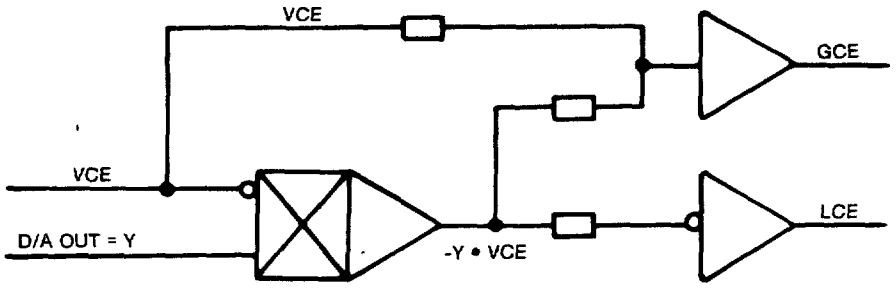
where Y is proportional to the count in the counter. Also it is seen that

$$GCE = VCE - (y)\,(VCE).$$

Therefore, when Y equals 1, LCE is equal to VCE and GCE equals zero. When Y equals zero, LCE equals zero and GCE equals VCE. The counter and D/A converter are connected such that when the counter is full, Y is zero, and when the counter is empty, Y equals 1. For mix operation, Y is some where between 0 and 1. Fig. 27 through 29, page 48, illustrate some typical fuel transfers.

Adjustments:

ADJUSTMENT	POTENTIOMETER	DESCRIPTION
GAS	R111	**Adjustment for inactive portion of GCE**
LIQ	R110	**Adjustment for inactive** portion **of LCE**



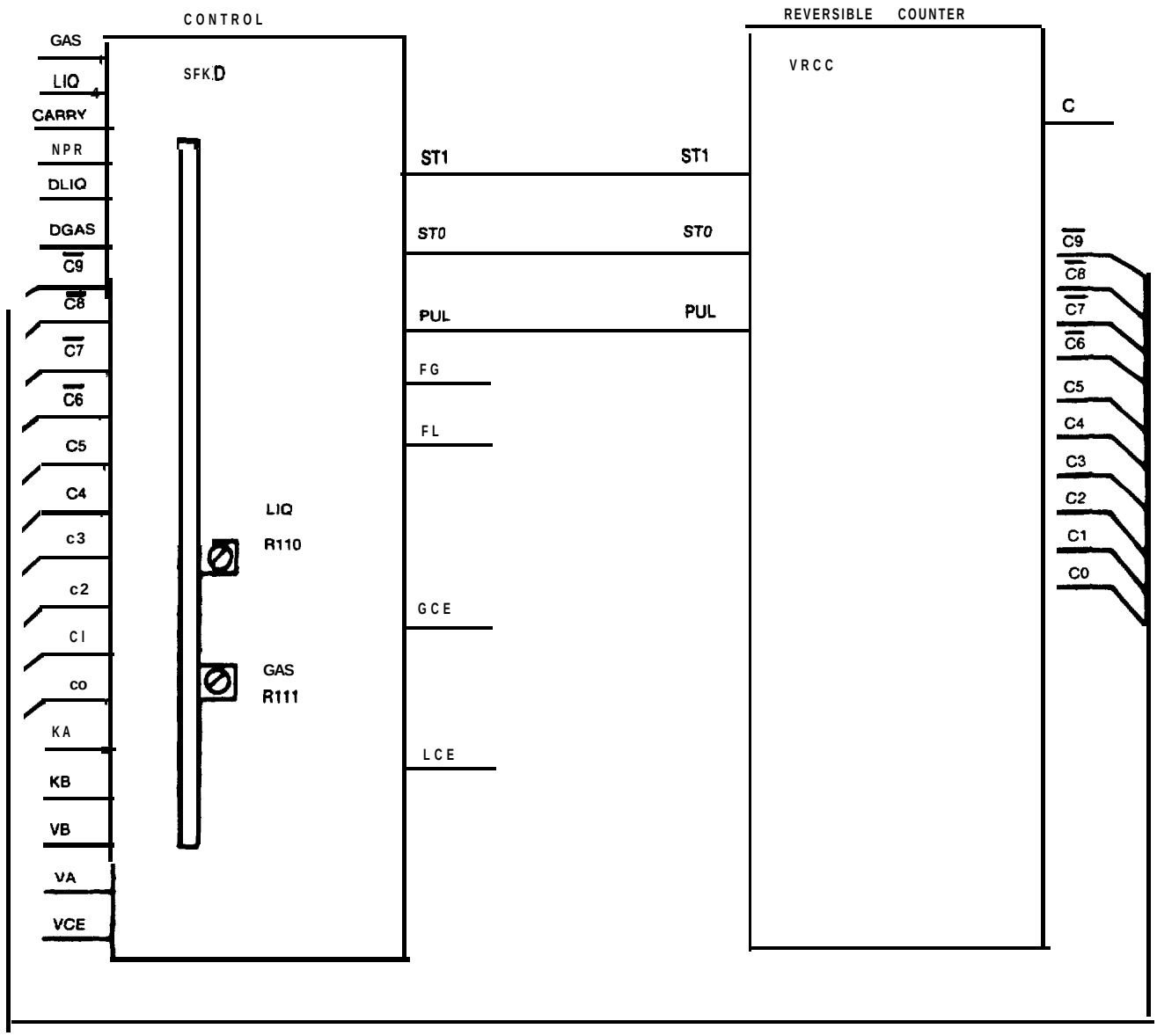


Fig. 25. Typical fuel splitter

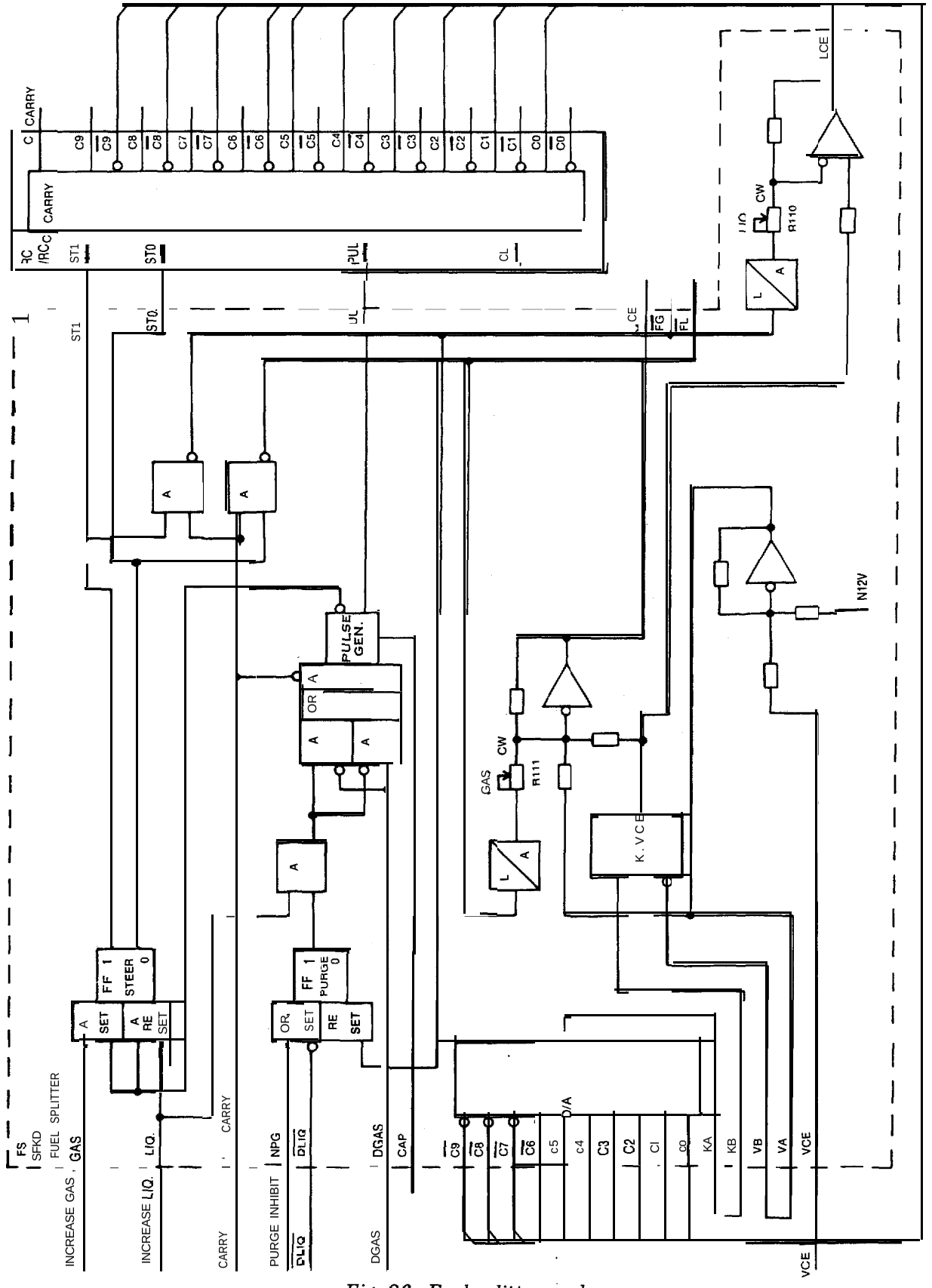


Fig. 26. Fuel splitter cards

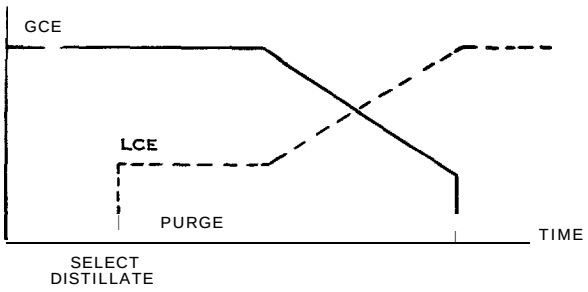


Fig. 27, Transfer from full gas to full distillate

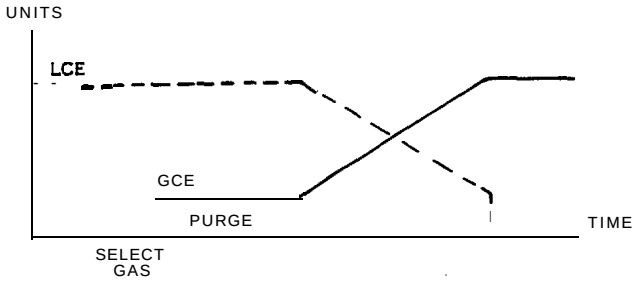


Fig. 28. Transfer from full distillate to full gas

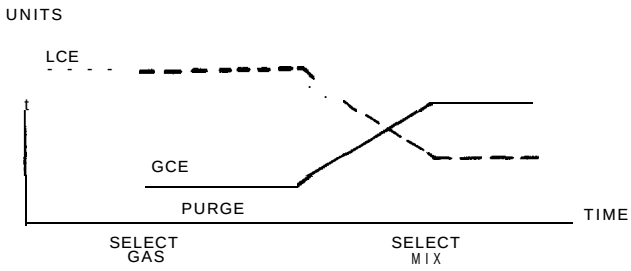


Fig. 29. Transfer from full distillate to mixture

C-8 LIQUID FUEL CONTROL (Optional)

(See Fig. 30, 31, page 49, 50)

The liquid control loop consists of (1) a fuel-flow detector card SFUA which receives a frequency proportional to fuel-flow rate from the sensor and (2) a regulator card SFKC which compares the feedback signal from SFUA with the product of the signals on inputs VCE and NHP and modulates the servovalve.

SFUA Card (See Fig. 30, 31)

The SFUA card contains two PR/A converters. Each of these converters receives a signal from a separate sensor and provides an analog signal at OUT 1 and OUT 2 which is proportional to frequency of signal from sensor. The largest signal from the two PR/A converters is selected and amplified to provide a negative analog signal out of FFN to be used as the fuel-flow feedback signal to SFKC regulator.

Different frequency ranges at input to PR/A converter can be processed by making appropriate jumpers between A, B, C and D for No. 1 PR/A and E, F, G and H for No. 2 PR/A converter. The following table gives required jumpers for various ranges.

Frequency Ranges, HZ	Jumpers
0 to 4290	No Jumpers
0 to 3340	A-D and E-H
0 to 2640	A-C and E-G
0 to 2180	A-B and E-F
0 to 1910	A-B-D and E-F-H
0 to 1660	A-B-C and E-F-G
0 to 1440	A-B-C and E-F-G-H

A comparator looks at the two analog signals from the PR/A converters and output ALARM becomes a "1" if they differ by more than a fixed percentage of the nominal signal. The selected high signal out of the converter is also compared to fuel-flow reference and gives a "1" out of TRIP if the feedback signal is greater than the reference and input 14HA is a "1." When input 14HA goes to a "0" output TRIP is inhibited.

Adjustments:

ADJUSTMENT	POTENTIOMETER
GAIN 1	R102
GAIN 2	R101
TRIP	R103

SFKC Card (See Fig. 30, 31)

The SFKC card contains the fuel-regulator amplifier, a multiplier, and servo-valve fault detector. The reference to the regulator amplifier is the product of the signals on inputs MP and NHP. Typically one reference input is received at VCE and an offset is added to remove the inactive portion of the signal. The offset signal is inverted and appears on pin VCEB. VCEB is connected to MP to provide one input to the multiplier. The regulator sums the reference signal from the multiplier with the feedback signal on FFN and modulates the servo valve.

Adjustments:

ADJUSTMENT	POTENTIOMETER	RANGE (volts d-c)
ZERO	R103	1.3 to 2.6
STAB	R104	
SPAN	R105	

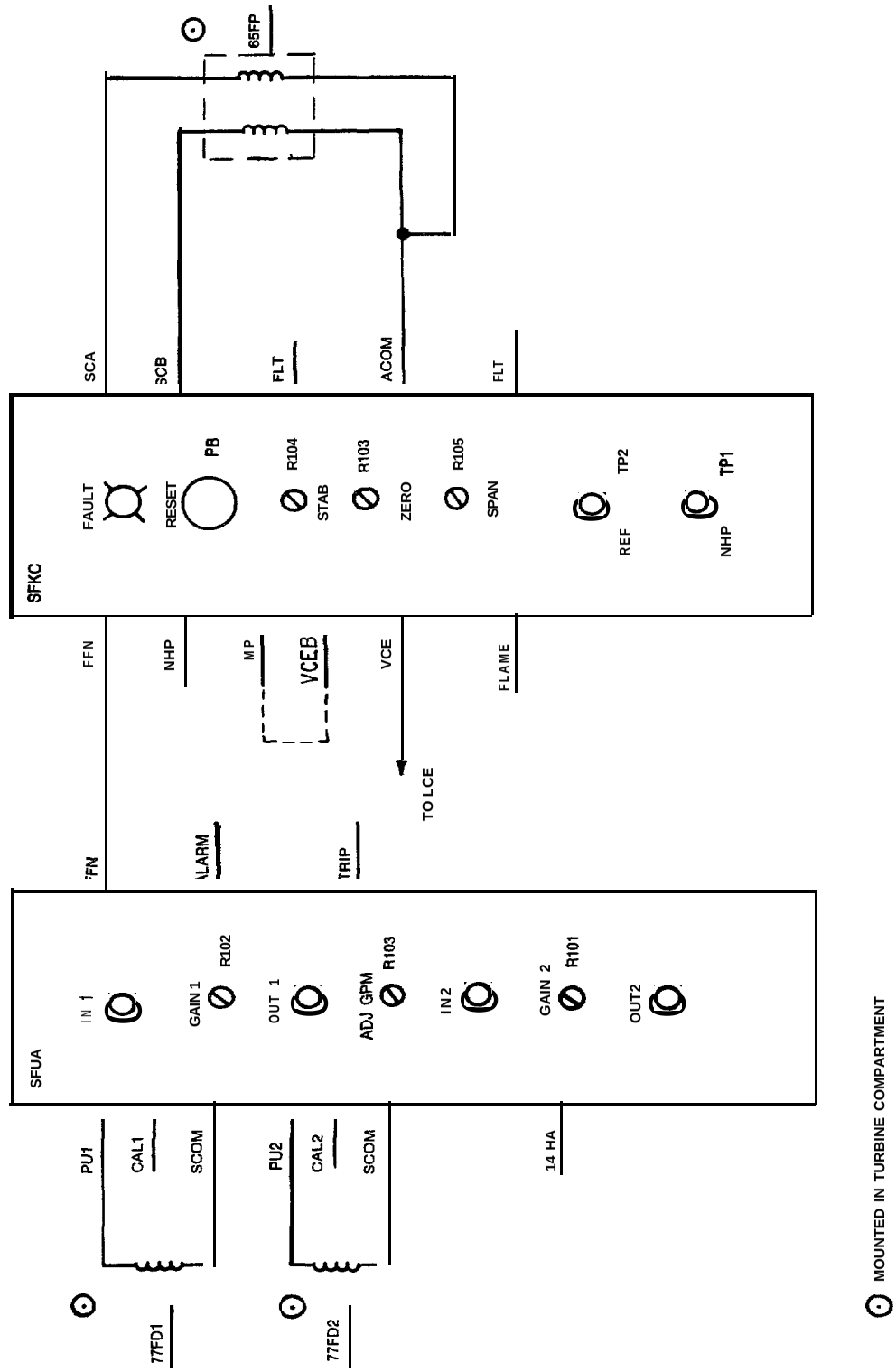


Fig. 30. Typical liquid fuel servo.

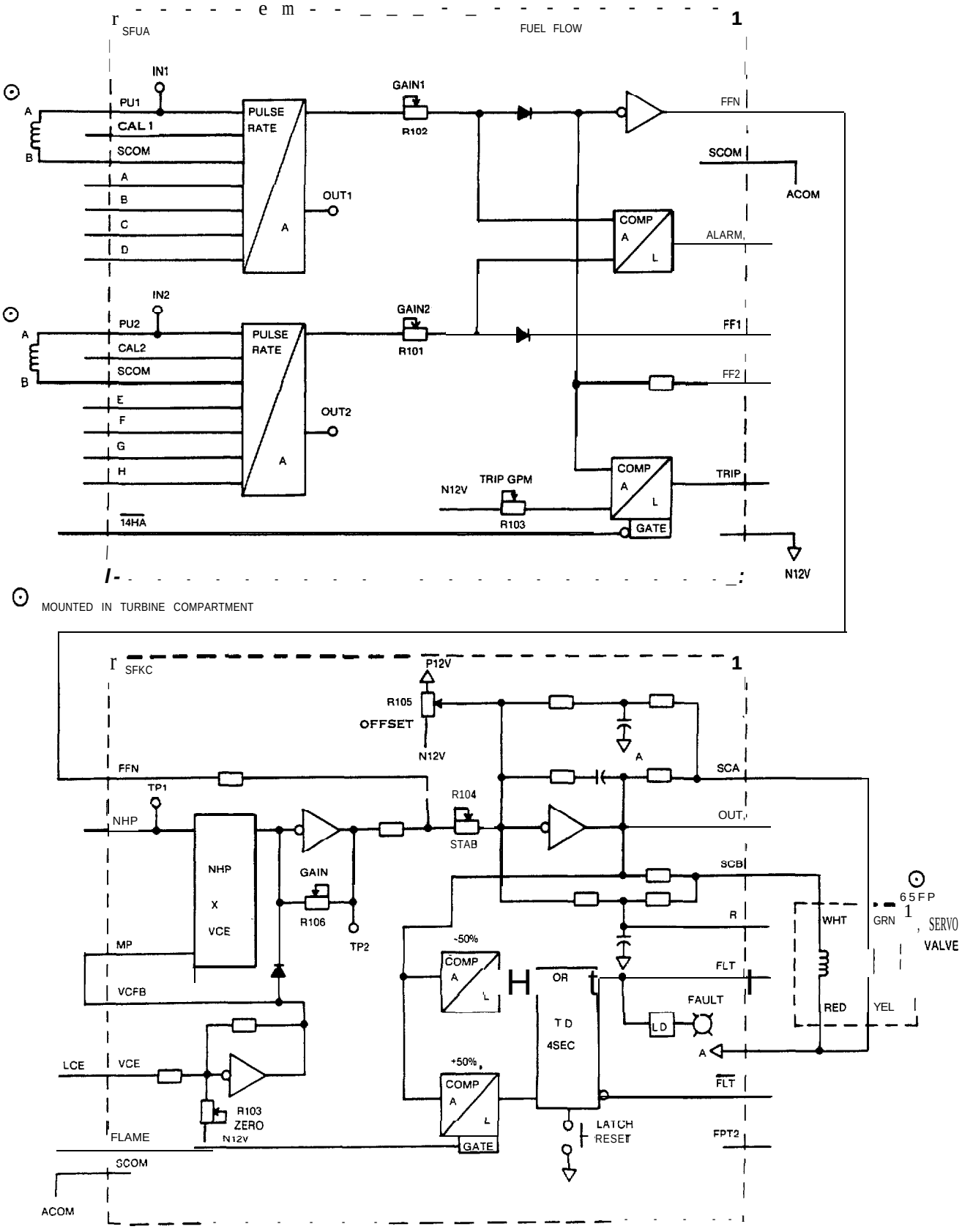


Fig. 31. Liquid fuel control cards

C-9 GAS CONTROL VALVE (Optional)

A typical diagram of the gas control-valve loop is shown in Fig. 33. The SSVG regulator card receives a feedback signal from SSVF and compares it to a fuel reference signal to modulate the servo valve. The feedback is developed using a linear, variable displacement transformer (LVDT) whose coupling co-efficiency is a function of the control-valve position. An oscillator of approximately 3 kHz provides the input for the LVDT. The amplitude of LVDT secondary varies with the position of the control valve. Two LVDT's and oscillators may be provided for increased reliability. An alarm is generated if the feedback circuit fails.

SOSG Card

The SOSG card contains an adjustable oscillator to supply a signal for the LVDT's. One SOSG card can drive a 200-mA RMS load at 6 to 8 volts. (This will drive up to 4 LVDT's.)

Adjustments:

ADJUSTMENT	POTEN-TIOMETER	RANGE
GAIN	R40	4.5 to 8.5 volts RMS
FREQ	R41	2350 to 3359 Hz.

SSVF Card (See Fig. 32, 33, pages 52, 53)

The SSVF card receives its input signal from the LVDT and detects the peak value of the signal and amplifies it to provide a negative feedback signal on POS for the SSVG regulator card. If either of the signals out of the peak detectors goes below a set threshold, output FBF becomes a "1." A minimum no fuel signal of 0.5 volts RMS is required from the LVDT's to prevent the alarm signal on FBF. A fuel-detector circuit is also provided to give a "1" on output FUEL when the feedback signal reaches a set threshold.

Adjustments:

ADJUSTMENT	POTENTIOMETER
FB ZERO	R60
FB SPAN	R62

SSVG Card (See Fig. 32, 33)

The SSVG card sums the feedback signal on FB with the reference signal on REF to modulate the servo valve. A signal on input VCE can be used as the reference by jumpering J2 to J1 and removing the signal on REF. A negative bias equivalent to 2 volts on REF is also summed with the reference to hold the gas control valve closed until the reference is greater than 2 volts. A voltage may be put on ST1 and ST2 to impose a bias directly on the gas control valve.

The card contains a comparator circuit which will give a "1" on FLTS and illuminate light FLTS if the absolute value of the output voltage stays above 3.5 volts for an extended period of time. A comparator circuit is also provided to detect an open or shorted servo-valve coil. Output FLTC will become a "0" for an open or shorted coil.

Adjustments:

ADJUSTMENT	POTENTIOMETER
GAIN	R90

C-10 STOP/SPEED RATIO VALVE (Optional)

When a gas fuel system is used, a stop/speed ratio valve is required to maintain the pressure upstream of the gas control valve at a value that is a function of turbine speed. (See Fig. 34, 35, pages 54, 55)

SSVD Card (See Fig. 35)

The SSVD receives a signal on P1 that is proportional to pressure. This signal is compared with the speed signal on NHP and provides an output on TRD. An offset and gain is provided on the NHP signal. The signal on TRD is the reference for the servo-valve regulator on the card. A "1" on input SR will force TRD to zero volts.

The feedback signal for the regulator is derived using an LVDT as was done in the gas control valve. Only one LVDT is used in this case.

Alarm output FLT will be a "1" and FAULT light will be on for the following:

(1) If TPC goes more positive than -1.2 volts indicating loss of feedback.

(2) If input D is not between -1.2 and -2.8V and input INH is a "0."

(3) When two SSVD's are used to drive the servo valve, C of each card can be tied to SCB of the other card to detect a faulty valve or circuit. For this case, an alarm will occur when the signals on SCB and C are of opposite sense. The alarm will be indicated on the card that SCB is more positive than C.

Adjustments:

ADJUSTMENT	POTENTIOMETER
ZERO	R95
GAIN	R92
PRESS SPAN	R93
NHP OFFSET	R96
STAB	R90

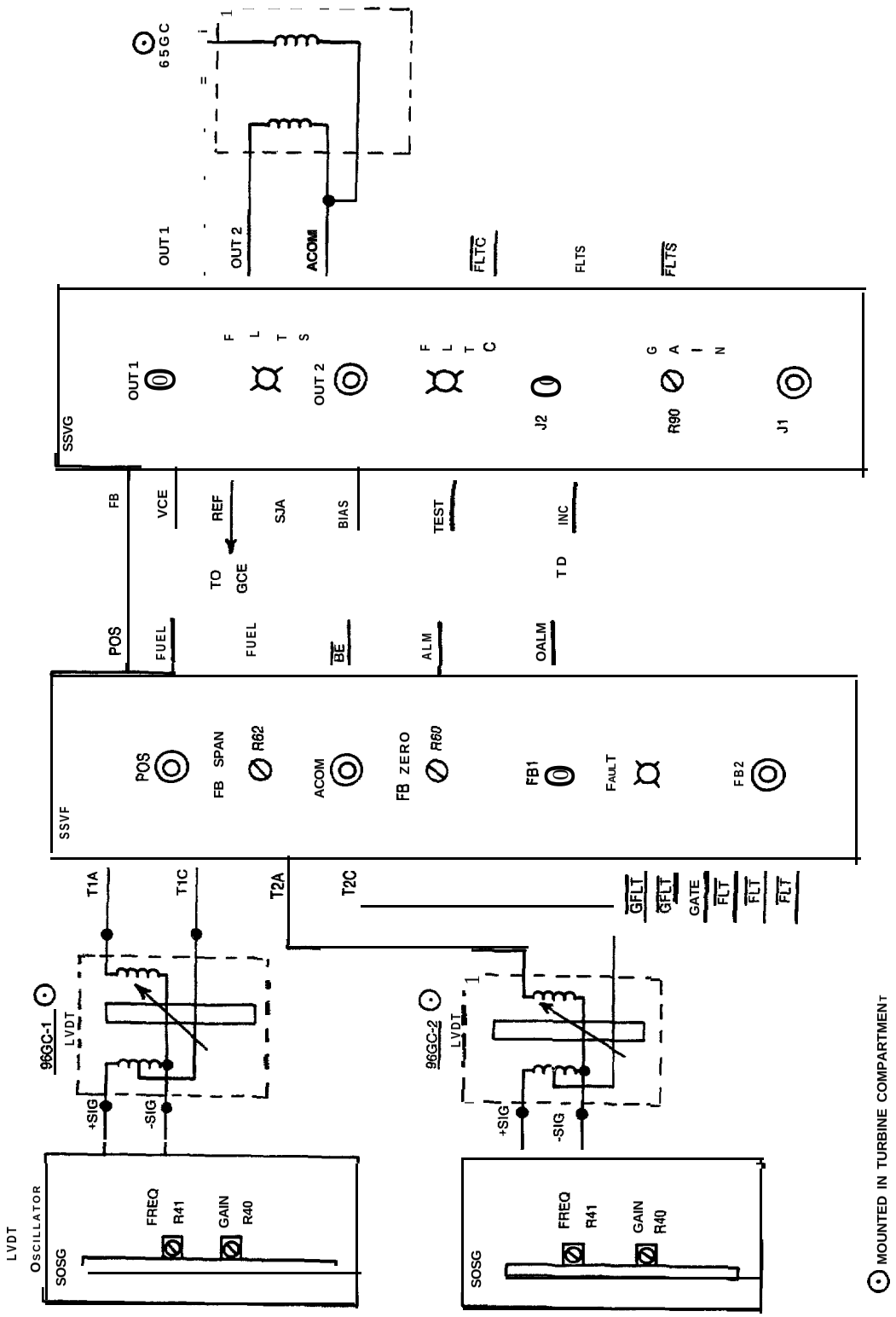


Fig. 32. Typical gas control valve

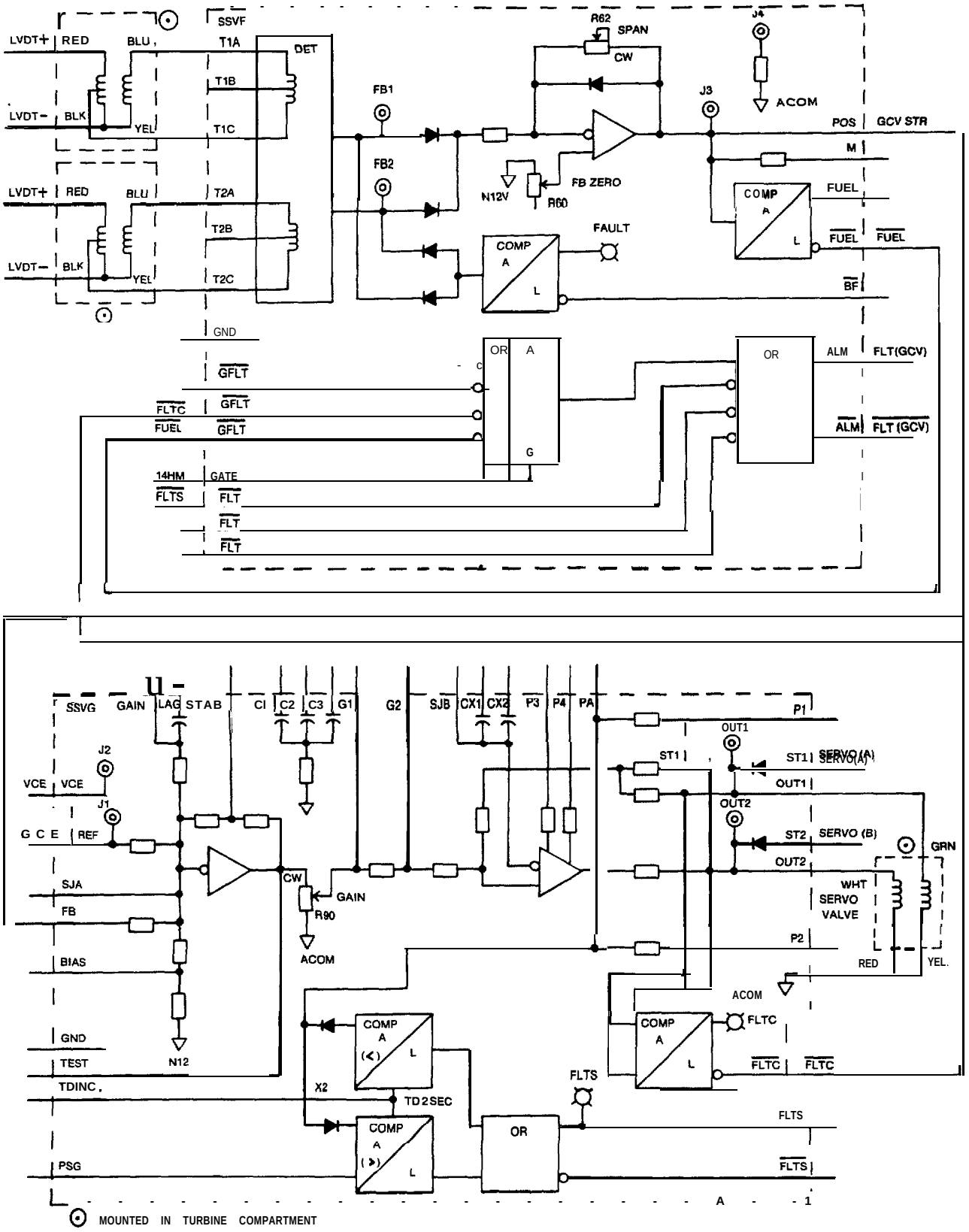


Fig. 33. Gas fuel control loop

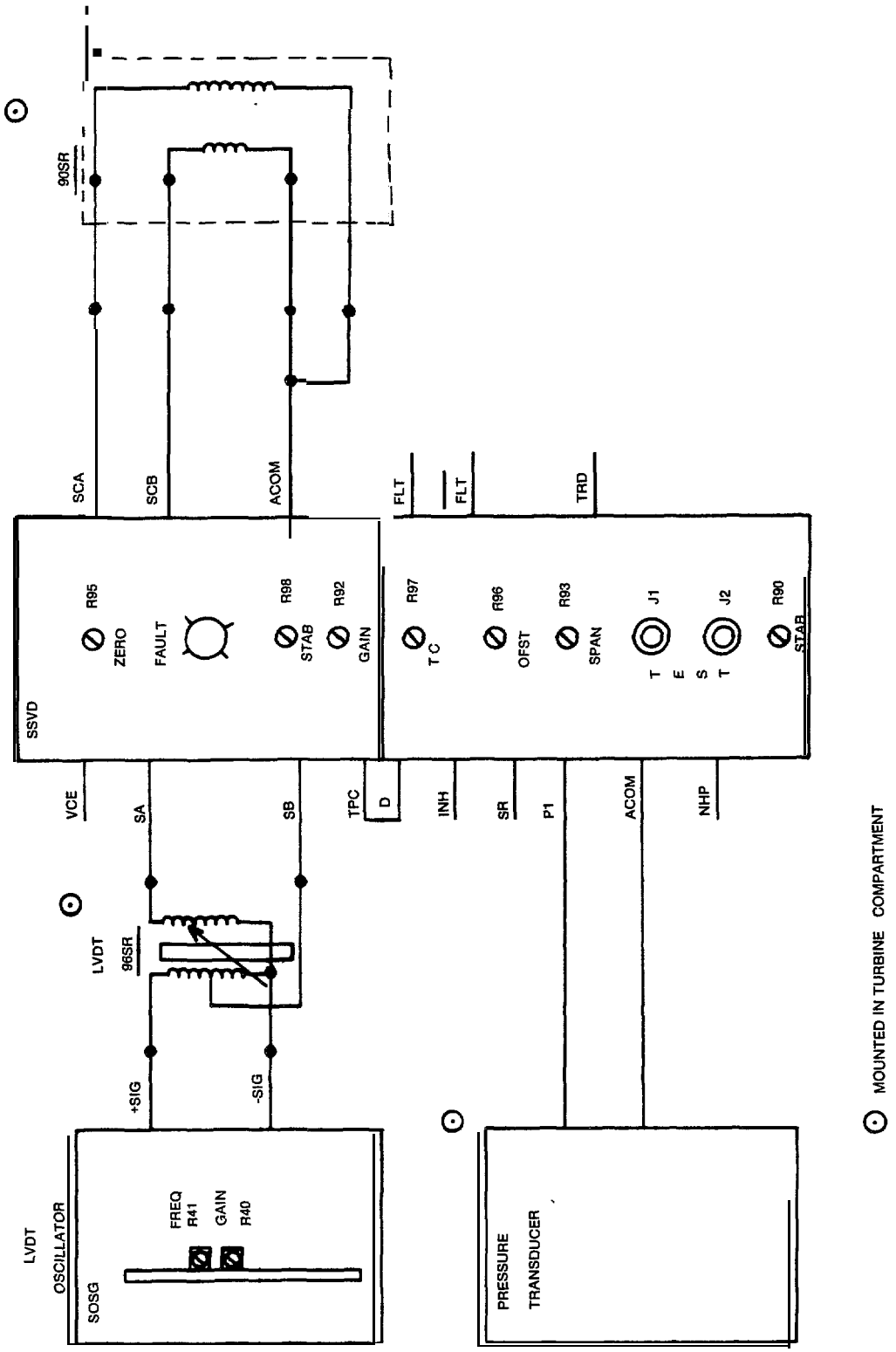


Fig. 34. Typical speed ratio valve servo

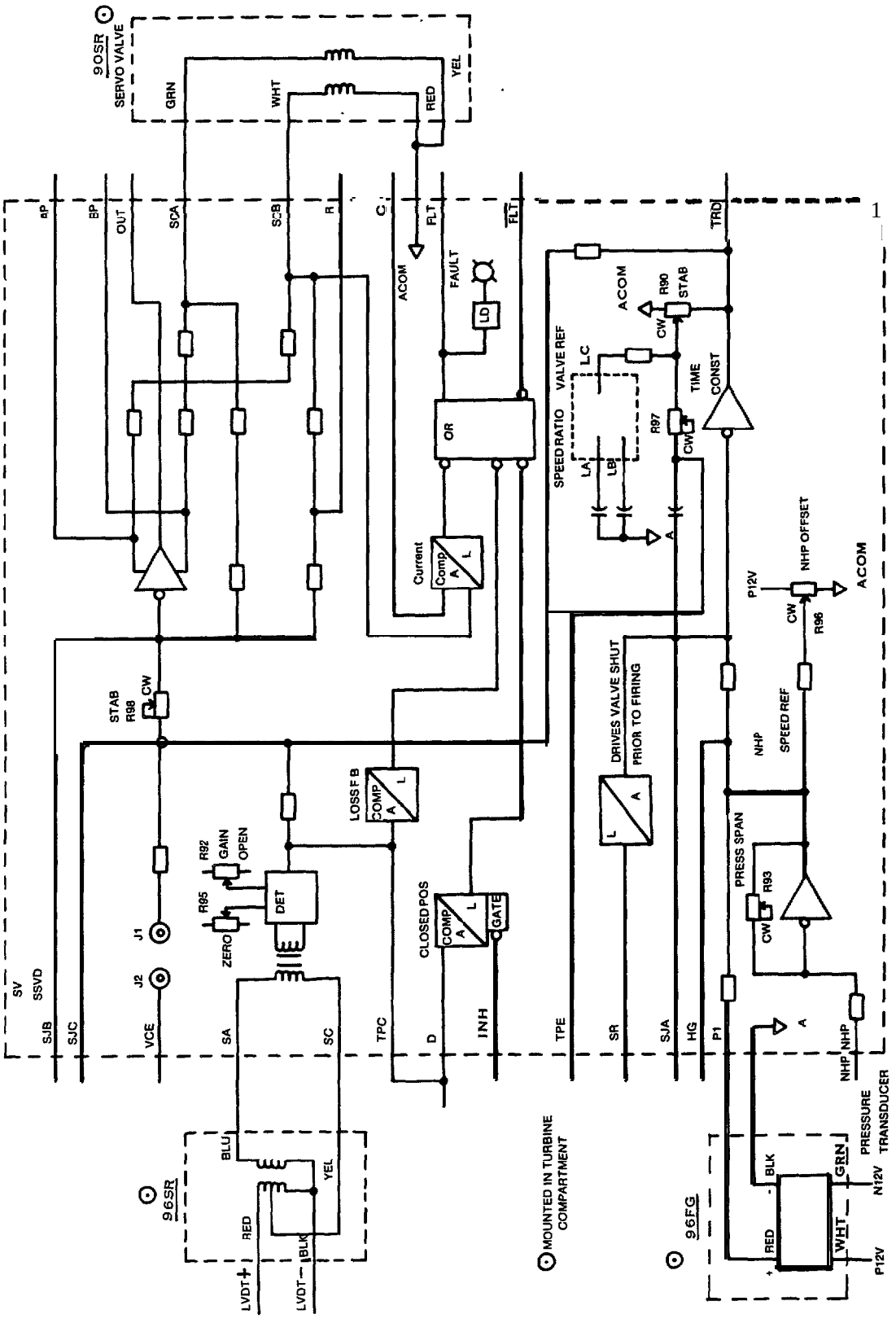


Fig. 35. Speed ratio card

SECTION D
PROTECTION

D-I "4" CIRCUIT

The focal point of the protection system is the VMPA card which generates the "4" signal and the shutdown bias to the servo valves. (Due to the magnitude of its importance, a detailed description will be presented.) The symbol for the VMPA card is shown in Fig. 36. To help eliminate some confusion, a "★" is placed near those pins not normally used in industrial applications. A typical protection system illustrating the "4" bus is shown in Fig. 37.

The VMPA serves the dual function of monitoring input signals as per-missives to start the turbine and as protective signals to trip the turbine. The "4" signal outputs are a function of a flip-flop which is set upon receipt of a start command when all permissives have been satisfied. With the flip-flop set, the "4" signal outputs permit the turbine to be started and operated. The flip-flop is reset when an abnormal operating condition is detected by protective monitoring devices. Resetting the flip-flop trips the turbine.

The VMPA functions, inputs and outputs, are discussed below. The discussion is referenced to Fig. 36 by numbers shown inside a circle.

Sum "1" Gate

The sum "1" gate (refer to ① on symbol) generates the "start-check" permissive leg to set the flip-flop. The permissive is satisfied when all inputs to the sum "1" gate are "1" 's. Once the flip-flop is set, this permissive is no longer required.

An "AND" expander point, SAE, is provided to expand the sum "1" gate by the use of LGXA cards.

Inputs: 14X, $\overline{\text{14HMX}}$, 12HT-1, 12HT-2, 12HT-3, 12LT-1, 12LT-2, 12LT-3, 14HSX (signal inverted), SAE.

Output: $\overline{\text{STCK}}$ • goes to logic "0" when all inputs to sum "1" gate are "1" 's.

Majority Logic

Two majority logic elements are used-one for HP overspeed ②, and another for LP overspeed ③.

The output of the HP logic will be a "I" as long as two of the three inputs-are "0." However, when two of the three become "1" 's, the output becomes a "0" indicating the unit should be tripped. The LP logic is "OR" 'ed with a speed signal "AND" circuit which bypasses the majority logic output during start-up. When in the run mode, the majority logic is active and operates in the same manner as the HP logic. The output of the LP logic is inverted giving a "0" for a safe condition and a "1" for a trip.

HP Logic Inputs: 12HT-1, 12HT-2, 12HT-3

HP Logic Output: Internal output to the start/run gate.

LP Logic Inputs: 12LT-1, 12LT-2, 12LT-3, 14LS, 14HSX

LP Logic Output: 12LT (connected to LT input of the start/run gate)

Sum "0" Gate

The sum "0" gate ④ monitors trip protective devices. The gate outputs a "1" when all inputs are "0" 's. This "1" output is a permissive to operate condition. If any input goes to a "1" state, the output will become a "0" and trip the turbine. Therefore, any unused input must be tied to common to allow the turbine to run. The output of the sum "0" gate is connected to the output of the start/run gate logic to form a hardwired "OR", Approximately 0.7 msec after the trip input signal is initiated, the trip bus will become a "0" to trip the turbine.

Inputs: T01, T02, T03, T04, T05, T06, T07, T08, T09, T10, T11, T12, T13.

Outputs: Internally tied to output of start/run gate logic. Combined output operates a time delay circuit, which is connected to the trip bus. This output also drives the $\overline{\text{PS}}$ output, which is a logic "0" when the sum "0" gate and the start/run gate are satisfied.

Start/Run Gate

The start/run gate consists of a gate for start-up ⑤ and a gate for running ⑥. The outputs of these two gates are "OR" 'ed to supply an output which is hard wire "OR" 'ed with the output of the sum "0" gate previously described.

The startup portion is an "AND" gate with "OR" expansion capabilities. This gate must be satisfied prior to and during the start-up sequence. The "OR" expander portion is used to maintain the output during the firing period. This gate is de-energized when its inputs change state, normally upon detection of flame (28FDX). The output has a 0.5 second time delay on transition from a "1" to a "0" to allow the run gate to be energized and maintain turbine operation. The output also has a 40 millisecond time delay on transition from a "0" to a "1" to insure tripping the turbine on loss of flame.

Inputs: $\overline{\text{14HSX}}$ from internal inverter, $\overline{\text{28FDX}}$, ST, or EXP.

outputs: Internal connection to "OR" gate with run portion and inverted output "$\overline{\text{SM}}$" for indication of start-up mode satisfied.

The run mode portion of start/run gate provides monitoring of protective trip signals which must be bypassed during start-up. This run gate is an "AND" gate used for monitoring HP overspeeds, LP overspeeds, and hydraulic dump pressure. These signals are monitored for permissible state after flame has been established. A trip state on any one of the inputs, including loss of flame, will trip the turbine.

Inputs: Internal HP overspeed signal from majority logic, "LT" normally connected to output "12LT" from LP majority logic, "$\overline{\text{63HD}}$," "$\overline{\text{28FDX}}$."

Outputs: Internal connection to ["OR") gate with start-up portion, and inverted output "RM" for indication of run mode satisfied.

Master Flip-Flop

The state of the master flip-flop ⑦ determines whether the unit may run or is tripped. Setting the FF permits the unit to operate. Resetting the FF trips the turbine.

The FF is set by a four input "AND" gate. This gate sets the FF only when the start-up checks are satisfied, lube oil pressure is established, the "TRIP BUS" is satisfied, and a start signal has been received.

The FF is reset when the "TRIP BUS" becomes dissatisfied (logic "0" state). The "TRIP BUS" can be driven to the trip state by several methods. Detection of a trip condition by the sum "0" gate will drive the "TRIP BUS" to the trip state. Dropping out the start/run gate will also trip the turbine. Loss of lube oil pressure, 63QT, or loss of power as detected by internal sensors will also cause a trip. The "TRIP BUS" trips the turbine on normal shutdowns when input TG becomes a logic "1" as a result of external logic sequencing. Operating the emergency stop push button trips the unit by disconnecting 28V power from input "ESTOP" and triggering a voltage sensor circuit connected to the "TRIP BUS."

Inputs: $\overline{\text{START}}$, ESTOP, 63QT, TG, and internal outputs from the sum "1" gate, sum "0" gate, start/run gate, and power-supply voltage sensing circuits.

outputs: $\overline{\text{4}}$, 4 . Logic outputs for sequencing.
4RD . relay driver output to energize a relay when the FF is set.
SRT . servo trip bias to drive control valves to their proper shut-down state when the turbine is tripped.

Hydraulic Pump Circuit

The hydraulic solenoid may be controlled by this driver output through a relay, ⑧. Two "AND" functions are "OR" 'ed. The lower "AND" provides an output prior to firing, and the top "AND" provides an output during the run mode. All inputs on the "AND" 's must be "0" 's to enable. Both "AND" circuits are controlled by the master flip-flop output and the $\overline{\text{HDX}}$ input.

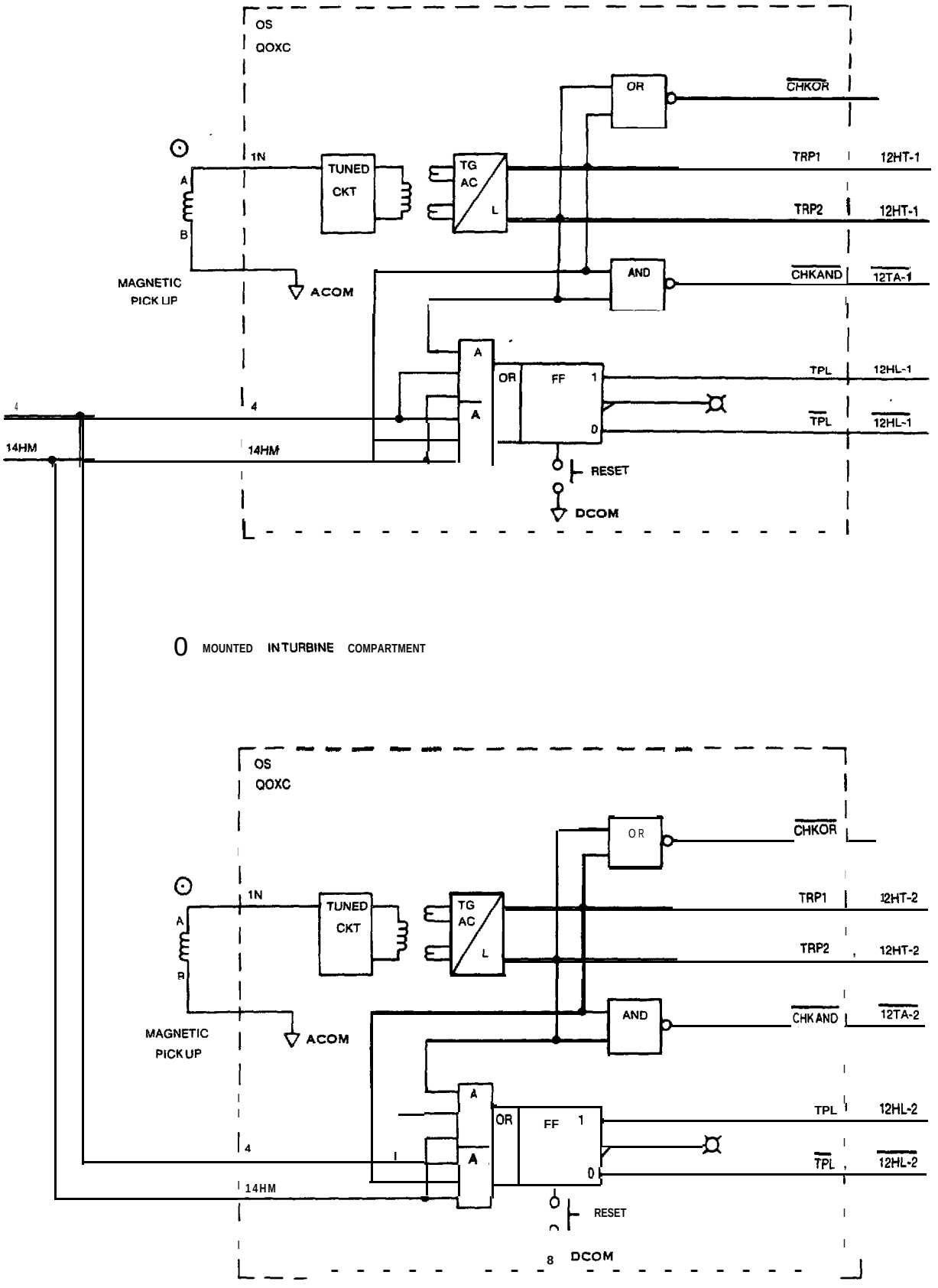


Fig. 39. Overspeed curds

D-3 OVERTEMPERATURE TRIP AND ALARM

The overtemperature protection system consists of two (four optional) separate channels. Each channel looks at the averaged output of thermocouples placed in the turbine exhaust duct. The over-temperature system includes cards SOTH, SOTG and the temperature averaging cabinet. The averaging cabinet is the same one used in the temperature control loop and will not be discussed here. (See Fig. 40 for diagram of over-temperature protection system).

SOTH Card: (See Fig. 41, page 64)

The SOTH card contains two identical circuits. Only the circuit associated with channel A will be discussed. The SOTH receives the compensated average signal from the averaging cabinet between pins 1NP and 1NN. This signal is amplified and appears at OTA. The amplified signal is also summed with the signal on AB, SJ, and internally generated NHP bias and reference signals. If OTA, SJ, NHP bias and AB exceed the reference, a "1" is generated on TP and TRIP lamp is on. The circuit is latching and may be set by depressing the reset push button. HR may be used to alter the range of the trip adjustment.

Adjustments:

ADJUSTMENT	POTENTIOMETER
OFFSET	R73
GAIN	R70
TRIP	R72

SOTG Card: (See Fig. 42, page 65)

The SOTG card receives the OTA and OTB signal from the SOTH card. These signals are summed with AB and NHP bias, compared to a reference signal, and generate a "1" on EA if either exceeds the reference. This card also generates an ambient bias signal to be used to droop the alarm and trip points. This ambient bias circuit will accept the same type signals as the STKD card.

The SOTG card also performs three circuit performance checks: (1) It compares the ambient bias generated in the temperature regulator card, STKD, with the internally generated ambient bias and annunciates an alarm if they differ by more than a predetermined value; (2) it compares the high of OTA, OTB, and TX with the average of OTA, OTB, and TX and annunciates an alarm if they differ by more than a predetermined value; (3) it also generates a system fail **trip** signal if both OTA and OTB drops below 200 F after the warm-up period, or if power is lost to the temperature averagingcabinet.

Adjustments:

ADJUSTMENT	POTENTIOMETER
AB OFFSET	R106
AB GAIN	R105
ALARM	R104
TEMP DIFF	R102

O-4 VIBRATION PROTECTION

A vibration protection system may consist of up to five trip and alarm (optional) channels. A trip on any channel will trip the turbine. A trip is latching and must be reset by push button or remote signal. Velocity vibration sensors provide the input signals. Each channel has circuitry to test for an open or shorted sensor. A typical diagram is shown in Fig. 43, page 66.

SVDC Card (See Fig. 44, page 67)

The SVDC card receives the signal from the sensor between pins VIB and ACOM. The peak value of this signal is detected and compared to a reference set by R1. If the peak value of the signal exceeds the reference a "1" is latched up on output T and a light is illuminated. The latching circuit may be reset via the local reset push button or a "1" on input RST. The setpoint reference may be monitored between TP3 and TP4. The peak value (VIB LEVEL) of the input signal may be monitored between TP1 and TP2. If the sensitivity of the sensor used is 0.2 volts peak for 1 inch/sec. pin GAIN is connected to ACOM to double the gain of the detector.

A sensor fault detector is provided to detect a shorted or open sensor. This is a latching circuit and must be reset by local push button or remote signal. The trouble light is illuminated and output "A" becomes a "1" for a faulty sensor. **The** following table lists the impedance required for no fault conditions:

Jumpers	No Fault **Impedance**
None	2.9K to 50K ohms
PIN 9 - P12V	80 to 800 ohms
PIN 6 - P12V	160 to 1.5K ohms

Adjustments:

ADJUSTMENT	POTENTIOMETER	RANGE
SET PT	R1	0.25 to 1.6 in/sec.

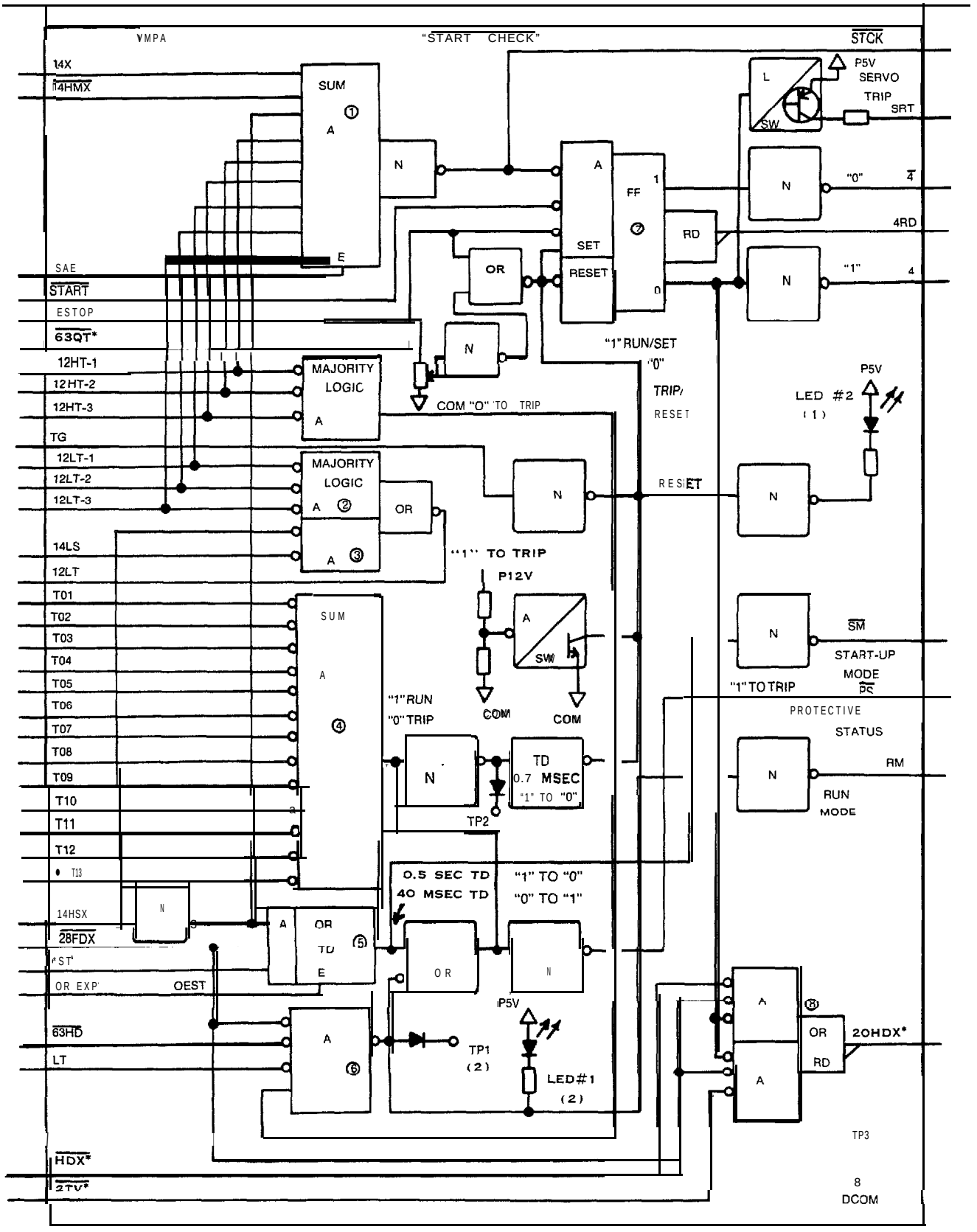
SVDE Card (Optional) (See Fig. 45, page 68)

The SVDE card contains three identical circuits. Only one of these circuits will be discussed. Typically, the input to the card is the 0 to 8 volts vibration level from the SVDC card and compares it to a setpoint reference. Input X1 is used for a O-to 8-volt signal. An input signal of 0 to 16 volts may be used by connecting J1 to J2 and receiving the signal on X2.

If the input signal reaches or exceeds the setpoint level, output A will become a "1" and alarm light will come on. If output L is tied to A, the alarm will latch up until reset via reset push button on card front, P12 volts on RS2 or "0" on RS1. If L is left open, the latch will reset when the input signal decreases 2% below alarm setpoint level. The setpoint level may be monitored at ALARM SETPT using a high-input-impedance meter.

Adjustments:

ADJUSTMENT	POTENTIOMETER	RANGE
ALARM SETPT	R50	0 to 8 volts on Xl 0 to 16 volts on X2



*- NOT USED FOR INDUSTRIAL APPLICATIONS

Fig. 36. VMPA card

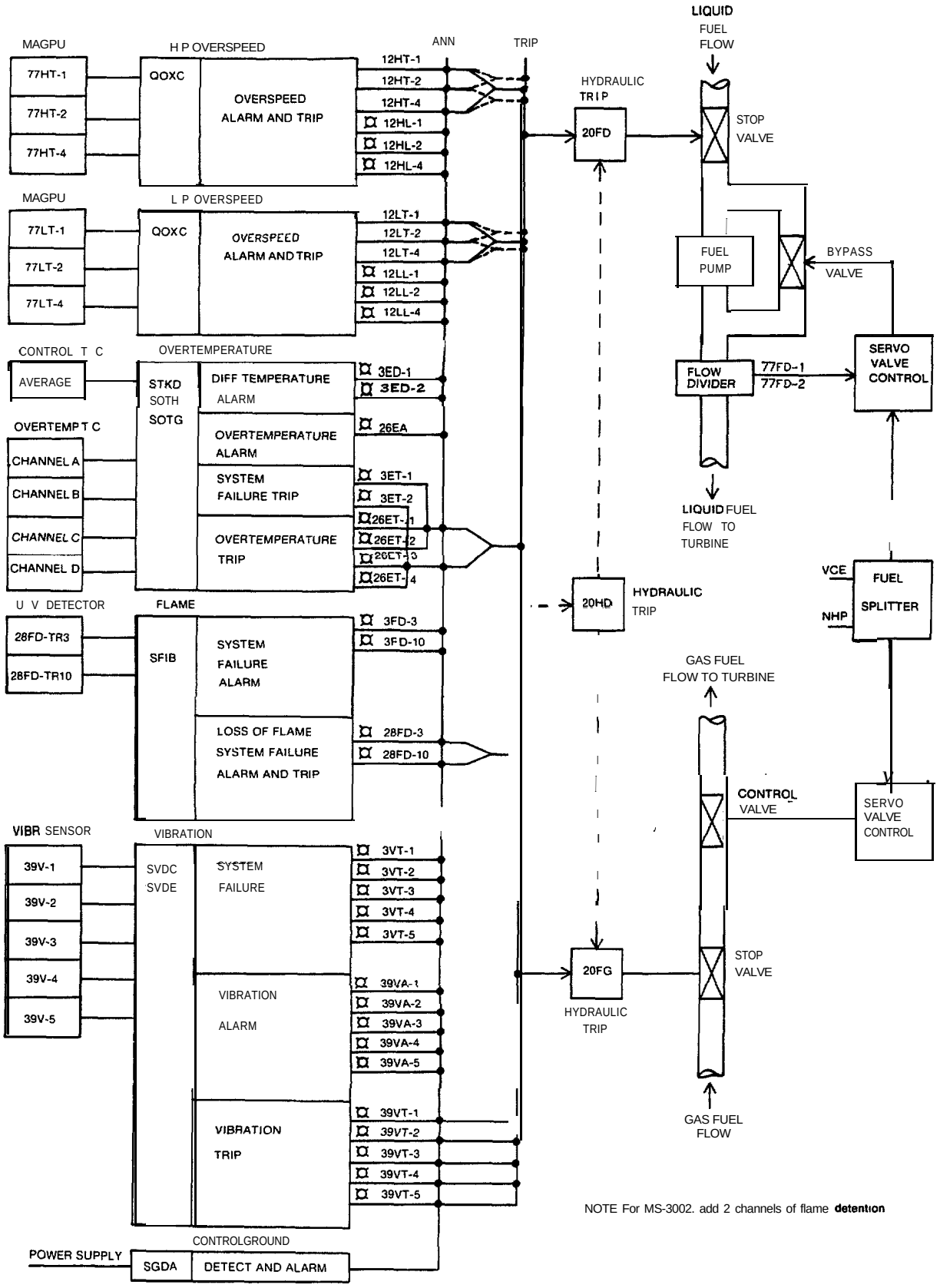


Fig. 37. Typical MS-5002 protection schematic

O-2 OVERSPEED

Overspeed protection is provided for both the LP and HP shaft. The LP and HP protection systerns are very similar, and each consists of two (three optional) separate sensing circuits to provide increased reliability. A diagram of a typical HP shaft protection is shown in Fig. 39.

QOXC Card (See Fig, 38, 39)

The QOXC card receives a frequency proportional to the speed of the turbine between pins IN and ACOM. Outputs TRP1 and TRP2 will be a "1" for the following conditions:

(1) Input speed is 10% above rated speed.

(2) Input speed is below 9% of rated speed when turbine is accelerating.

(3) Input speed is below 4% of rated speed on deceleration of turbine.

Output $\overline{\text{CHKOR}}$ is a "0" if either TRP1 or TRP2 is a "1." Output $\overline{\text{CHKAND}}$ is a "0" if both TRP1 and TRP2 are "1." However the primary purpose of the circuit is to generate a "1" on outputs TRP1 and TRP2 when the turbine is operating over a predetermined speed. A latching circuit is provided to hold the trip indication signal on TPL. Output TPL will latch in the "1" state if either TRP1 or TRP2 is a "1" and inputs 14HM and 4 are "1." The push button on the card front must be depressed to clear the latch.

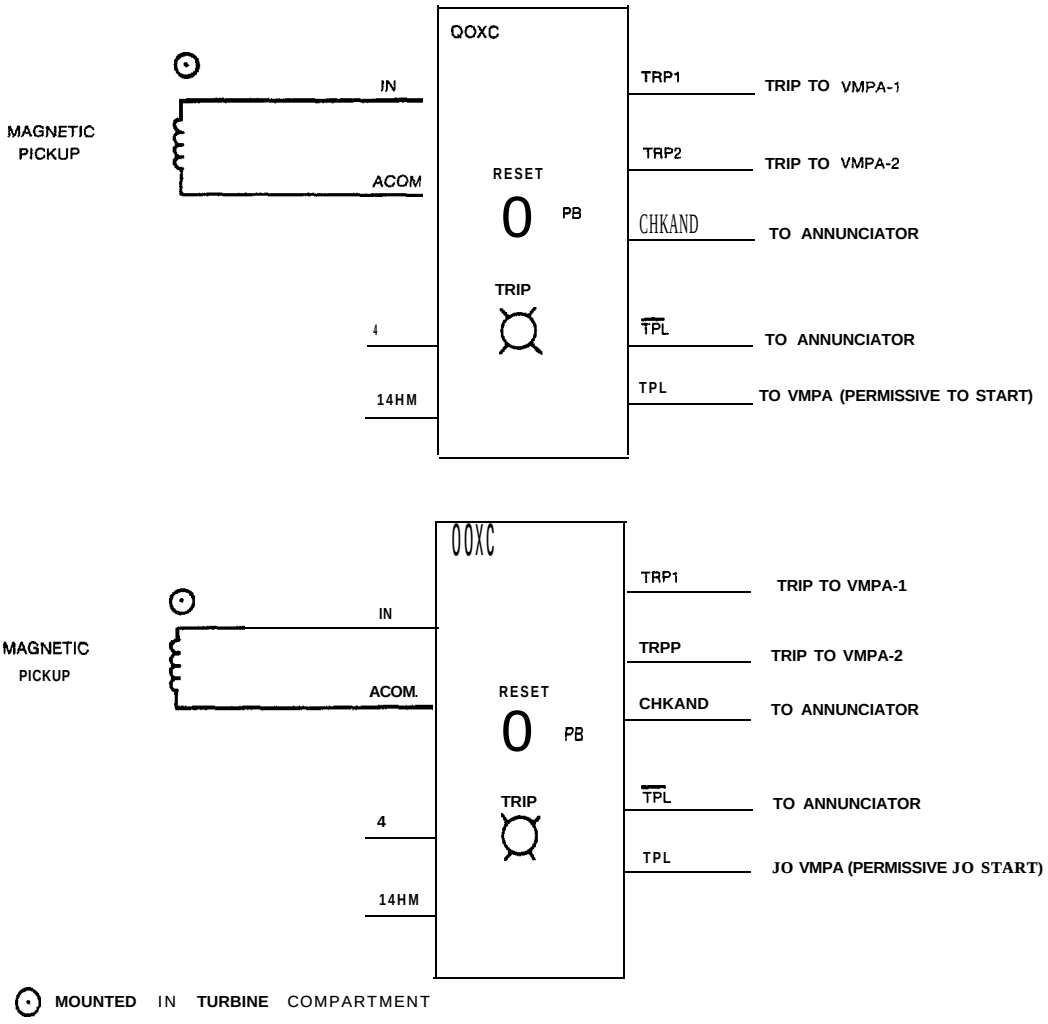


Fig. 38. Typical overspeed trip

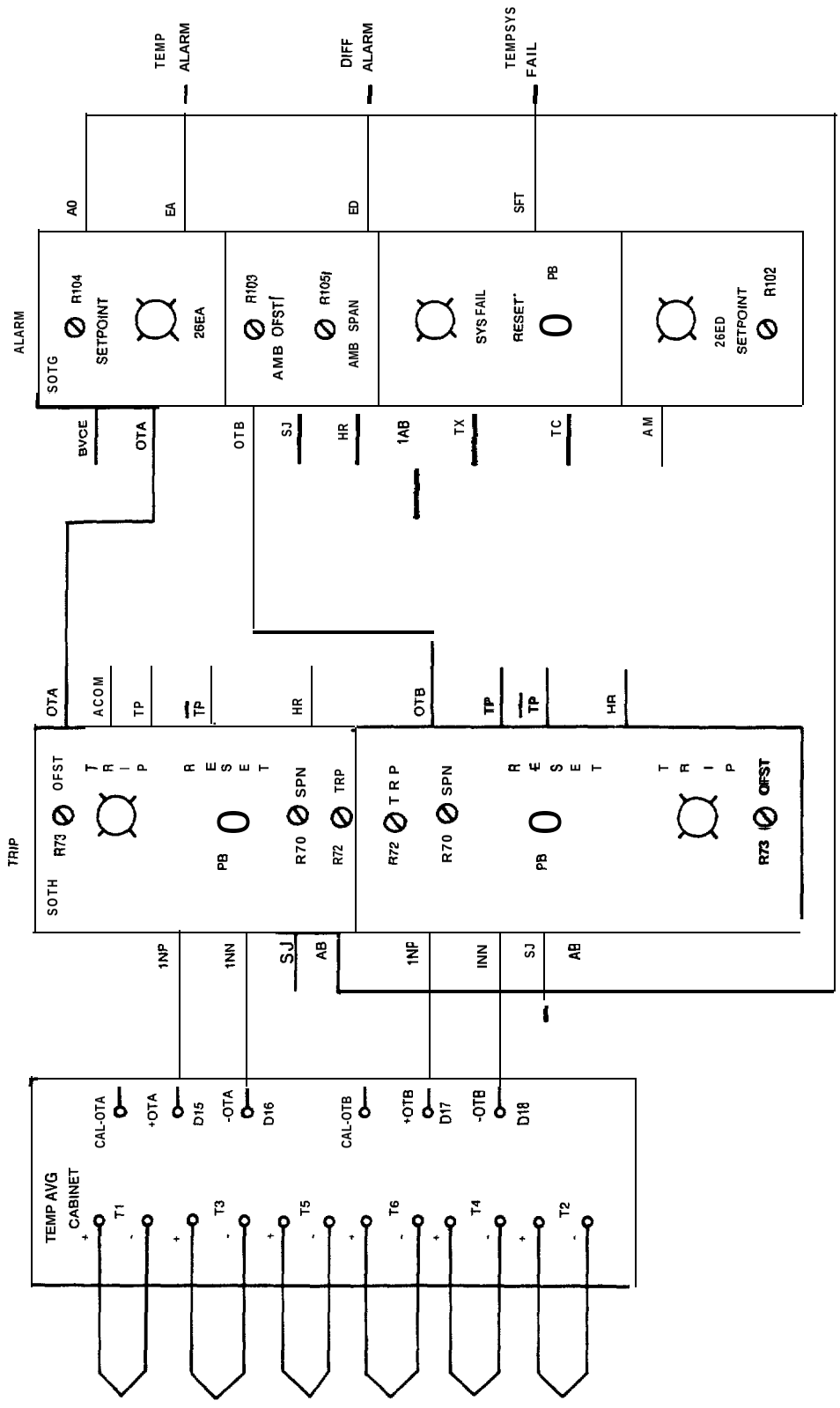


Fig. 40. Overtemperature alarm and trip (Protection system)

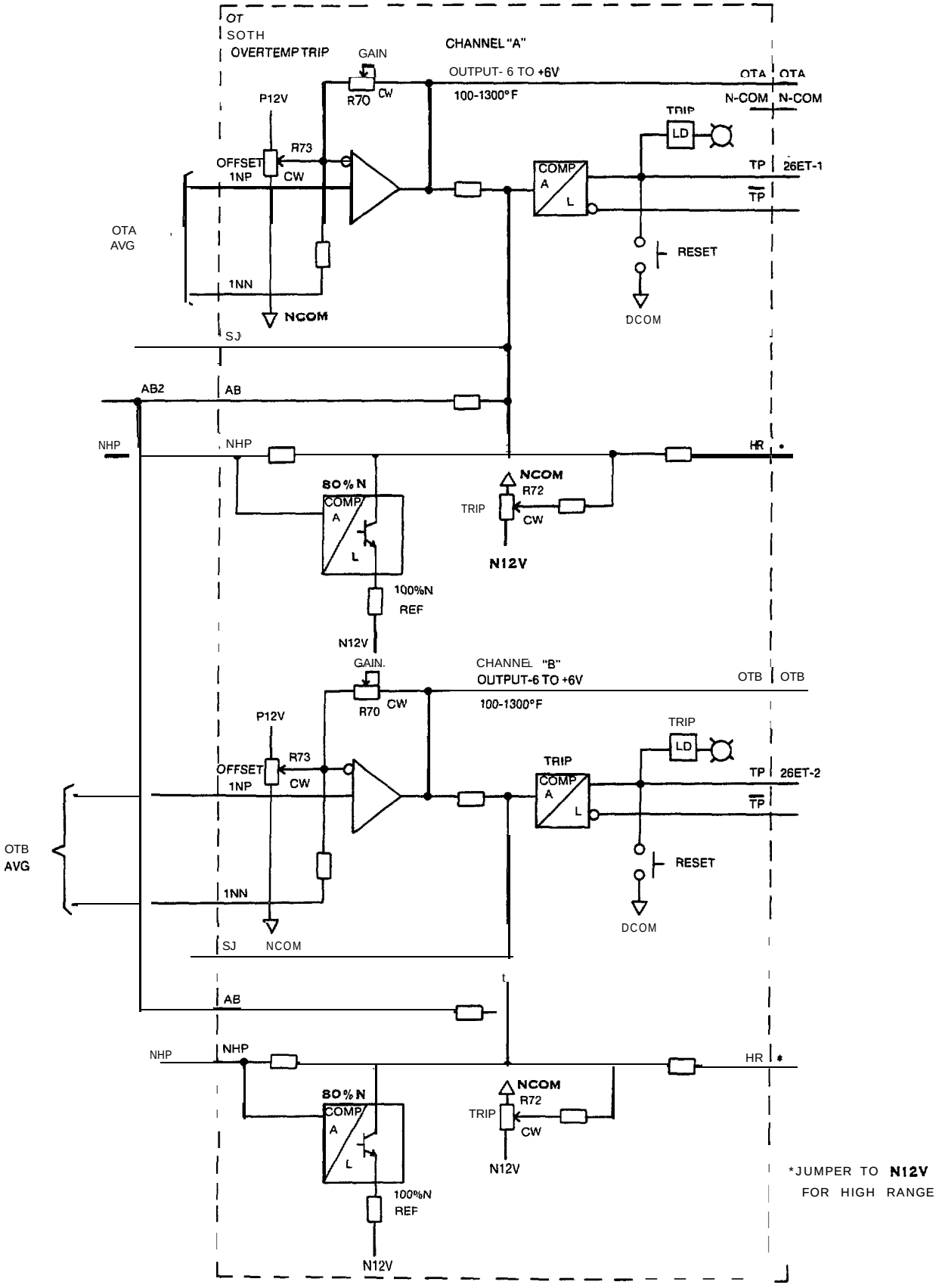


Fig. 41. Overtemperature trip card — SOTH

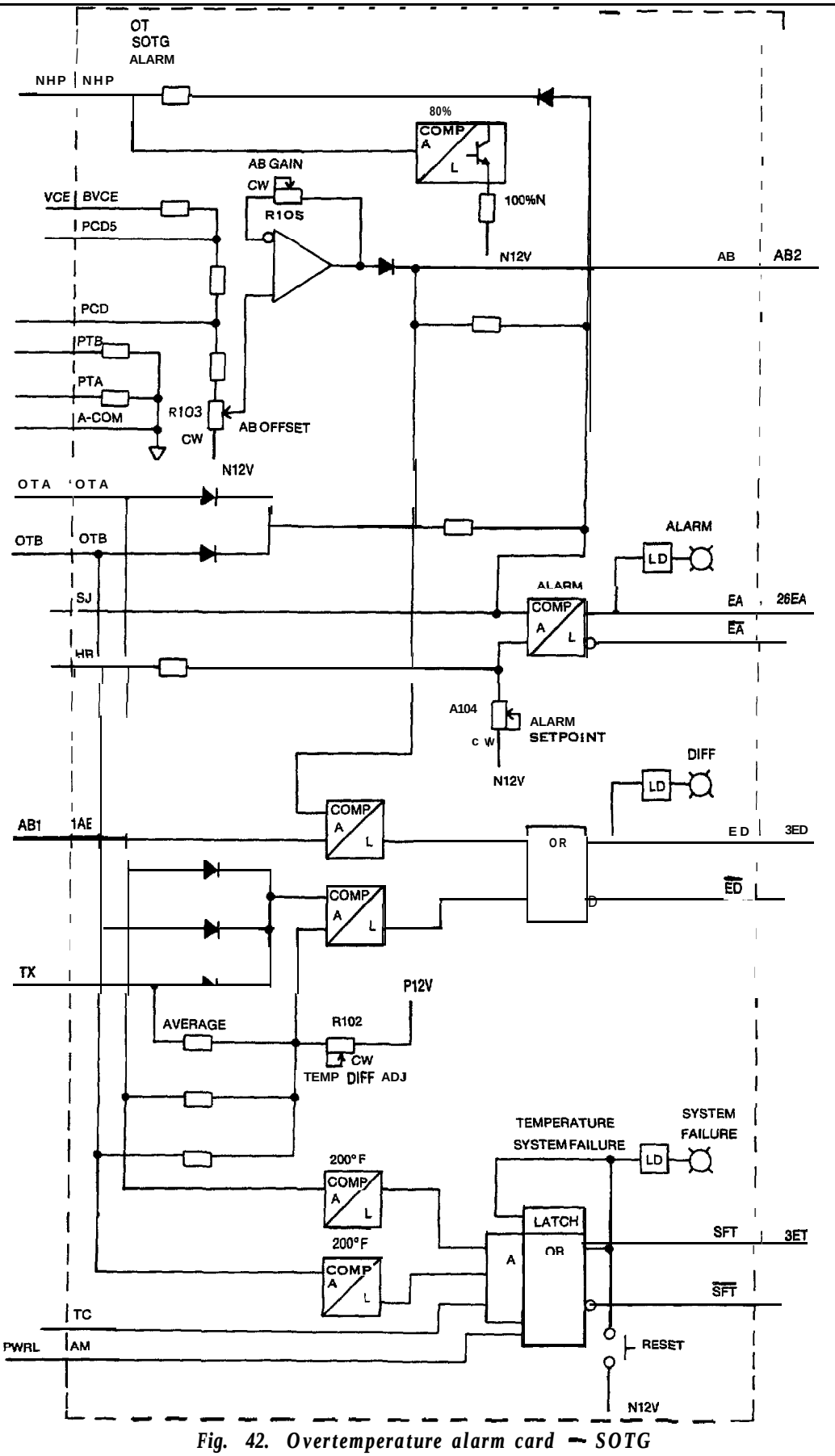


Fig. 42. Overtemperature alarm card — SOTG

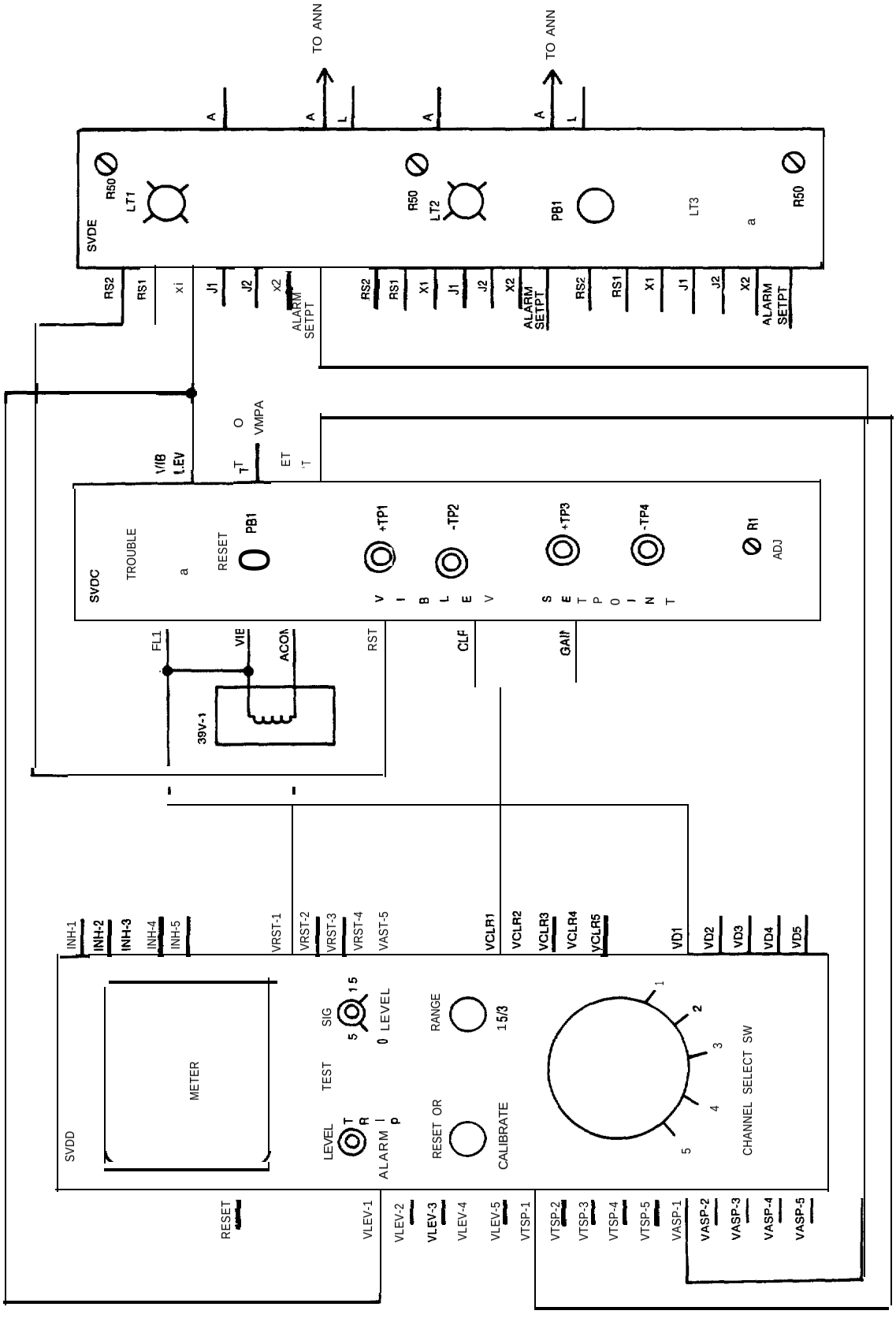


Fig. 43. Vibration protection system (Alarm and trip)

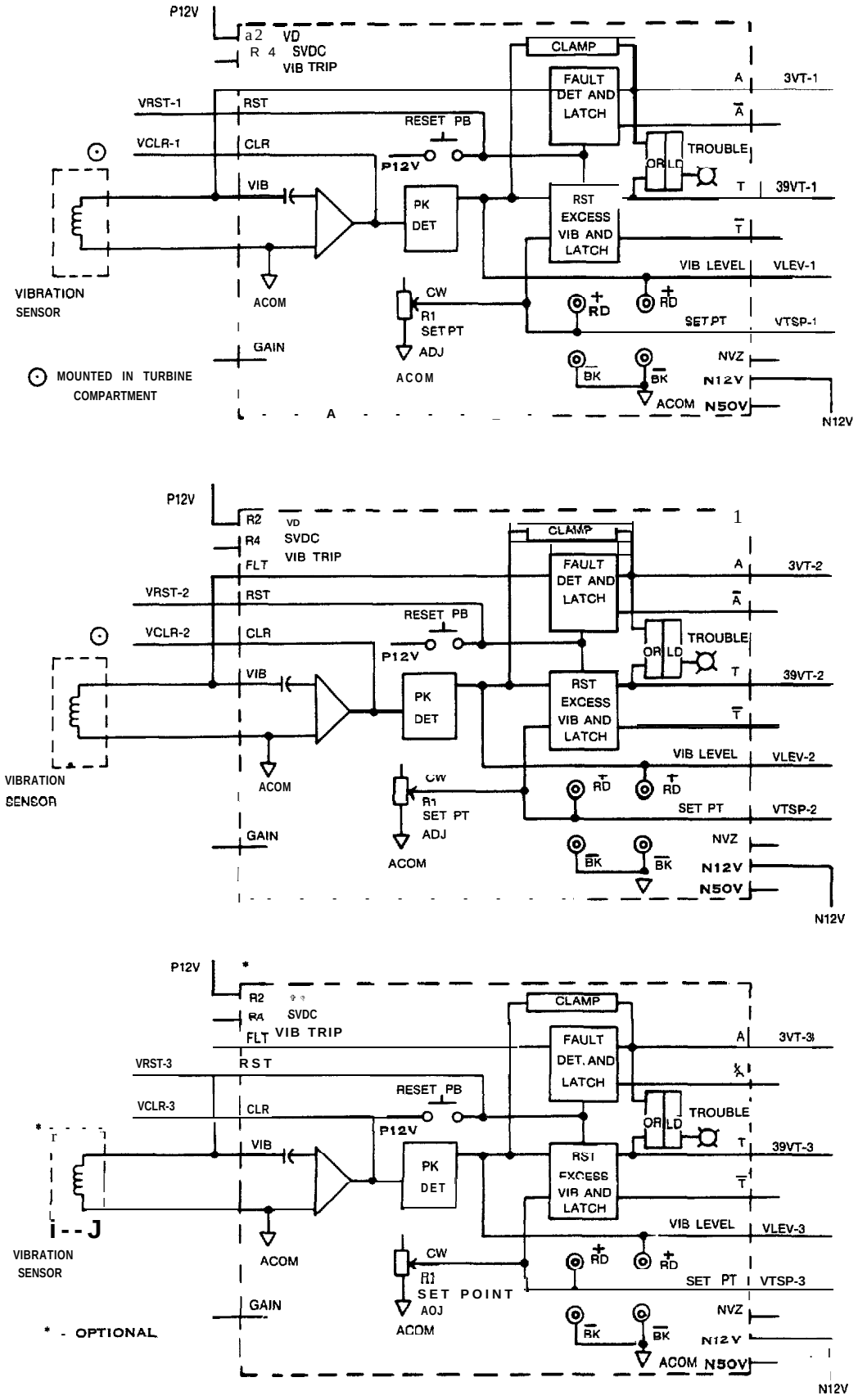


Fig. 44. Vibration protection card — SVDC

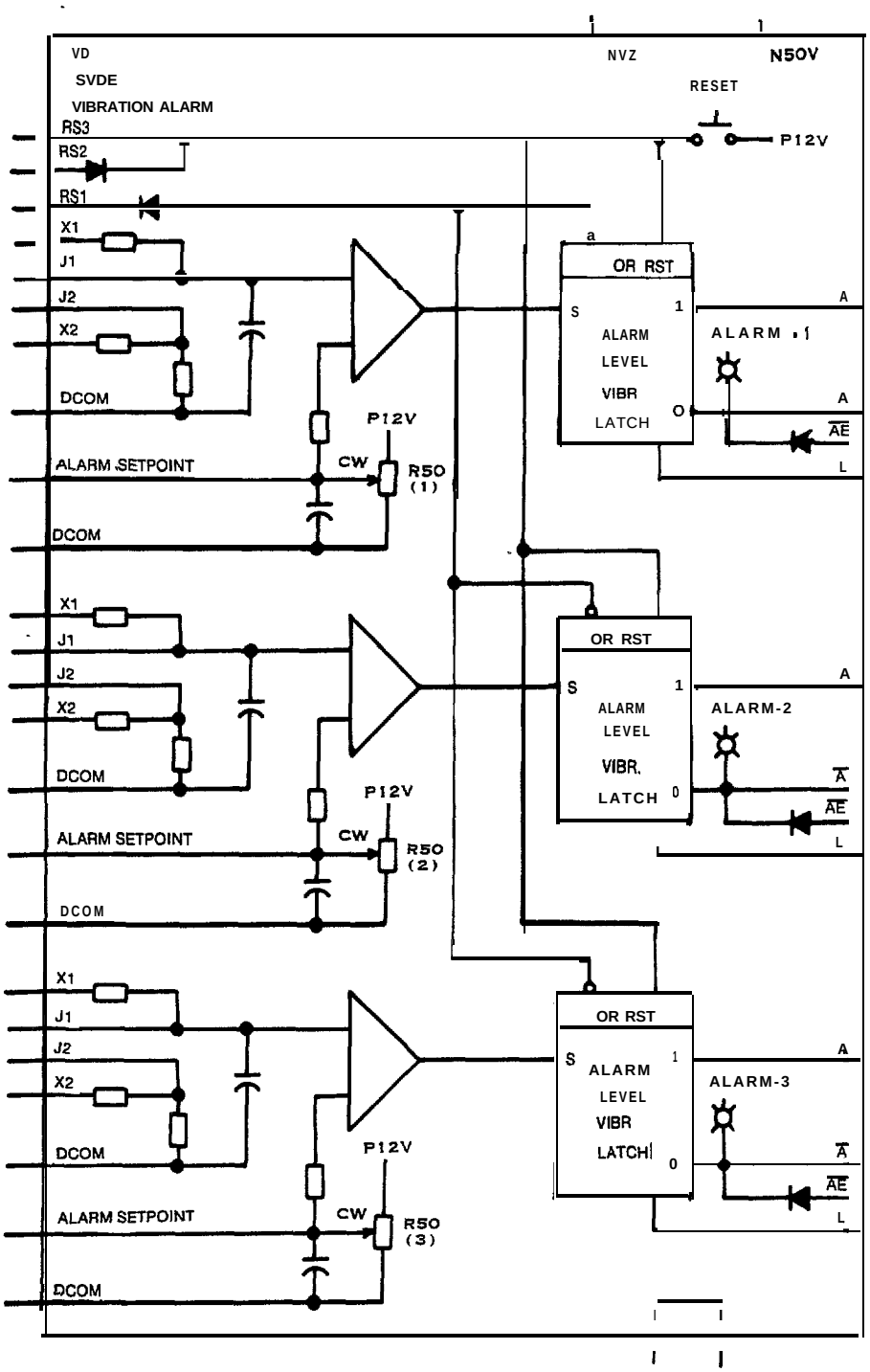


Fig. 45. Vibration alarm card — SVDE

SVDD Card (Optional) (See Fig. 46, page 70)

The SVDD card provides a means of checking and adjusting the SVDC card. The "channel selector" switch on SVDD selects the SVDC card to be checked. The SVDD calibrator card has four output signals. (1) an inhibit signal which may be used to prevent tripping the system when the calibrate push button is depressed, (2) a clear signal to clear trip at completion of test, (3) a reset signal to reset latched trips, (4) a 100-hertz signal with adjustable magnitude to simulate the sensor signal.

The inputs to the calibrator are the vibration level and setpoint level from the SVDC card. The "meter selector switch" determines which of these will be displayed on the SVDD meter. The "calibrate" push button must be depressed to connect the SVDD inputs and outputs to the SVDC card. Any trips in a SVDC card will be reset upon depressing or releasing the calibrate push button.

Adjustments:

ADJUSTMENT	POTENTIOMETER	RANGE
LEVEL	R56	0 to 0.7 volts RMS

D-5 FLAME DETECTOR

A typical flame detection system is shown in Fig. 47. Each channel of the flame detection system receives an input from a separate ultra-violet scanner. Either one of two systems is presently being used: (1) McGraw - Edison scanner and SFIB detection card; (2) Honeywell scanner and detection system. For the Honeywell system, refer to the Honeywell publication, in Volume II of the instruction book. The SFIB system is discussed below.

SFIB Card (See Fig. 47, 48, pages 71, 72)

The SFIB card detects the presence of flame by looking at the current drawn by the ultra-violet scanner. A 1 K-Hertz square wave between pins F11 and F12 is fed to the scanner. When the scanner detects presence of flame, the current out of F11 increases. When this current reaches a level set by R49, output FLM will become a "0" indicating presence of flame provided no-fault condition is indicated by a "1" on FLT. FLT will be a "1" if sensor is open.

Adjustments:

ADJUSTMENT	POTENTIOMETER
SENS	R49

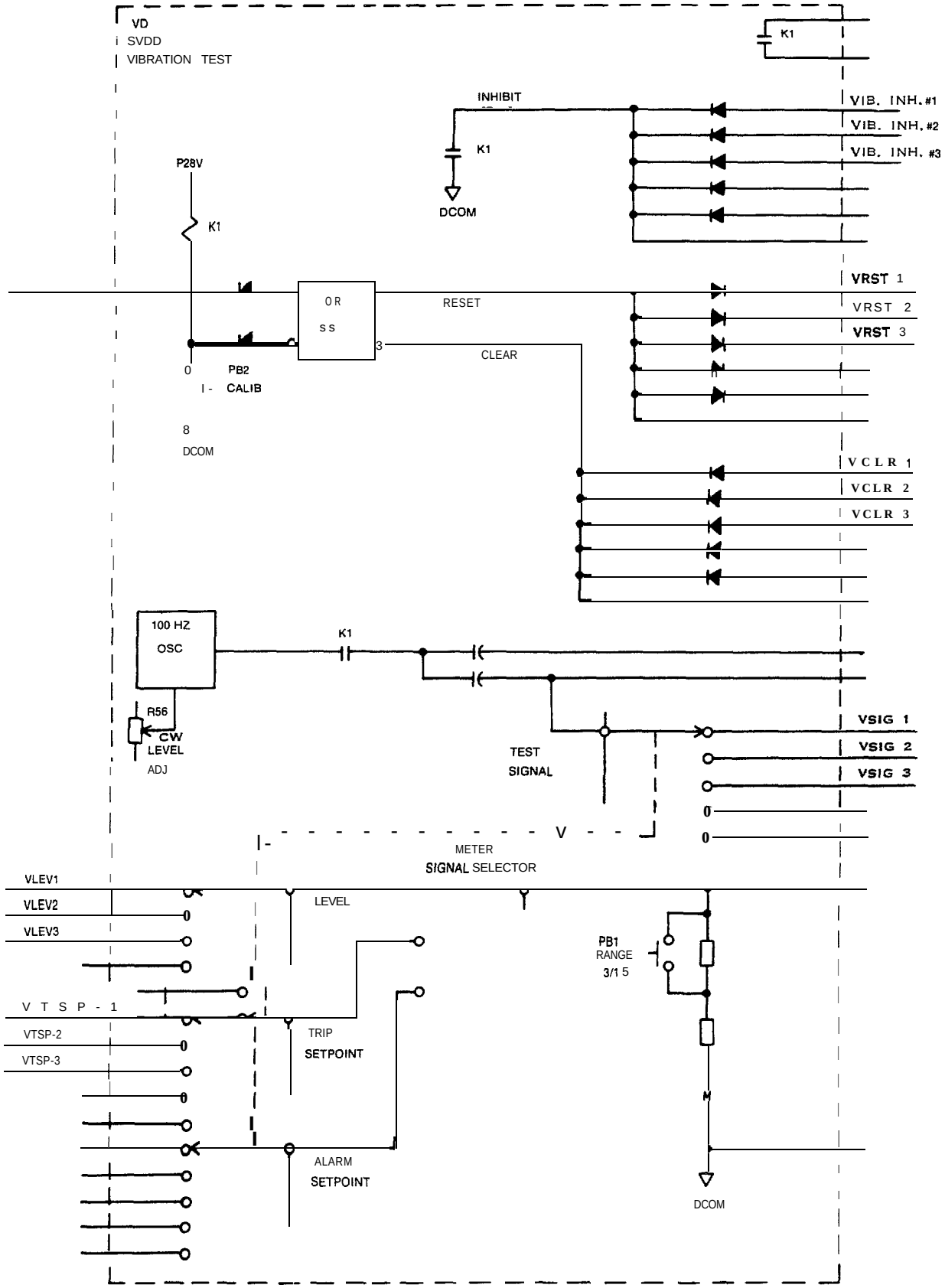


Fig. 46. Calibration card —SVDD

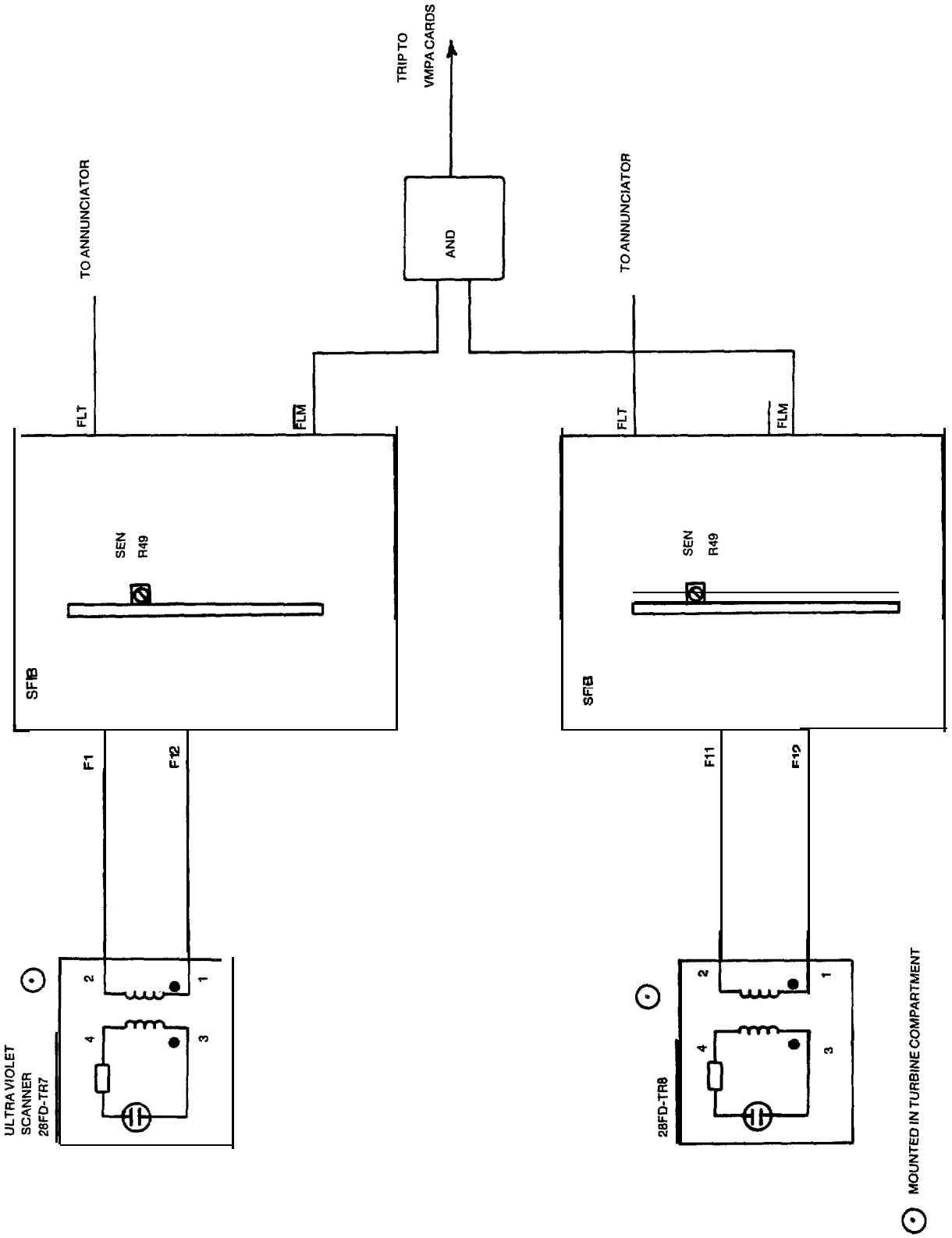


Fig. 47. Typical flame detection system

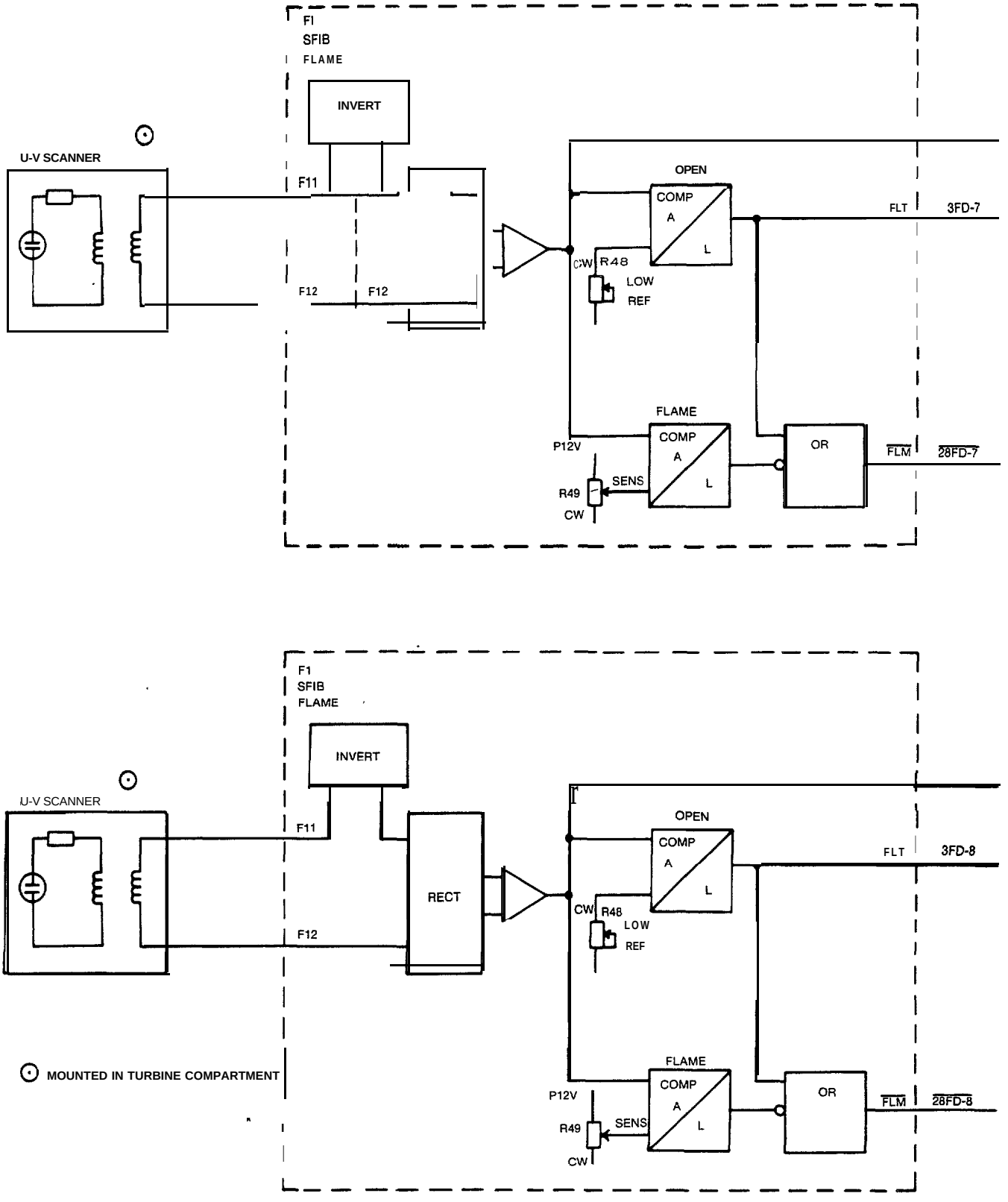


Fig. 48. Flame detector card — SFIB

D-6 GROUND DETECTORS (See Fig. 49)

Ground detector circuits are provided for indication of a ground on the P28, P5.3, P12, N12, ACOM and P125-volt busses. A ground on any of these busses will be annunciated. The following table gives the impedance to ground required for detection.

Bus	Impedance for Fault
P28	*2.0 - 3.3* MEG Ohms
P12	*0.82 - 1.5* **MEG** Ohms
P5.3	*390 - 680* K Ohms
N12	*7.5 - 15* K Ohms
ACOM	*7.5 - 15* K Ohms
P125	**Nominal - 5 K Ohms**

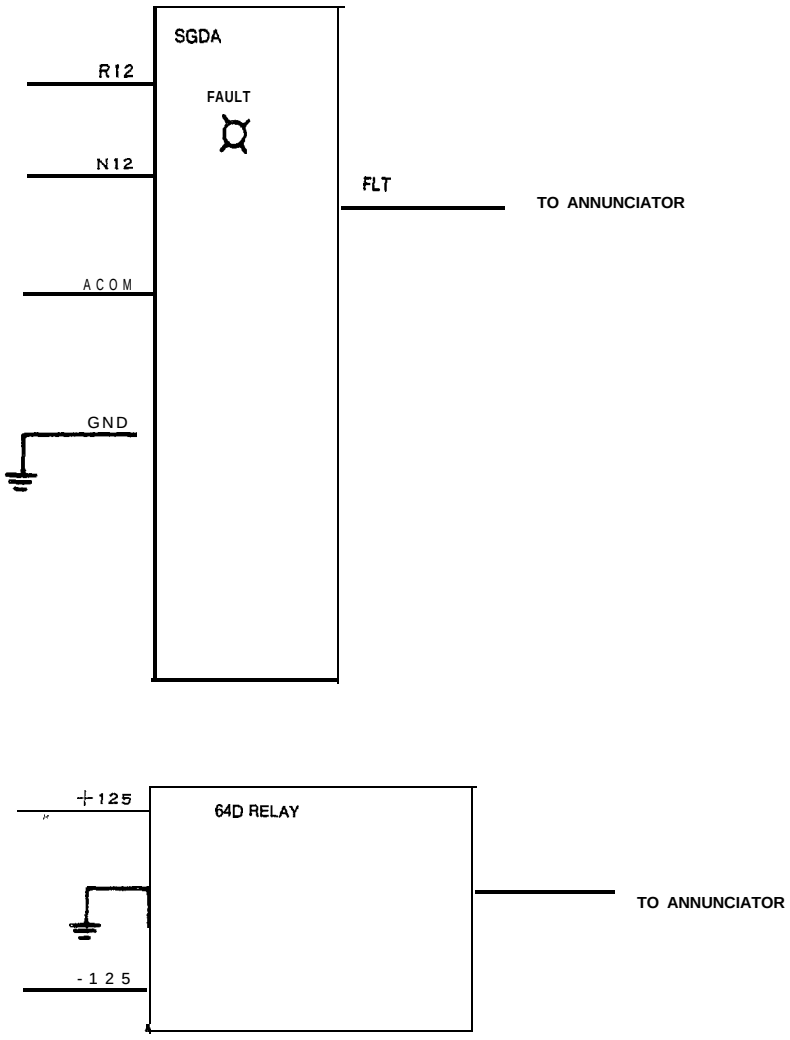


Fig. 49. Ground detectors

SECTION E
POWER-SUPPLY SYSTEM

A block diagram of the power-supply system is given in Fig. 50, page 79. The system is broken down into the following three functions (parts):

(a) Source selection
(b) D-c bulk supply
(c) Final regulator

E-I GENERAL DESCRIPTION

(a) Source Selection

The source selectors allow the d-c bulk supply to utilize power from one of as many as three sources. As illustrated in Fig. 51, page 80, these sources are:

(1) d-c battery
(2) commercial a-c
(3) inductor alternator

The standard configuration is to use one source, namely the d-c battery. In this case, no source selection modules are **needed;** the d-c bulk source is connected directly to the battery. The source selectors are available as options to satisfy special case requirements. The operation of the source selectors is almost obvious when reference is made to Fig. 52, page 81. Source selector IC3603A166A has two possible inputs, d-c battery and/or a-c commercial power. The battery input has:

(1) fusing, 20 amperes
(2) filtering for transcient voltage sup pression.
(3) lamp (DSI) for "power on" indication
(4) diode **to** isolate the battery source from other sources

The a-c commercial input has:

(1) fusing, 20 amperes
(2) transformer isolation with taps
(3) rectification with filtering
(4) lamp (DSZ) for "power on" indication
(5) diode to isolate the source from other sources

Source selector IC3603A167A is used only for an inductor alternator source. This module performs the following:

(1) fusing, 20 amperes
(2) autotransformer with taps
(3) rectification and filtering
(4) pre-regulation to 120 volts d-c
(5) current limit
(6) fusing, 15 amperes
(7) lamp for "power on" indication
(8) **crowbar,** to short circuit output if the voltage exceeds 135 volts d-c

The source with the highest voltage automatically takes over and supplies the current.

(b) D-c Bulk Supply

Basically, the d-c bulk supply performs the following:

(1) accepts a nominal 125 volts d-c voltage
(2) converts to a-c (350 to 450 hertz)
(3) transforms voltage to appropriate levels to feed the P28V, P12V, N12V, and P5V power-supply circuits
(4) **provides** isolation from the primary (i.e. the 125-volt d-c circuitry)
(5) rectifies and filters the above voltages
(6) feeds the P28V bus directly from the d-c bulk supply
(7) feeds the P12V, N12V, P5V busses via the final regulator as illustrated in Fig. 53, page 82.

The main features of the d-c bulk supply are illustrated in Fig. 52. Working from left to right in Fig. 52:

(1) on-off switch
(2) fusing, 20 amperes
(3) thyrector, for spike suppression
(4) inductor, to isolate switching spikes (generated in bulk supply) from getting back onto **the source bus**
(5) 11,800 microfarads of capacitance to allow the power supplies to ride through a 65-msec source voltage dip
(6) isolating diode, to prevent a voltage dip on the source from discharging the capacitance of step (5)
(7) 100-microfarad capacitor, to provide a-c coupling *to* ground
(8) fusing, 15 amperes, current limit (fast) type

E-I GENERAL DESCRIPTION (Continued)

(9) transformer; the 125 volts d-c (nominal) is applied to the center of the primary, with each end alternately switched to the return 125 volts d-c, via switching transistors. The secondaries are rectified and filtered to provide d-c voltages to the final regulator and the P28V bus (refer to Fig. 53)

(10) The EPSU card provides control of the switching transistors of step (9), and provides control power to EPSV. The EPSU is in turn controlled via pulses from the EPSV card.

(11) The EPSV card is a coarse P28V voltage regulator; it senses the output voltage on the P28V secondary winding, compares it to a reference and sends out control pulses to EPSU (step [10]). EPSV also provides a shunt load control (power transistor) to ensure continuous current in the filter inductor (minimum of three amperes).

A photograph of the bulk supply is given in Fig. 54, page 83.

(c) Final Regulator

The functions of the final regulator are illustrated in Fig. 53, page 82, and are as follows:

(1) Module IC3601A250B (including EPSW):

- Steps down a coarsely regulated voltage, P12X, to a finely regulated voltage, P12V;
- provides current limit;
- provides current and voltage jacks for measuring purposes;
- provides overvoltage protection by means of a crowbar circuit (i.e. by firing an SCR).

(2) Module IC3601A250A: same as Step (1) except P5V bus is controlled instead of the P12V bus.

(3) Card EPSX: same as Step (1), except N12V bus is controlled instead of the P12V bus and overvoltage protection is via a diode instead of an SCR (provides protection against being pulled positive).

(4) Card EPSY:

- provides an outside, fine voltage, control loop on the P28V bus, i.e. it trims the coarse voltage regulator EPSV.
- provides current limit control via a "STOP" signal to EPSV.
- provides current and voltage jacks for measuring purposes.

(5) Card SBMB (Optional): This card monitors the output voltages on the four busses and provides a latching alarm signal (generally taken to the annunciator) if any of the voltages exceeds certain limits (low/high). The alarm reset push buttons (four) are on the card front, along with the LED (four) alarm indicators.

A photograph of the final voltage regulator is given in Fig. 55, page 83.

E-2 APPLICATION INFORMATION

(a) Source Selection

(1) The d-c battery input characteristics are as follows:

- Voltage range - 99-134 volts d-c
- The battery usually consists of 56 cells, floated at 2.4 V/C;
- No load - 2.0 V/C, discharge = 1.75V/C

(2) The commercial a-c input characteristics are as follows:

- Nominal volts +5%, -10%; site to site variation of nominal is 130-105 volts RMS.
- Frequency - 50/60 Hz; a-c is sometimes grounded.
- The commercial a-c input is taken to a transformer which has five primary taps of approximately B-volt voltage steps. The primary nominal voltages are:

Winding	Volts In	Volts out (X1-X2 Winding)
H1-H2	111	100
H1-H3	116	100
H1-H4	122	100
H1-H5	128	100
H1-H6	135	100

At a particular installation the nominal voltage level should be determined. The next higher voltage windings should be used as the input. This is done to avoid the possibility of developing over 140 volts output. Voltages in excess of 140 volts can damage relays and auxiliary equipment which are supplied by the output.

E-2 APPLICATION INFORMATION (Continued)

- Output voltage-determined by input voltage and transformer winding used. Nominal peak no-load output voltage will be approximately 140 volts d-c. *The* output regulation at an output current of 15 amperes is approximately 15 volts.
- Output current-Maximum is 1'7 amperes.

(3) The inductor alternator input characteristics are as follows:

Range of rated frequency (frame size l-7) = 12-3 kHz

Nominal volts at 100% speed= 135 volts

Machine-to-machine variation at 100% speed, volts = +10%

Voltage and frequency proportional to speed

Range of speed for each turbine = 80-110%

Nominal operating range, generator drives = 100%

Nominal operating range, pumper drives = 80-105%

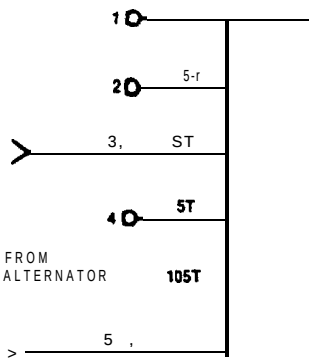
Nominal operating range, marine drives = 80-105%

Overspeed test = 110%

Full load (16 amperes) regulation at 0.85pf = 20%

The input from the inductor alternator is fed to an autotransformer, per Fig. 51.

The autotransformer is provided with three 5% taps. If the output voltage of the converter at TB1-4 falls below 120 volts d-c or is greater than 225 volts d-c, the autotransformer taps should be changed. The transformer is wired for a nominal open circuit transformation ratio of 1.09. If the output voltage is low, move wire from 3 to 4; if the output voltage is high, move wire from 3 to 2.



The current limit is set to 16 amperes (EPSA, R61, bottom)

The voltage output is set to 120 volts d-c on TB1(1) to TB1(2) (EPSA, R62, top).

The crowbar voltage is set to 125 volts d-c SCBN, R3).

(b) Bulk Supply

(1) Input voltage range is 95 to 140 volts d-c

(2) Input current is 2.1 to 7.3 amperes at 95 volts d-c and 1.75 to 6.1 amperes at 140 volts d-c.

(3) Outputs are as follows:

Terminal Points	Voltage (8%-100% Load)	No Load Voltage	Rated Current
P30	29.5-31	31	15 Amp
P12X	18-20	34.8	6 Amp
N12X	18-21	32.7	1 Amp
P5X	10-13	20.8	6 Amp
P11	11-11.5		30 Amp

(4) The current signal across P30V to P28X (O.l-ohm shunt), is 0.3 volts d-c to 1.5 volts d-c for a zero to 15-ampere load on the P28V bus. A shunt regulator maintains a minimum to give continuous current even when the load on the P28V is zero

(5) The supplies will not dip during a diesel start (or equivalent), defined as follows: dip to 60 volts, back *to 80* volts in 65 msec.

(6) Card EPSU (Primary supply control)

Functions:
- primary transistor base drive,
- primary current limit
- control power supply
- protection - if base and collector high, base drive re moved and LED indication

Inputs:
- collector volts (CA, CB)
- emitter currents. EA1-3, EB1-3, SCA, SCB
- pulses (control) from coarse regulator card EPSV (time ratio control)
- +51-volt zener input for control power (feeds inverter)

outputs:
- Base drive
- Control power supply to EPSV, 13-volt, 5 kHz
- LED indication for failure

E-2 APPLICATION INFORMATION (Continued)

(7) Card EPSV (P28V coarse regulator)

Functions:
- coarse voltage regulator on 28-volt supply output-voltage and providing control pulses to EPSU (above)
- control to shunt current regulator to ensure +12, -12, +5.3 stay up
- provides control power to other cards via P11 (11 volts d-c) which does not disappear when inverter shuts down

Inputs:
- control power ACC1, ACC2, from EPSU
- voltage feedback, AC1, AC2, corn.
- stop-logic, "1" shutdown inverter "0" starts inverter
- Reg - input from fine voltage regulator EPSY.
- Signal bias operation of course regulator
- Current bias operation of course regulator
- Current signal across shunt P30, P28X

outputs:
- Pulses to EPSU (time ratio control)
- control supply P11, 11 volts
- S corn - return for P11
- SH - base drive to shunt current regulator

(c) Final Regulator

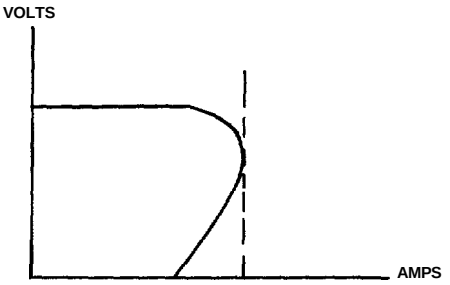
(1) The final regulator has the following output characteristics:

Volts	+28V*	+12V	-12V	+5.3V
Bus	P28V	P12V	N12V	P5V
Return	P COM	A COM	A COM	D COM
Rated Current	15A	6A	1A	6A
Current Limit	17A	7A	1.2A	7A
Ripple UP-p	250 mV	10 mV	10 mV	100 mV
Reg. (LD)	300 mV	30 mV	30 mV	150 mV
Drift (Temp)	—	60 mV	60 mV	40 mV
100% Current Shunt-Signal	1.5V	1V	2V	1V

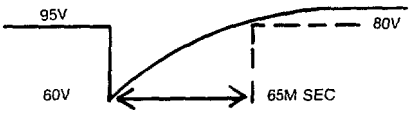
*For input of 105 to 140 volts d-c; for input down to 95 volts, reduce 28V load to 7A or accept P28V drop to 26V.

(2) An SCR is used on each of the P12V and P5V busses and is called a "crowbar" circuit. Its purpose is to monitor the busses and protect the load equipment from overvoltage. The P12V bus crowbar operates at 15.5 to 19.5 volts; the P5V bus crowbar operates at 8-10.5 volts; the N-12V bus has a diode to protect against positive going voltages; the P28V has no crowbar, since its load is not as voltage sensitive as the other busses.

(3) The supplies have a current limit foldback as follows:



(4) The supplies will not dip during a diesel start (or equivalent), defined as follows:



(5) A-c voltage, 350-450 Hz, 60 volts RMS is available from the bulk supply; for a-c loading, reduce P28V volt-amps by an equivalent amount; non-standard.

(6) Jacks on card fronts, for both current and voltage signal.

(7) Monitors over/under voltage on all supplies, and +28-volt supply overcurrent monitoring with LED and logic annunciation.

(8) EPSY (P28 fine regulator)

Functions:
- fine voltage regulation
- current limit

Inputs:
- P11 - control supply
- P28 - 28V bus, for ring bus monitor
- P30-P28X - for current limit signal
- P30-PCOM raw d-c supply
- sense - remote voltage sensing

E-2 APPLICATION INFORMATION (Continued)

outputs:
- Regulation -fine voltage regulation bias to EPSV.
- $\overline{CL}$ - current limit logic signal "0" = CL; to be used with monitor
- STOP - to shut off inverter on CL "1" = inverter OFF

(9) EPSW (P12V, P5V regulator)

Functions:
- -regulate +5.3 volts or +12 bus
- current limit on +5.3 volts or +12 bus
- crowbar +5.3 volts or +12 bus

Inputs:
- P28X - control power
- P12X or P5X - power
- V sensing - remote sensing

outputs:
- Gate to crowbar SCR
- V_m - to monitor
- base drive to series transistor regulator

(10) EPSX (12V regulator)

Functions:
- Regulate - 12 volts
- Current limit

Inputs:
- N12X - power
- V_m - (+12-volt supply) control power
- V_s - remote sensing

outputs:
- V_m - monitor voltage
- N12V - output power

(11) SBMB (Bus Monitor)

Functions:
- monitor under/over voltage on four supplies detection at 10-15% error.
- annunciate with LED (1st fault indication)
- signal logic; 1 = alarm (4)
- Monitor current limit on EPSY (28V regulator), $\overline{CL}$ = 0 = current limit condition
- Local reset PB and remote PB

Inputs:
- V_m for each supply (monitor voltage)
- P11; control power from EPSV
- $\overline{CL}$ - CL from +28 regulator

Outputs:
- $\overline{STOP}$ — "0" for normal condition; may be used to shut down inverter.
- Logic signals - alarm (4); 1 = alarm

E-3 TROUBLESHOOTING GUIDE

These instructions are written to aid the troubleshooter in isolating power supply troubles. This is a guide only, and cannot be applied blindly without some knowledge of troubleshooting procedures and of the power supplies, themselves. It is recommended that the troubleshooter, if not already familiar, briefly review the general description of the power supply system before proceeding with these instructions.

(a) **Symptom:** P5V or P12V bus out of limits, but all other busses correct.

Action:

(1) It's possible that the "crowbar" protection circuit has been energized due to external voltage spiking. Verify that external instrumentation, etc., is not disturbing the bus. Remove power for several seconds (to allow SCR to reset) and reapply power.

(2) If fault still present, remove power, replace EPSW, reapply power.

(3) If fault still present, verify that input to final regulator module (P12X or P5X) is as defined in application section E-2b, (3); if not, the bulk supply is suspect. (i.e. the diode-filter section of the bulk supply).

(4) If step (3) was correct, remove power, isolate all loading on the power supply in trouble, reapply power. If the power supply is correct now, the trouble is probably external to supply, either excessive load or overvoltage from other source.

(5) If power supply is still not correct, module IC3601A250A (P5V) or IC3601-A250A (P12V) is suspect.

E-3 TROUBLESHOOTING GUIDE (Continued)

(b) Symptom: N12V bus is out of limits, but other busses are correct.

Action:

(1) Suspect EPSX; remove power, replace EPSX.

(2) If bus still not correct, isolate load and check N12V bus. If the bus returns to normal, the trouble is probably external to the power supply system.

(3) If bus still not correct, measure voltage on N12X (input to EPSX) and verify that it is as defined in application section E-2b, (3). If not, the bulk supply is suspect (i.e. diode-filter section).

(c) Symptom: P28V bus out of limits, but the other busses correct.

Action:

(1) EPSY is suspect; remove power, **replace** EPSY.

(2) If bus is still not correct, isolate the load and check P28V bus. If the bus returns to normal, the trouble is probably external to the power supply.

(3) If bus still not correct, **measure the voltage** on P28X, and verify that it is as defined in application section E-2b, (3); if not, the bulk supply is suspect; i.e. diode rectifier output is suspect.

(d) Symptom: All power supply busses are low, but EPSU light emitting diode (LED) is off, and bulk supply making a noise (hum).

Action:

(1) Check that bulk supply input is 95 volts d-c or greater.

(2) Check that the "STOP" signal (with respect to PCOM), labeled interconnections between bulk supply and regulator, is a logic "zero."
Use an oscilloscope if available. This signal should stay at zero volts. If the P28V supply is running into current limit, it will bounce between "zero" and approximately 2 volts d-c. Also check the voltage signal across the current jacks of the EPSY card and compare to the nominal current limit signal of 1.7 volts d-c.

(3) If the trouble is still not determined, remove power, isolate the P28V load, reapply power and check for correct bus voltages. If the busses are still low, jumper "STOP" to "PCOM." If the bus voltages are now correct, EPSY is suspect. If the bus voltages are still low, bulk supply cards EPSU, EPSV are suspect.

(4) If the **busses** became correct when the P28V load is isolated, suspect P28V supply current limit trouble or primary current limit trouble. But P28V current limit was **checked** in step (2). Therefore, primary current limit, that is EPSU, is suspect **(change** out EPSU).

(e) Symptom: EPSU light emitting diode "LED" is on; bulk **supply will** normally be shut down, i.e. will not be making a noise (hum).

Action:

(1) Remove power and check all power transistors on bulk supply heat sink HS1, HS2.

(2) If transistors are correct suspect EPSU and replace.

(f) Symptom: Bulk supply not humming, EPSU "LED" is off, but indicating light, DSI is on.

Action:

(1) Remove power and check fuses; replace any bad fuses.

(2) Remove EPSU, EPSV and check power transistors on HS1 and HS2; replace any bad transistors.

(3) Apply power and check for battery voltage from CA to H-COM and from CB to H-COM (located on TB2). Battery voltage must be between 95-145 volts d-c. If this voltage is not present, suspect bad power transistor or some other low impedance path on the front **end.** Remove power, isolate and remove this low impedance path.

(4) If battery voltage is correct on CA, CB, check for +51 volts d-c between pins 11, 51 in the EPSU slot.

(5) If the problem has not been identified to this point, remove power, plug in new EPSU and old EPSV cards and check system. If operation is still ***not*** correct, remove power, replace EPSV, and try again.

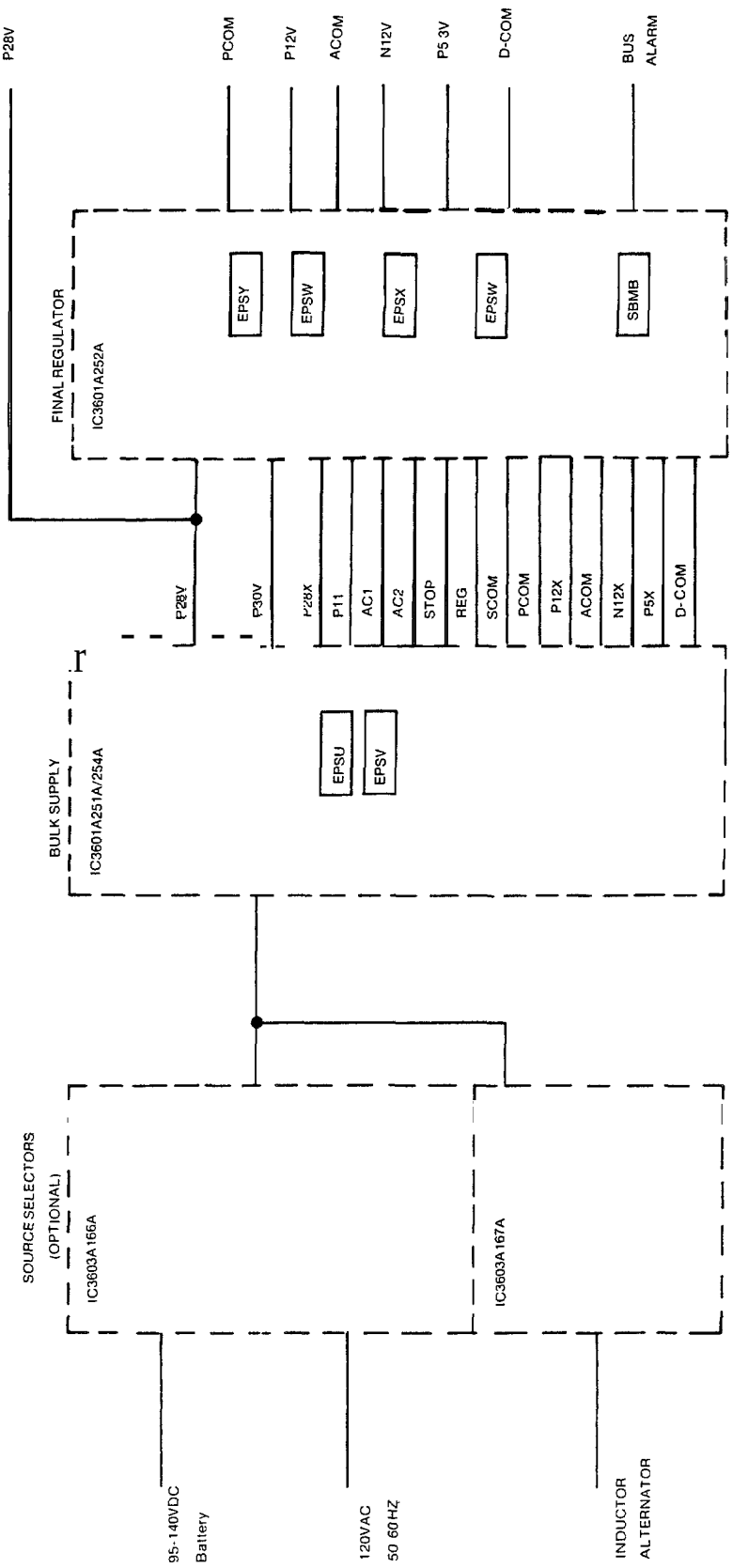


Fig. 50. MARK *II power supply configuration*

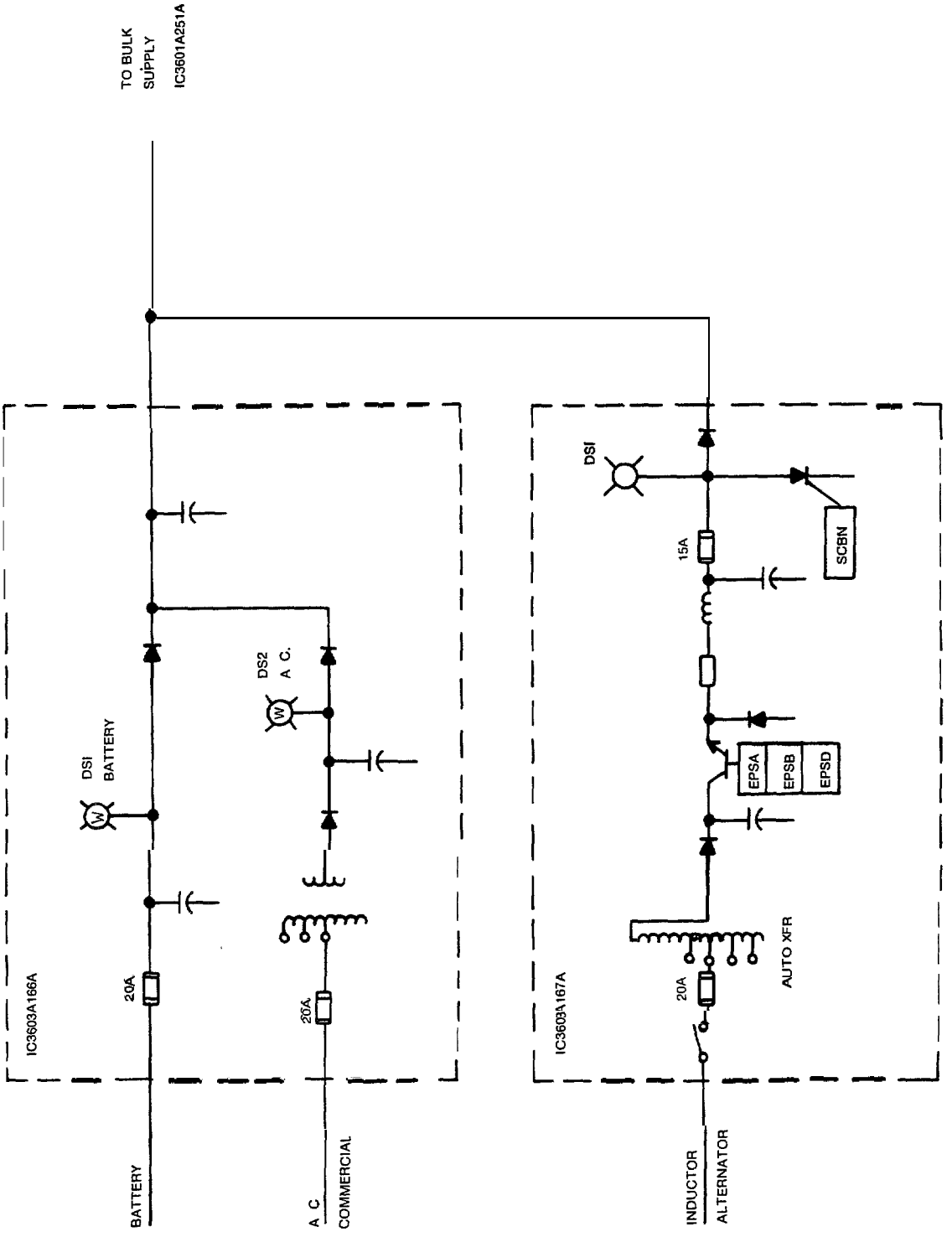


Fig. 51. Source selector for MARK II power supply (when used)

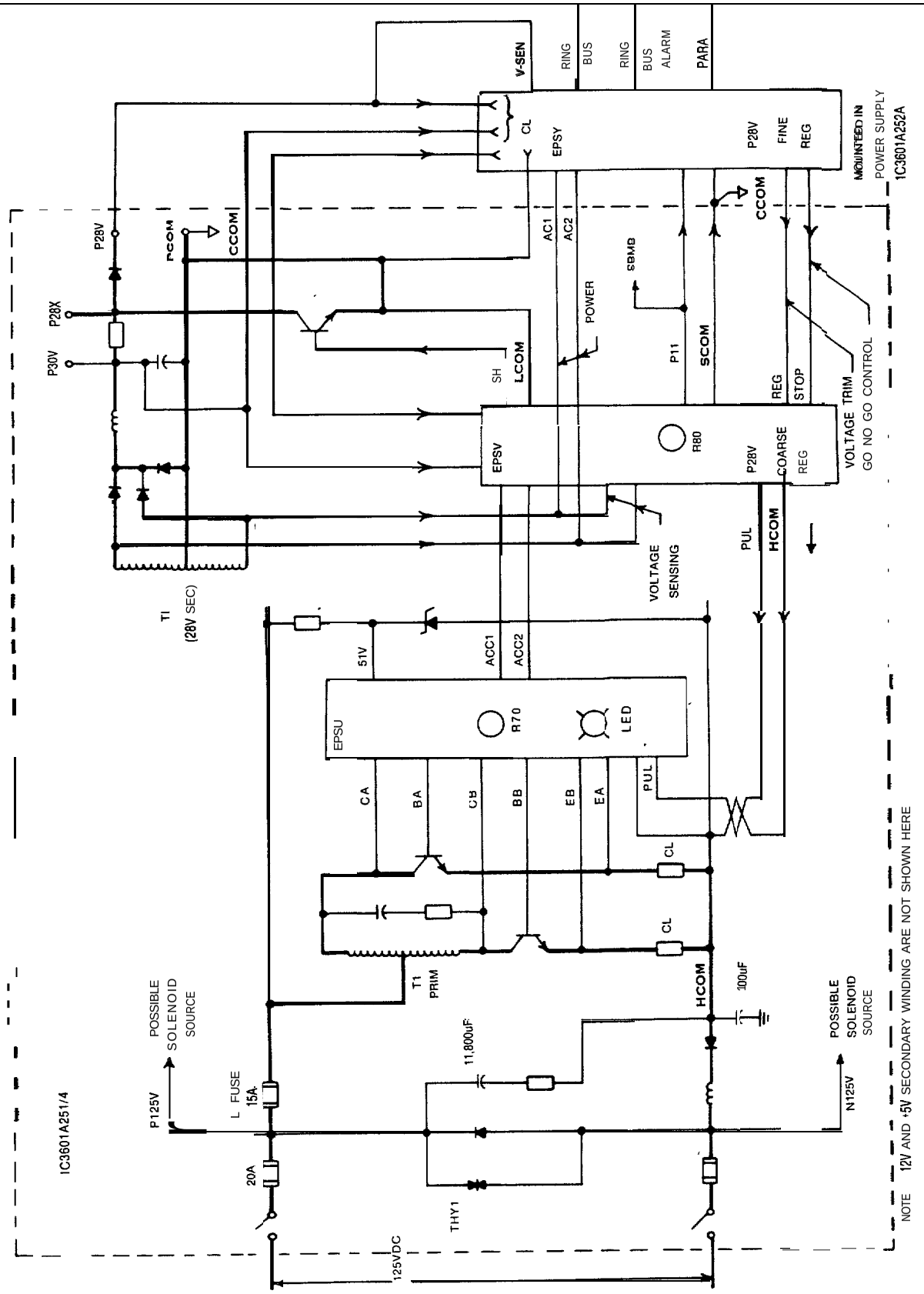


Fig. 52. D-c bulk power supply

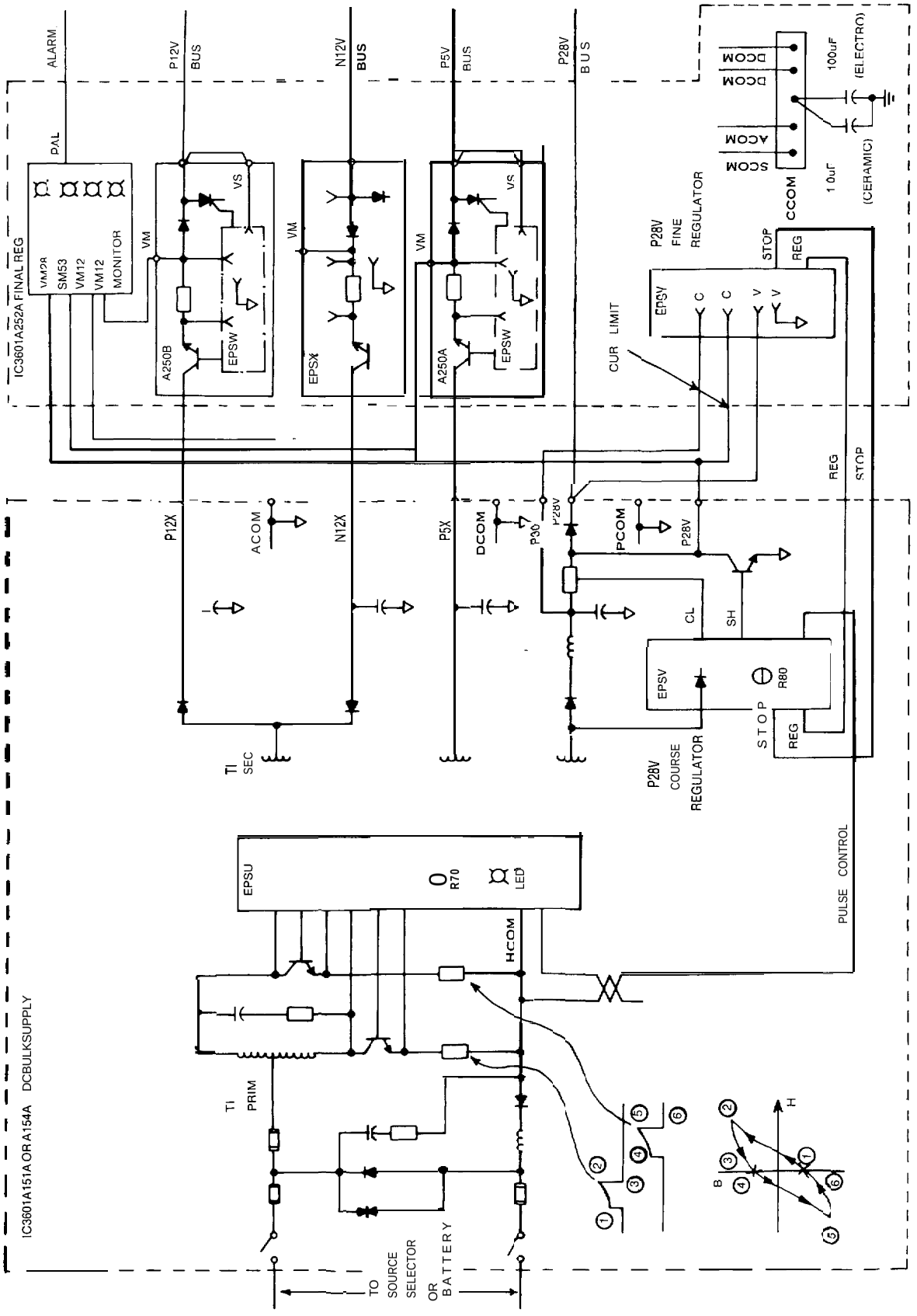


Fig. 53. D-c bulk supply and final regulator

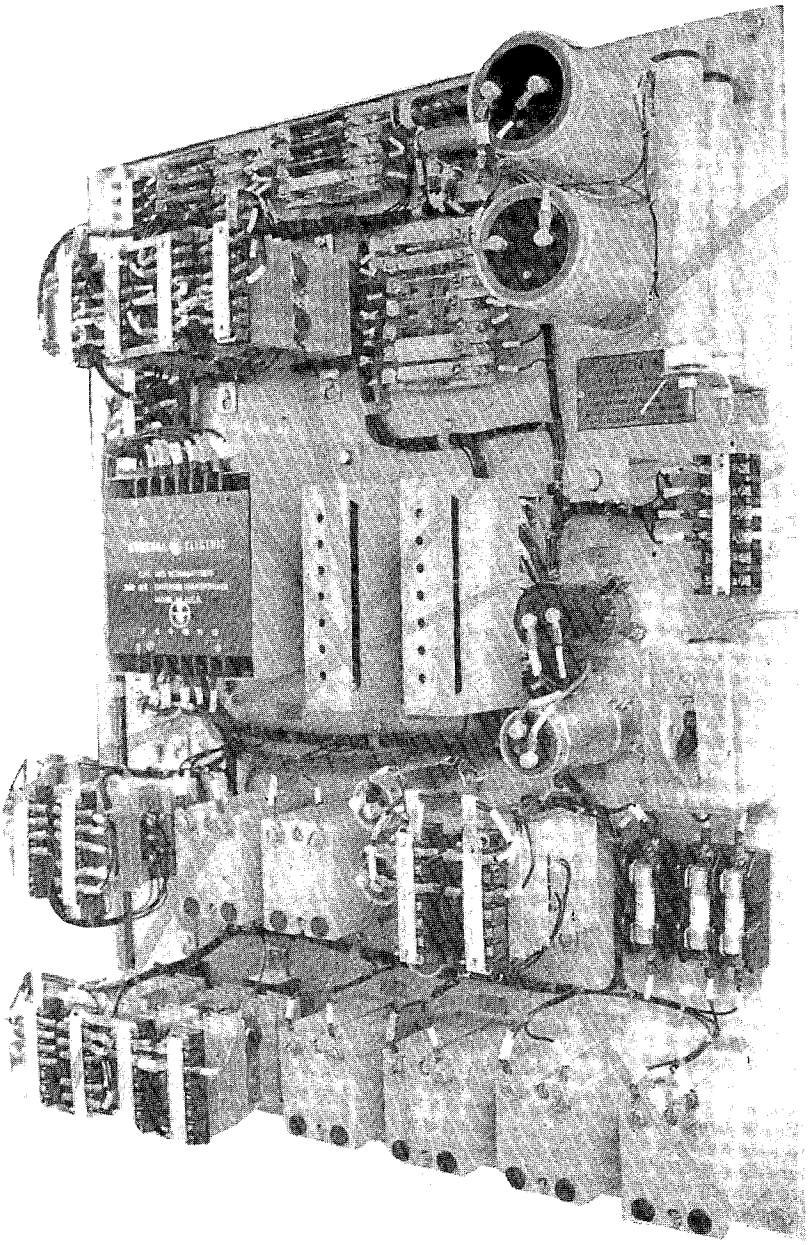


Fig. 54. Bulk power supply

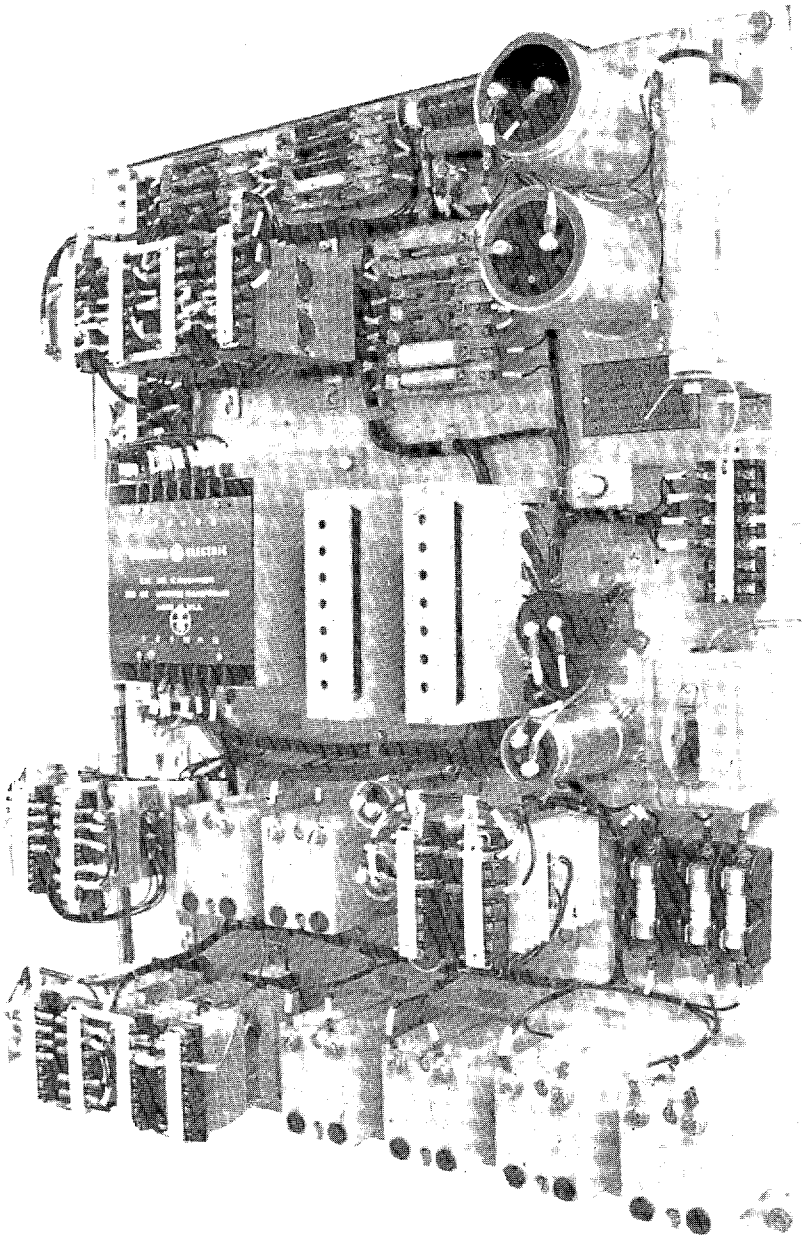


Fig. 54. Bulk power supply

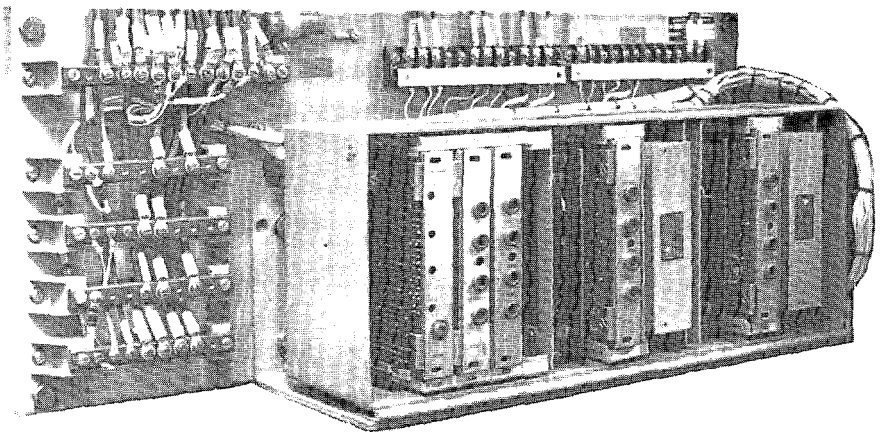


Fig. 55. Voltage regulator